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Introduction

In 3GPP, there are multiple northbound API-related specifications (e.g. APIs for Service Capability Exposure Function (SCEF) functionalities defined in 3GPP TS 23.682 [2], API for the interface between MBMS service provider and BM-SC defined in 3GPP TR 26.981 [5]). To avoid duplication and inconsistency of approach between different API specifications, 3GPP has considered the development of a common API framework (CAPIF) that includes common aspects applicable to any northbound service APIs.

The present document specifies the functional model, procedures and information flows needed to support the CAPIF, and the guidelines for consistent northbound API (service and CAPIF APIs) development in 3GPP.

NOTE: It is possible to use the CAPIF defined common aspects for other APIs as well, apart from northbound APIs.

1 Scope

The present document specifies the architecture, procedures and information flows necessary for the CAPIF. The aspects of this specification include identifying architecture requirements for the CAPIF (e.g. registration, discovery, identity management) that are applicable to any service APIs when used by northbound entities, as well as any interactions between the CAPIF and the service APIs themselves. The common API framework applies to both EPS and 5GS, can be hosted within a PLMN or SNPN, and is independent of the underlying 3GPP access (e.g. E-UTRA, NR).

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document.*

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 23.682: "Architecture enhancements to facilitate communications with packet data networks and applications".

[3] 3GPP TS 23.501: "System Architecture for the 5G System; Stage 2".

[4] 3GPP TS 23.502: "Procedures for the 5G System; Stage 2".

[5] 3GPP TR 26.981: "MBMS Extensions for Provisioning and Content Ingestion".

[6] 3GPP TS 32.240: "Telecommunication management; Charging management; Charging architecture and principles".

[7] ETSI GS MEC 011 (V1.1.1): "Mobile Edge Computing (MEC); Mobile Edge Platform Application Enablement".

[8] ETSI GS MEC 009 (V1.1.1): "Mobile Edge Computing (MEC); General Principles for Mobile Edge Service APIs".

[9] OMA-ER_Autho4API-V1_0-20141209-A: "Authorization Framework for Network APIs".

[10] OMA-TS-REST_NetAPI_Common-V1_0-20180116-A: "Common definitions for RESTful Network APIs".

[11] OMA-TS-NGSI_Registration_and_Discovery-V1_0-20120529-A: "NGSI Registration and Discovery".

[12] 3GPP TS 33.122: "Security Aspects of Common API Framework for 3GPP Northbound APIs".

[13] 3GPP TS 23.435: "Procedures for Network Slice Capability Exposure for Application Layer Enablement Service".

[14] IETF RFC 6749 (October 2012): "The OAuth 2.0 Authorization Framework".

[15] 3GPP TS 23.434: "Service Enabler Architecture Layer for Verticals (SEAL); Functional architecture and information flows".

[16] 3GPP TS 33.501: "Security architecture and procedures for 5G System".

[17] 3GPP TS 28.541: "Management and orchestration; 5G Network Resource Model (NRM); Stage 2 and stage 3".

[18] 3GPP TS 28.532: "Management and orchestration; Generic management services".

[19] 3GPP TS 28.538: "Management and orchestration; Edge Computing Management".

[20] 3GPP TS 32.254: "Telecommunication management; Charging management; Exposure function northbound Application Programing Interfaces (APIs) charging".

3 Definitions and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

API: The means by which an API invoker can access the service.

API invoker: The entity which invokes the CAPIF or service APIs.

API invoker profile: The set of information associated to an API invoker that allows that API invoker to utilize CAPIF APIs and service APIs.

API exposing function: The entity which provides the service communication entry point for the service APIs.

API exposing function location: The location information (e.g. civic address, GPS coordinates, data center ID) where the API exposing function providing the service API is located.

API category: A name used to group service APIs by the service domain in which they are used (e.g., IoT, V2X). The CAPIF core function maintains a list of API category names.

API management function: The entity which enables the API provider to perform administration of the service APIs.

API publishing function: The entity that enables the API provider to publish the Service APIs information in order to enable the discovery of APIs by the API invoker.

Authorization function: The entity which issues access tokens to the API invokers after successfully authenticating the resource owner and obtaining authorization.

CAPIF administrator: An authorized user with special permissions for CAPIF operations.

Common API framework: A framework comprising common API aspects that are required to support service APIs.

Designated CAPIF core function: The CAPIF core function which is configured as the serving CAPIF core function for interconnection.

Network Slice Info: Network slice information of service API which include a list of network slice related identifiers, as described in clause 8.4 of 3GPP TS 23.435 [13]).

Northbound API: A service API exposed to higher-layer API invokers.

Onboarding: One time registration process that enables the API invoker to subsequently access the CAPIF and the service APIs.

Resource: The object or component of the API on which the operations are acted upon.

Resource owner: A UE user or an MNO subscriber capable of granting access to a protected resource related to the invoked API via resource owner function.

Resource owner function: The entity that enables the authorization for resource access and managing and revoking authorization for resource access.

Resource owner-aware northbound API access: An API invocation scenario where the API invoker needs an authorization from the resource owner.

Service API: The interface through which a component of the system exposes its services to API invokers by abstracting the services from the underlying mechanisms.

Serving Area Information: The location information for which the service APIs are being offered to.

CAPIF provider domain: A domain that contains an instance of CAPIF core function and may contain API provider domains and API invokers. The CAPIF provider could be a PLMN, SNPN or 3rd party. Throughout this document, PLMN trust domain is often used as the typical deployment of a CAPIF provider domain however SNPN trust domain or 3rd party trust domain are applicable as well.

PLMN trust domain: The entities protected by adequate security and controlled by the PLMN operator or a trusted 3rd party of the PLMN.

SNPN trust domain: The entities protected by adequate security and controlled by the SNPN operator or a trusted 3rd party of the SNPN. **3rd party trust domain:** The entities protected by adequate security and controlled by the 3rd party.

Application management client: The application developers utilize the CAPIF APIs using an application management client as an API invoker to obtain the service APIs information to implement the application program. Such application programs are hosted on cloud, edge or on a UE.

Hosted applications: The hosted applications (which are programmed to utilize the service APIs) as an API invoker invoke the CAPIF APIs and service APIs as per the application business logic.

Channel Aggregator Platform: The Channel Aggregator aggregates the APIs from one or more southbound CAPIF providers with the intention to re-expose such APIs or expose value-added APIs developed using the APIs from the southbound CAPIF providers to the northbound side Application management client or Hosted applications. The Channel Aggregator Platform as API invoker invokes the CAPIF APIs and service APIs as per its business logic.

For the purposes of the present document, the following terms and definitions given in 3GPP TS 32.240 [6] apply:

Offline charging
Online charging

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

5GS	5G System
AEF	API Exposing Function
AF	Application Function
AMF	API Management Function
APF	API Publishing Function
API	Application Program Interface
AS	Application Server
BM-SC	Broadcast Multicast Service Centre
CAPIF	Common API Framework
CRUD	Create, Read, Update, Delete
DDoS	Distributed Denial of Service
E-UTRA	Evolved Universal Terrestrial Radio Access
EPS	Evolved Packet System
ETSI	European Telecommunications Standards Institute
GS	Group Specification
IP	Internet Protocol
MBMS	Multimedia Broadcast and Multicast Service
MEC	Multi-access Edge Computing
NEF	Network Exposure Function
NGSI	Next Generation Service Interfaces
NR	New Radio
OMA	Open Mobile Alliance

OAM	Operations, Administration and Maintenance
OWSER	OMA Web Services
PC	Protocol Converter
PLMN	Public Land Mobile Network
REST	REpresentational State Transfer
RNAA	Resource owner-aware Northbound API Access
RO	Resource Owner
ROF	Resource Owner Function
RPC	Remote Procedure Call
SCEF	Service Capability Exposure Function
SCS	Service Capability Server
SNPN	Stand-alone Non-Public Network
UDDI	Universal Description, Discovery and Integration
URI	Uniform Resource Identifier
WSDL	Web Services Description Language

4 Architectural requirements

4.1 General

4.1.1 Introduction

This subclause specifies the general requirements for CAPIF architecture.

4.1.2 Requirements

[AR-4.1.2-a] The CAPIF shall provide mechanisms (e.g. publish service APIs, authorization, logging, charging) to support service API operations.

[AR-4.1.2-b] The CAPIF shall enable API invoker(s) to discover and communicate with service APIs from the API providers.

[AR-4.1.2-c] Reference points between CAPIF and external applications shall be provided as APIs.

[AR-4.1.2-d] Reference points internal to CAPIF may be provided as APIs.

4.1.3 Requirements for supporting 3rd party API providers

[AR-4.1.3-a] The CAPIF shall provide mechanisms (e.g. publish service APIs, authorization, logging, charging) to support service API operations from trusted 3rd party API providers.

[AR-4.1.3-b] The CAPIF shall enable API invoker(s) to discover and communicate with service APIs from trusted 3rd party API providers.

4.2 Service API publish and discover

4.2.1 Introduction

This subclause specifies the service API publish and discover related requirements.

4.2.2 Requirements

[AR-4.2.2-a] The CAPIF shall provide a mechanism to publish the service API information to be used by the API invokers to discover and subsequently invoke the service API.

[AR-4.2.2-b] The CAPIF shall provide a mechanism for the API invokers to discover the published service API information as specified in [AR-4.2.2-a] according to the API invokers' interest.

[AR-4.2.2-c] The CAPIF shall provide a mechanism to restrict the discovery of the published service API information by the API invokers, based on configured policies.

[AR-4.2.2-d] The CAPIF shall provide a mechanism to configure policies to restrict the discovery of the published service API information.

[AR-4.2.2-e] The CAPIF shall provide mechanism to support Serving Area Information related to service APIs.

4.2.3 Requirements for 3rd party API providers

[AR-4.2.3-a] The CAPIF shall provide a mechanism to publish the service API information of the 3rd party API providers.

4.3 Security

4.3.1 Introduction

This subclause specifies the security related requirements for API invokers.

4.3.2 Requirements

[AR-4.3.2-a] The CAPIF shall provide mechanisms to hide the topology of the PLMN trust domain from the API invokers accessing the service APIs from outside the PLMN trust domain.

[AR-4.3.2-b] The CAPIF shall provide mechanisms to authenticate API invokers prior to accessing the service APIs.

[AR-4.3.2-c] The CAPIF shall provide mechanisms to authenticate API invokers upon the service API invocation.

[AR-4.3.2-d] The CAPIF shall provide mechanisms to authorize API invokers to access the service APIs.

[AR-4.3.2-e] The CAPIF shall provide mechanisms to validate authorization of the API invokers upon the service API invocation.

[AR-4.3.2-f] The CAPIF shall provide mechanisms for mutual authentication between the CAPIF and the API invoker.

[AR-4.3.2-g] The CAPIF shall provide mechanisms to control the service API access for every API invocation.

[AR-4.3.2-h] The communication between the CAPIF and the API invoker shall be confidentiality protected.

[AR-4.3.2-i] The communication between the CAPIF and the API invoker shall be integrity protected.

[AR-4.3.2-j] The CAPIF shall provide mechanisms to authenticate the service API publishers to publish and manage the service API information.

[AR-4.3.2-k] The CAPIF shall provide mechanisms to authorize the service API publishers to publish and manage service API information.

[AR-4.3.2-l] The CAPIF shall provide mechanisms to validate authorization of the service API publishers to publish and manage service API information.

[AR-4.3.2-m] The CAPIF shall provide mechanisms to enable that upon authorization revocation, the API invoker and the API Exposing Function are notified about the revocation of previously granted authorization.

4.3.3 Additional requirements for 3rd party API provider

[AR-4.3.3-a] The CAPIF shall provide mechanisms to hide the topology of the 3rd party API provider trust domain from the API invokers accessing the service APIs from outside the 3rd party API provider trust domain.

[AR-4.3.3-b] The CAPIF shall provide authorization mechanism for service APIs from the 3rd party API providers.

[AR-4.3.3-c] The CAPIF shall provide data confidentiality (across API providers) for data (e.g. logging, charging) related to service APIs from multiple API providers.

4.4 Charging

4.4.1 Introduction

This subclause specifies the charging related requirements for the usage of service APIs.

4.4.2 Requirements

[AR-4.4.2-a] The CAPIF shall support online and offline charging for service APIs usage.

[AR-4.4.2-b] The CAPIF shall provide mechanisms to record the usage (e.g. invocation count) of the service APIs for charging purpose, on a per API invoker basis.

[AR-4.4.2-c] The CAPIF shall provide mechanisms to record timestamp of the service API invocation.

[AR-4.4.2-d] The CAPIF shall provide mechanisms to record the service API related information, e.g. API location.

4.4.3 Requirements for 3rd party API providers

[AR-4.4.3-a] The CAPIF shall support online and offline charging for 3rd party API providers' service APIs usage.

[AR-4.4.3-b] The CAPIF shall provide mechanisms to query charging related information of the 3rd party service APIs by the authorized users.

4.5 Operations, Administration and Maintenance

4.5.1 Introduction

This subclause specifies the OAM aspects including performance monitoring, fault monitoring, policy configurations, and certain lifecycle management aspects such as monitoring the running status of service APIs and related operations.

4.5.2 Requirements

[AR-4.5.2-a] The CAPIF shall provide mechanisms to monitor the status of service APIs, e.g. starting and stopping access of the service APIs.

[AR-4.5.2-b] The CAPIF shall provide mechanisms to monitor and report the performance of the service APIs.

[AR-4.5.2-c] The CAPIF shall provide mechanisms to monitor and report the fault information about the service APIs.

[AR-4.5.2-d] The CAPIF shall provide mechanisms to record change events of service APIs, e.g. service APIs relocation.

[AR-4.5.2-e] The CAPIF shall provide mechanisms to configure policies related to service APIs.

4.5.3 Requirements for 3rd party API providers

[AR-4.5.3-a] The CAPIF shall provide mechanisms to configure policies related to 3rd party service APIs by the authorized users.

[AR-4.5.3-b] The CAPIF shall provide mechanisms to monitor faults, performance and status of the 3rd party service APIs by the authorized users.

4.6 Service API invocation monitoring

4.6.1 Introduction

The CAPIF includes monitoring functions. This enables API provider to monitor service API invocations, to determine critical aspects such as system load, API usage information, uncover potential overload and attacks (e.g. DDoS) conditions.

4.6.2 Requirements

[AR-4.6.2-a] The CAPIF shall provide mechanisms to capture service API invocation events and make them available to service API provider.

[AR-4.6.2-b] The CAPIF shall provide mechanisms to notify events related to overload and threat conditions (e.g. system load, resource usage information).

[AR-4.6.2-c] The CAPIF shall provide mechanisms to allow service API provider to apply monitoring filters based on criteria such as API invoker's ID and IP address, service API name and version, invoked operation, input parameters, and invocation result.

4.7 Logging

4.7.1 Introduction

The CAPIF supports the ability to log events and store the corresponding logs. This enables the API providers to use the logs for the purpose of tracing back and statistical analysis.

The following events in CAPIF are supported for logging:

- Service API invocation events;
- API invoker onboarding events; and
- API invoker interactions with the CAPIF (e.g. authentication, authorization, discover service APIs).

4.7.2 Logging events related to service API invocations

[AR-4.7.2-a] The CAPIF shall provide mechanisms for service API invocation event logging and storage functionality.

[AR-4.7.2-b] The service API invocation log shall be stored for a configurable time period, according to the service API provider's policy.

[AR-4.7.2-c] The service API invocation log shall be stored securely, and shall only be accessed by CAPIF administrators of the service API provider.

4.7.3 Logging events related to API invoker onboarding

[AR-4.7.3-a] The CAPIF shall provide mechanisms for API invoker onboarding event logging and storage functionality.

[AR-4.7.3-b] The API invoker onboarding log shall be stored at least for the duration during which the onboarding is valid.

[AR-4.7.3-c] The API invoker onboarding log shall be stored securely, and shall only be accessed by CAPIF administrators.

4.7.4 Logging events related to API invoker interaction with the CAPIF

[AR-4.7.4-a] The CAPIF shall provide mechanisms for the event logging of API invoker interactions with the CAPIF (e.g. authentication, authorization, discover service APIs).

[AR-4.7.4-b] The API invoker interactions log shall be stored for a configurable time period.

[AR-4.7.4-c] The API invoker interactions log shall be stored securely, accessed only by CAPIF administrators.

4.8 Auditing service API invocation

4.8.1 Introduction

The CAPIF includes auditing capabilities. This enables the service API provider to identify illegal service API invocations e.g. by querying the service API invocation log.

4.8.2 Requirements

[AR-4.8.2-a] The CAPIF shall provide mechanisms to query the service API invocation log, by CAPIF administrators.

4.9 Onboarding API invoker

4.9.1 Introduction

This subclause specifies the requirements related to onboarding API invoker to the CAPIF.

4.9.2 Requirements

[AR-4.9.2-a] The CAPIF shall provide the capability to onboard new API invokers.

[AR-4.9.2-b] The CAPIF shall support granting an API invoker's request to onboard with the CAPIF administrator.

[AR-4.9.2-c] The CAPIF shall support offboarding an API invoker from the CAPIF.

[AR-4.9.2-d] The CAPIF shall support updating an API invoker's API list e.g., subsequent to discovery of new API(s).

4.10 Policy configuration

4.10.1 Introduction

This subclause specifies the policy configuration related requirements.

4.10.2 Requirements

[AR-4.10.2-a] The CAPIF shall support policy configurations (e.g. related to the protection of platforms and network, specific functionalities exposed, message payload size or throughput).

4.11 Protocol design

4.11.1 Introduction

In order for the CAPIF to be common across all present and future API invokers for various usages and purposes, a minimum common protocol stack model is necessary so that all API invokers that use the common-framework-based

API need to support only one and the same set of protocols, e.g. security layer protocol(s). Extensibility of this model allows evolution and re-use.

4.11.2 Requirements

[AR-4.11.2-a] The CAPIF shall support a minimum common protocol stack model common for all API implementations to be based on.

[AR-4.11.2-b] The CAPIF shall support a common security mechanism for all API implementations to provide confidentiality and integrity protection.

[AR-4.11.2-c] The CAPIF shall be extensible to support different protocol stack models, including related security mechanisms, in addition to the minimum common protocol stack model.

NOTE: Potentially, Stage 3 needs to consider all CAPIF APIs for protocol extensibility.

[AR-4.11.2-d] CAPIF APIs and associated information flows shall be extensible to support vendor-specific functionality.

4.12 Interconnection between the CAPIF providers

4.12.1 Introduction

Two organizations with a business relationship that have each deployed CAPIF may need to interoperate to allow API invokers in each trust domain to utilize service APIs from both CAPIFs as illustrated in figure 4.12.1-1.

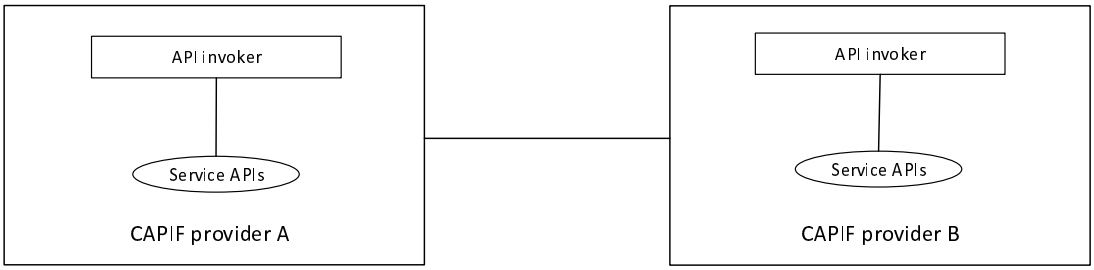


Figure 4.12.1-1: Interconnection between the CAPIF providers

NOTE: From each CAPIF provider's perspective the other CAPIF provider is a 3rd party.

4.12.2 Requirements

[AR-4.12.2-a] The CAPIF shall provide mechanisms to enable the API invokers of the CAPIF provider to discover and invoke the service APIs of the 3rd party CAPIF provider.

[AR-4.12.2-b] The CAPIF shall provide mechanisms to enable a CAPIF provider to publish, retrieve unpublish and update service APIs in a 3rd party CAPIF provider.

[AR-4.12.2-c] The CAPIF shall provide mechanisms for authentication, authorization, authorization revocation for access to the service APIs exposed by a 3rd party CAPIF provider.

4.13 Identities

4.13.1 Introduction

This subclause specifies the identities related requirements.

4.13.2 Requirements

[AR-4.13.2-a] The CAPIF shall support uniform addressing (e.g. identity) for communication within the same trust domain or from the 3rd party trust domain.

[AR-4.13.2-b] The CAPIF shall support identities for uniquely identifying each service API.

[AR-4.13.2-c] The CAPIF shall support identities for uniquely identifying each API invoker.

4.14 API provider domain interactions

4.14.1 Introduction

This subclause specifies the API provider domain interactions related requirements.

4.14.2 Requirements

[AR-4.14.2-a] The CAPIF shall enable interactions between multiple API exposing functional entities within the same trust domain.

[AR-4.14.2-b] The CAPIF shall enable interactions of multiple API exposing functional entities between trust domains.

4.15 Dynamic routing of service API invocation

4.15.1 Introduction

This subclause specifies the dynamic routing of service API invocation related requirements.

4.15.2 Requirements

[AR-4.15.2-a] The CAPIF shall provide a mechanism to support the dynamic routing of service API invocation.

4.16 Registering API provider domain functions

4.16.1 Introduction

This subclause specifies the requirements related to registration of API provider domain functions on the CAPIF core function.

4.16.2 Requirements

[AR-4.16.2-a] The CAPIF shall provide the capability to register API provider domain functions.

[AR-4.16.2-b] The CAPIF shall support the capability to update the registration information of the API provider domain functions.

4.17 Resource owner-aware northbound API invocation

4.17.1 Introduction

This subclause specifies requirements related to the resource owner-aware northbound API invocation.

4.17.2 Requirements

[AR-4.17.2-a] The CAPIF shall support applications on the UE acting as an API invoker and applications on a server acting as an API invoker.

[AR-4.17.2-b] The CAPIF shall support the authentication of the resource owner.

[AR-4.17.2-c] The CAPIF shall provide mechanisms to manage (provide, obtain, revoke) resource owner authorization for API invoker(s) to access (API provider domain hosted) resources owned by resource owners.

[AR-4.17.2-d] The CAPIF shall provide a mechanism to enable API invoker on a UE to access resources of another UE or group of UE(s) with resource owner authorization.

[AR-4.12.2-e] The CAPIF shall enable API invokers to access (API provider domain hosted) resources via the service APIs exposed by a 3rd party CAPIF provider with resource owner authorization.

5 Involved business relationships

5.1 Basic CAPIF business relationships

Figure 5.1-1 shows the typical business relationships in CAPIF.

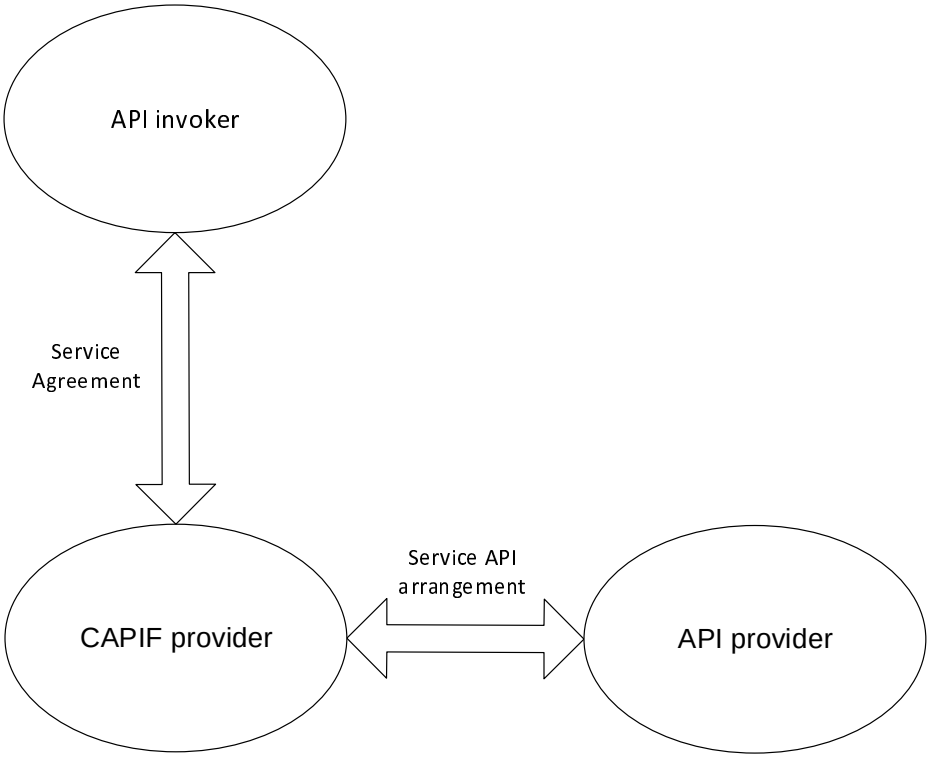


Figure 5.1-1: Business relationships in CAPIF

The API invoker has service agreement with a CAPIF provider and consumes the CAPIF APIs and service APIs. There are various API invokers like application management client (used by application developers, application service provider), hosted applications (on cloud, edge or UE), and channel aggregator (who aggregates the CAPIF APIs and/or the service APIs). For more details about these API invoker roles, please refer to Annex F.

The API provider hosts one or more service APIs and has a service API arrangement with CAPIF provider to offer the service APIs to the API invoker.

The CAPIF provider and the API provider can be part of the same organization (e.g. PLMN operator), in which case the business relationship between the two is internal to a single organization. The CAPIF provider and the API provider can be part of different organizations, in which case the business relationship between the two must exist.

5.2 CAPIF business relationships for RNAA

Figure 5.2-1 shows the CAPIF business relationships for the resource owner-aware northbound API access (RNAA).

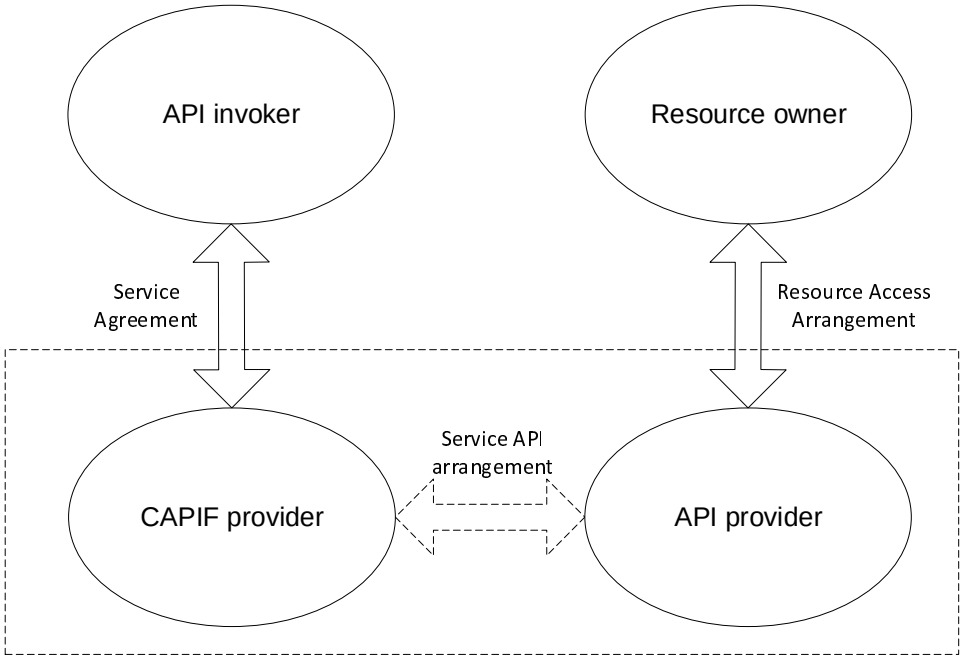


Figure 5.2-1: CAPIF business relationships for RNAA

The business relationships the API invoker, the CAPIF provider, and the API provider follow the description in the clause 5.1. In addition to them, the resource owner is an entity capable of granting access to a protected resource related to the resource exposed by the API provider. The API invoker and the resource owner can be the same entity or separate entities. In the current release, the resource owner is a user of a UE and can provide authorization information using the UE.

The CAPIF provider and the API provider can belong to the same organization (e.g. PLMN operator), in which case the service API arrangement is not required explicitly.

The CAPIF provider and the API provider (e.g. 3rd party CAPIF provider or 3rd party API provider) can belong to different organizations, in which case the service API arrangement is required. The specification of service API arrangement is out of scope of 3GPP.

6 Functional model

6.1 General

The Common API framework (CAPIF) functional architecture is described in this subclause. The CAPIF architecture is defined as service-based and interactions between the CAPIF functions are represented in two ways:

- A service-based representation, where CAPIF functions enable other authorized CAPIF functions to access their services;

- A reference point representation, where interactions between any two CAPIF functions (e.g. CCF, AEF) is shown by an appropriate point-to-point reference point (e.g. CAPIF-3).

The CAPIF functional architecture can be adopted by any 3GPP functionality providing 3GPP northbound service APIs.

Charging aspects of CAPIF framework are specified in 3GPP TS 32.254 [20].

NOTE 1: The terms “functional architecture” and “functional model” mean the same and have been used interchangeably in this specification.

NOTE 2: The functional model described in this specification applies to both PLMN(s) and to SNPN(s).

6.2 Functional model description

6.2.0 Functional model description for the CAPIF

Figure 6.2.0-1 shows the reference point based functional model for the CAPIF.

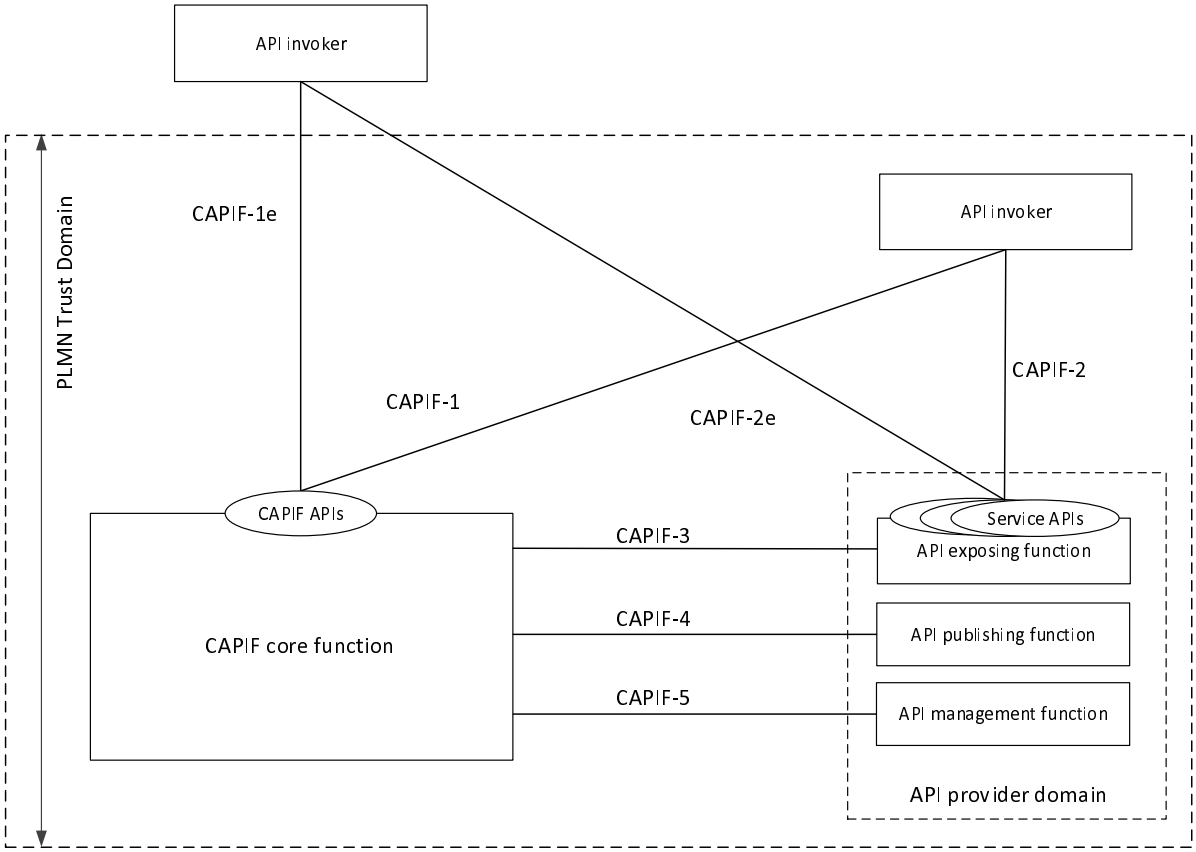


Figure 6.2.0-1: Functional model for the CAPIF

The CAPIF is hosted within the PLMN operator network (or even an SNPN). The API invoker is typically provided by a 3rd party application provider who has service agreement with PLMN operator. The API invoker may reside within the same trust domain as the PLMN operator network.

In a reference point based model, the API invoker within the PLMN trust domain interacts with the CAPIF via CAPIF-1 and CAPIF-2. The API invoker from outside the PLMN trust domain interacts with the CAPIF via CAPIF-1e and CAPIF-2e. The API exposing function, the API publishing function and the API management function of the API provider domain (together known as API provider domain functions) within the PLMN trust domain interacts with the CAPIF core function via CAPIF-3, CAPIF-4 and CAPIF-5 respectively.

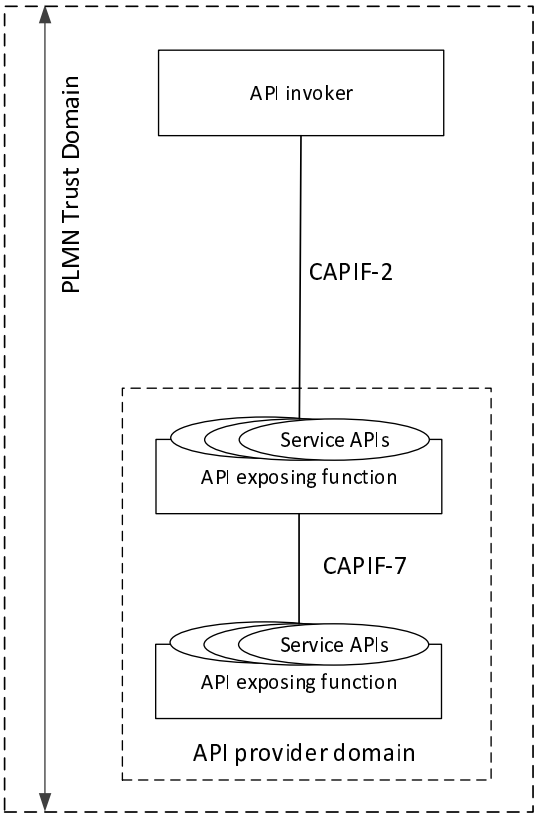


Figure 6.2.0-2: Functional model for interactions between API exposing functions

As illustrated in figure 6.2.0-2, the interactions between the API exposing functions within the PLMN trust domain is via CAPIF-7.

The CAPIF core function provides CAPIF APIs to the API invoker over CAPIF-1 and CAPIF-1e. The API exposing function provides the service APIs to the API invoker over CAPIF-2 and CAPIF-2e.

NOTE 1: The communication between the API exposing function and the CAPIF core function, between the API publishing function and the CAPIF core function and between the API management function and the CAPIF core function over CAPIF-3, CAPIF-4 and CAPIF-5 respectively can be API based.

The detailed information of the APIs provided by the CAPIF core function is specified in clause 10.

The security aspects of CAPIF reference points are specified in 3GPP TS 33.122 [12].

Figure 6.2.0-3 illustrates the CAPIF functional model using service-based interfaces.

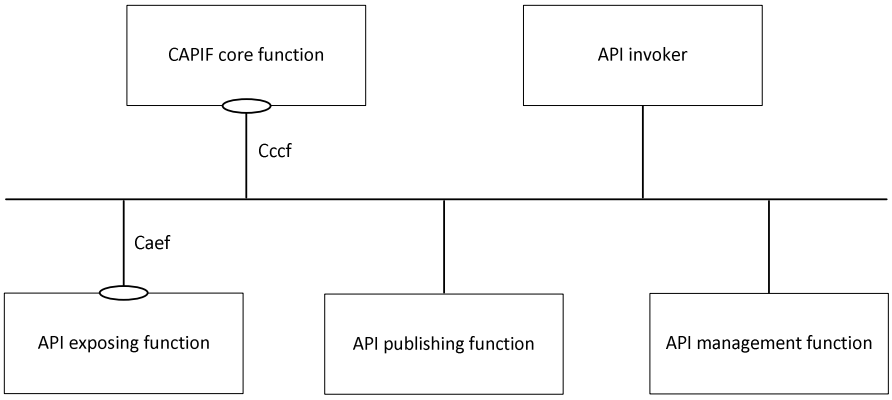


Figure 6.2.0-3: CAPIF functional model representation using service-based interfaces

Table 6.2.0-1 specifies the service-based interfaces supported by CAPIF.

Table 6.2.0-1: Service-based interfaces supported by CAPIF

Service-based interface	Entity	APIs offered
Cccf	CAPIF core function	Specified in subclause 10
Caef	API exposing function	Specified in subclause 11

6.2.1 Functional model description to support 3rd party API providers

Figure 6.2.1-1 shows the functional model for the CAPIF to support 3rd party API providers.

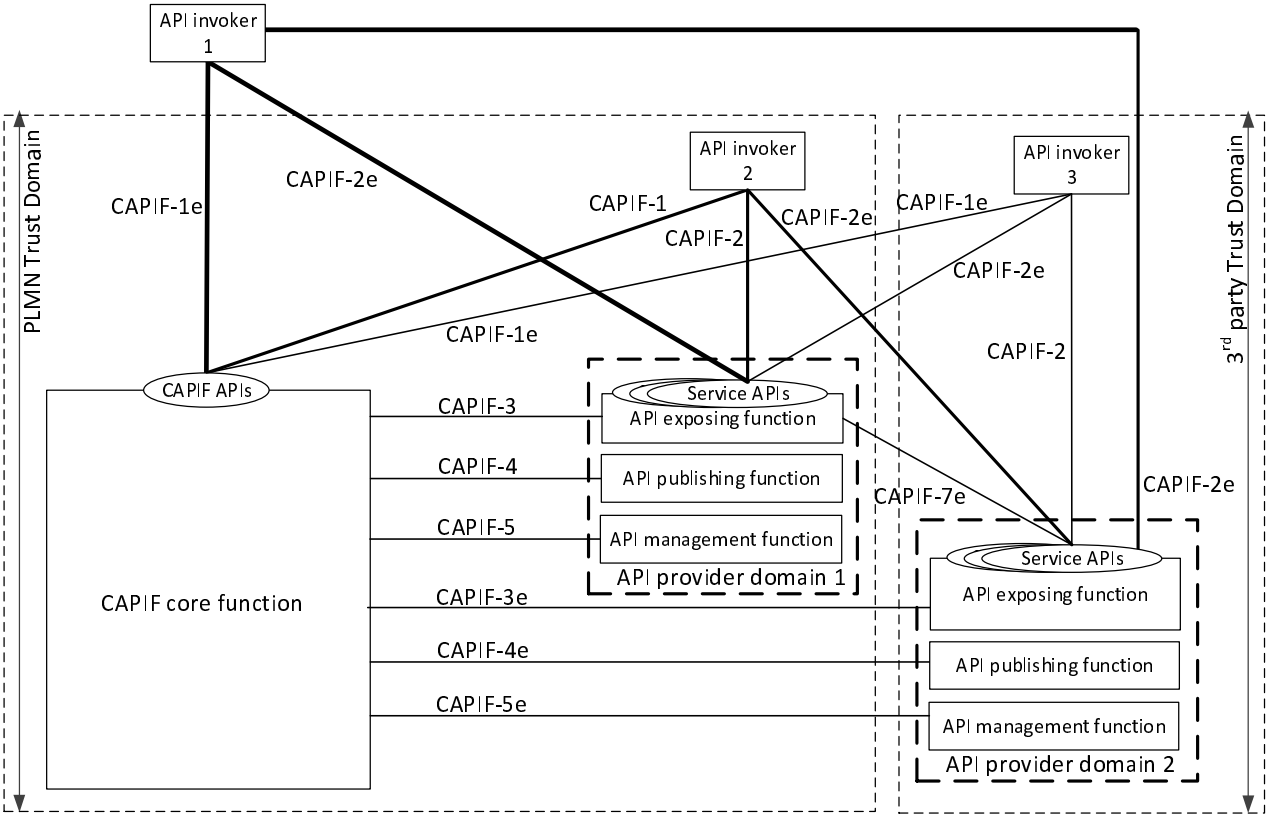


Figure 6.2.1-1: Functional model for the CAPIF to support 3rd party API providers

The CAPIF core function in the PLMN trust domain supports service APIs from both the PLMN trust domain and the 3rd party trust domain having business relationship with PLMN. The API invokers may exist within the PLMN trust domain, or within the 3rd party trust domain or outside of both the PLMN trust domain and the 3rd party trust domain. The API provider domain 1 offers the service APIs from the PLMN operator. The API provider domain 2 offers the service APIs from the 3rd party. When the 3rd party API provider is a trusted 3rd party of the PLMN, the API provider domain 1 may also offer the service APIs from the 3rd party.

The API invoker 2 within the PLMN trust domain onboards and interacts with the CAPIF core function via CAPIF-1, and invokes the service APIs in the PLMN trust domain via CAPIF-2 and invokes the service APIs in the 3rd party trust domain via CAPIF-2e. The API invoker 3 within the 3rd party trust domain onboards and interacts with the CAPIF core function via CAPIF-1e, and invokes the service APIs in the PLMN trust domain via CAPIF-2e and invokes the service APIs in 3rd party trust domain via CAPIF-2. The API invoker 1 from outside the PLMN trust domain and 3rd party trust domain, onboards and interacts with the CAPIF core function via CAPIF-1e and invokes the service APIs in the PLMN trust domain and the service APIs in the 3rd party trust domain via CAPIF-2e. The CAPIF core function consolidates in the API invoker profile the service API enrollment information for the service APIs in the PLMN trust domain and the service APIs in the 3rd party trust domain, as specified in clause 8.1.3.

The API exposing function, the API publishing function and the API management function of the API provider domain 1 within the PLMN trust domain interacts with the CAPIF core function via CAPIF-3, CAPIF-4 and CAPIF-5 respectively. The API exposing function, the API publishing function and the API management function of the API provider domain 2 within the 3rd party trust domain interacts with the CAPIF core function in the PLMN trust domain via CAPIF-3e, CAPIF-4e and CAPIF-5e respectively. The API exposing function within the PLMN trust domain and the 3rd party trust domain provides the service APIs to the API invoker, offered by the respective trust domains.

The interactions between the API exposing functions within the PLMN trust domain is via CAPIF-7 (not shown in the figure 6.2.1-1 for simplicity). The API exposing function within the PLMN trust domain interacts with the API exposing function in the 3rd party trust domain via CAPIF-7e.

NOTE 1: The communication between the API exposing function and the CAPIF core function, between the API publishing function and the CAPIF core function and between the API management function and the CAPIF core function over CAPIF-3/3e, CAPIF-4/4e and CAPIF-5/5e respectively can be API based.

The detailed information of the APIs provided by the CAPIF core function is specified in clause 10.

NOTE 2: The security aspects of CAPIF reference points are under SA3 responsibility and out of scope of the present document.

6.2.2 Functional model description to support CAPIF interconnection

Figure 6.2.2-1 shows the architectural model for the CAPIF interconnection which allows API invokers of a CAPIF provider to utilize the service APIs from the 3rd party CAPIF provider.

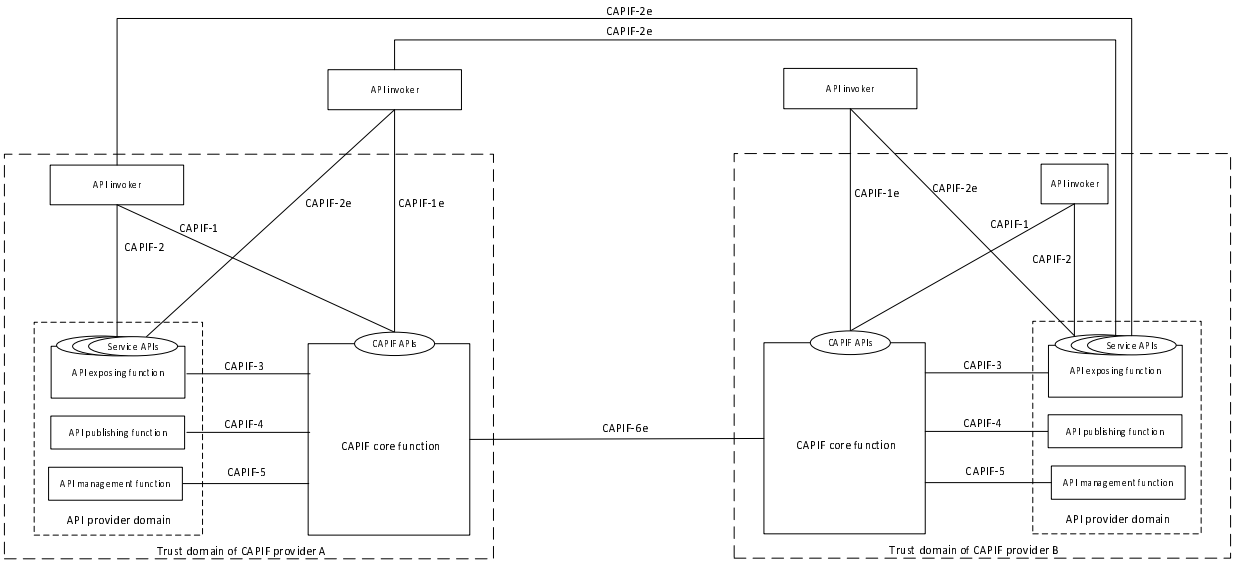


Figure 6.2.2-1: High level functional architecture for CAPIF interconnection with multiple CAPIF provider domains

Figure 6.2.2-2 shows the architectural model for the CAPIF interconnection within the same CAPIF provider domain, which allows API invokers of CAPIF core function 1 to utilize the service APIs from CAPIF core function 2, where both CAPIF core function 1 and CAPIF core function 2 are hosted within the trust domain of the CAPIF provider A.

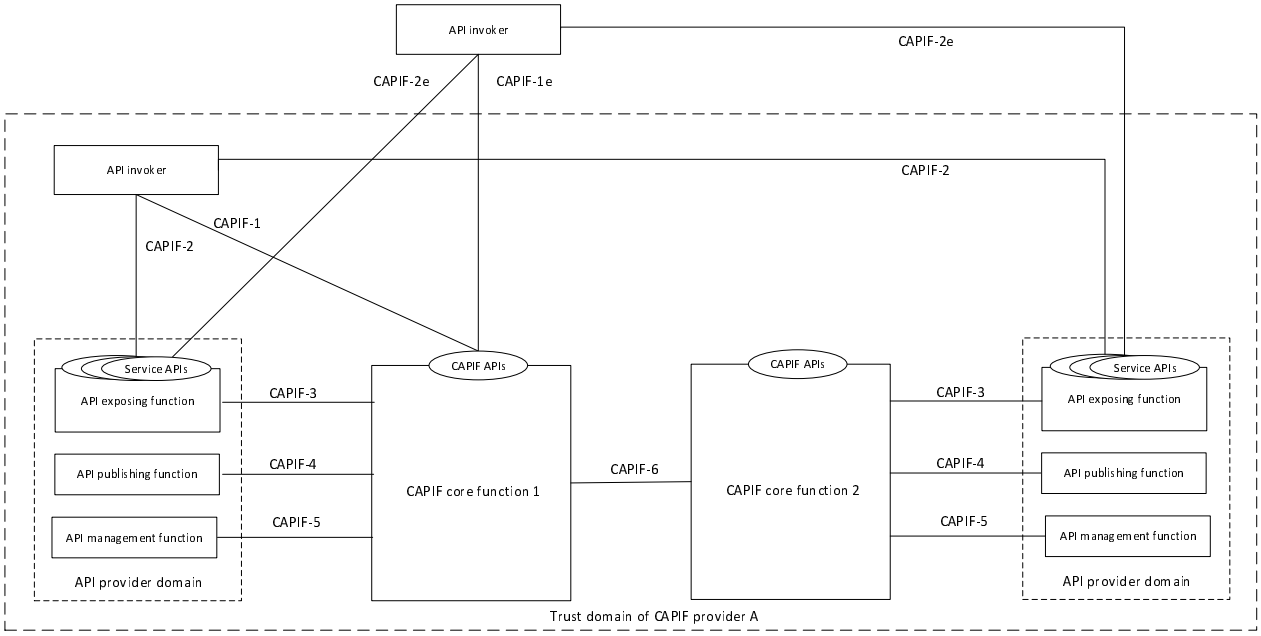


Figure 6.2.2-2: High level functional architecture for CAPIF interconnection within a CAPIF provider domain

The CAPIF provider A and CAPIF provider B host the CAPIF in their trust domains. A business relationship exists between the CAPIF providers.

The CAPIF providers in their respective trust domain hosts multiple CAPIF instances where each CAPIF instance consists of the CAPIF core function (local), the API provider domain and the API invokers. All interactions within the CAPIF instance is according to the functional model specified in clause 6.2.0.

When multiple CAPIF instances are deployed by a CAPIF provider there may be a hierarchy associated with the multiple CAPIF core function deployed which allows:

- the designated CAPIF core function of the CAPIF provider A to interconnect with the designated CAPIF core function of the CAPIF provider B; and

- within CAPIF provider A, one or more CAPIF core function interacts with the designated CAPIF core function 1.

The designated CAPIF core function of the CAPIF provider A provides the information about the CAPIF instances and service APIs deployed by the CAPIF provider A to the designated CAPIF core function of the CAPIF provider B and vice versa over CAPIF-6e reference point.

The CAPIF core function 2 of CAPIF provider A provides the information about the service APIs to the CAPIF core function 1 over CAPIF-6 reference point.

NOTE 1: Void

The API invokers may exist within the trust domain of CAPIF provider A, or within the trust domain of CAPIF provider B or outside of the trust domains of both CAPIF provider A and CAPIF provider B. The API invoker of a CAPIF provider is onboarded with the CAPIF core function in the corresponding trust domain of the CAPIF provider.

NOTE 2: For sake of simplicity, the service API interactions of API invokers of the CAPIF provider B are not shown. From each CAPIF provider's perspective the other CAPIF provider is a 3rd party.

One or more CAPIF core function can publish service APIs to the designated CAPIF core function over CAPIF-6 reference point and, also discover the service APIs from the designated CAPIF core function and vice versa over CAPIF-6 reference point.

The API invoker within the trust domain of CAPIF provider A interacts with the CAPIF core function of the CAPIF provider A via CAPIF-1 and discovers the service APIs of both CAPIF providers, and invokes the service APIs in the trust domain of CAPIF provider A via CAPIF-2 and invokes the service APIs in the trust domain of CAPIF provider B via CAPIF-2e. The API invoker from outside the trust domain of CAPIF providers, interacts with the CAPIF core function of th CAPIF provider A via CAPIF-1e and invokes the service APIs in the trust domain of the CAPIF providers via CAPIF-2e.

NOTE 3: The communication between the CAPIF core function of the CAPIF providers over CAPIF-6 or CAPIF-6e can be API based.

The detailed information of the APIs provided by the CAPIF core function is specified in clause 10.

NOTE 4: The security aspects of CAPIF reference points are under SA3 responsibility and out of scope of the present document.

NOTE 5: All interactions among entities within the CAPIF provider domains (regardless if CAPIF is deployed in a PLMN, SNPN or 3rd party network) are ruled by the functional model in clause 6.2.0, the support of 3rd party API providers is as in clause 6.2.1, whereas the interconnection among CCFs is according to this clause.

NOTE 6: RNAA for CAPIF interconnection is not specified in this release of the specification.

6.2.3 Functional model description to support RNAA

Figure 6.2.3-1 shows the architectural model for the RNAA which allows the resource owner to provide authorization to access their resources through Service API invocation.

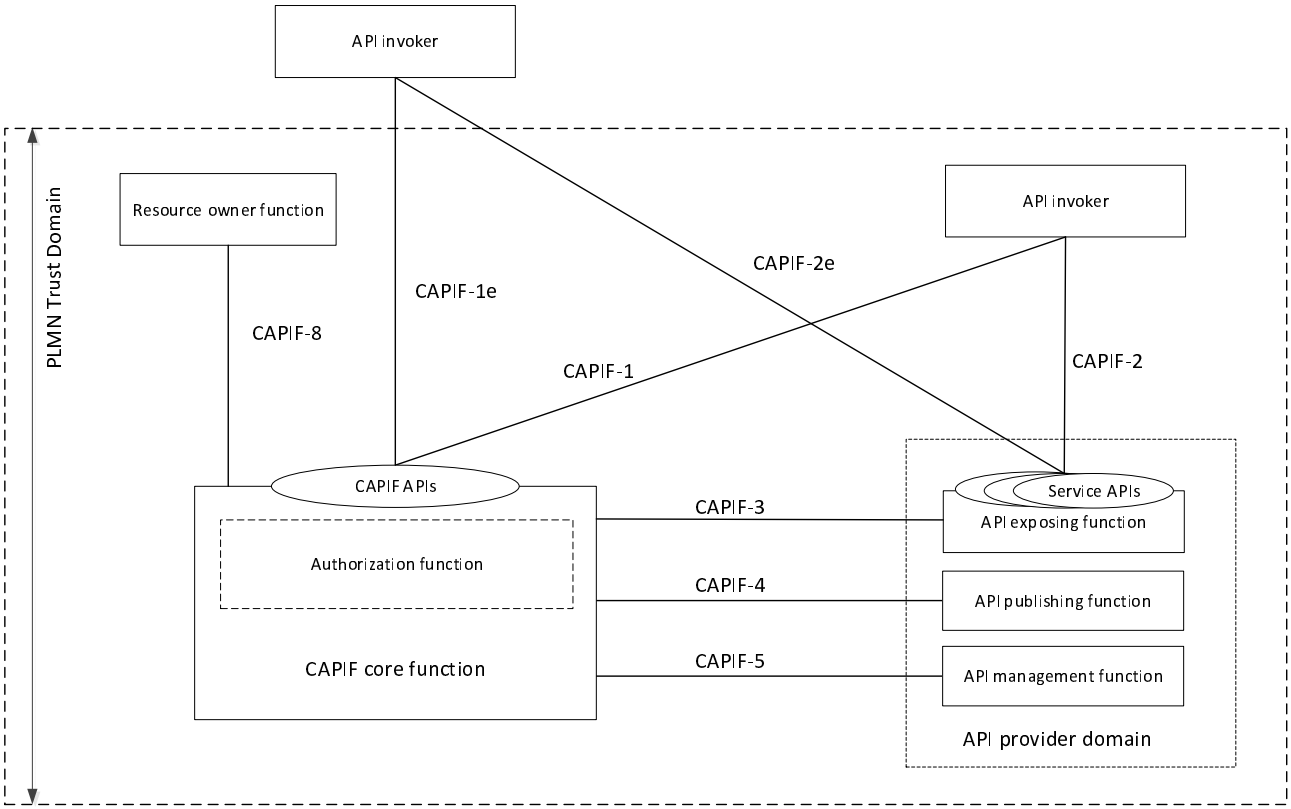


Figure 6.2.3-1: High level functional architecture for CAPIF supporting RNAA

The authorization function is an internal entity of the CAPIF core function and performs API invoker authorization based on the API invoker authorization policies available in the CCF and the RO authorization information obtained via the ROF.

The resource owner function interacts with the authorization function in the CAPIF core function via CAPIF-8. The authorization function in the CAPIF core function interacts with the resource owner function to obtain the resource owner authorization. The CAPIF core function (authorization function) uses resource owner authorization information obtained via the ROF as part of the processing of the authorization requests received from API invokers.

The API exposing function (e.g. NEF, SCEF) interacts with the authorization function in the CAPIF core function via CAPIF-3. The API exposing function acts as an authorization enforcement point according to the authorization token received from the API invoker.

NOTE 1: RNAA support is not dependant of the access network, i.e., the RNAA support is not restricted to 5G networks.

The API invoker interacts with authorization function in the CAPIF core function via CAPIF-1/CAPIF-1e.

NOTE 2: In the current release, 3rd party API providers (i.e., API providers outside the PLMN trust domain) are not supported for RNAA.

NOTE 3: The interaction between resource owner function and CCF over CAPIF-8 is not specified in the current release of the specification.

NOTE 4: The authorization information from the resource owner used by CCF (described in 3GPP TS 33.122 [12]) is independent from the user consent information used from user subscription data at UDM/UDR (described in Annex V of 3GPP TS 33.501 [16]). In the current release of 3GPP specifications, no synergy between CCF and UDM is specified.

The security aspects of CAPIF supporting RNAA are specified in 3GPP TS 33.122 [12].

6.3 Functional entities description

6.3.1 General

Each subclause is a description of a functional entity and does not imply a physical entity.

6.3.2 API invoker

The API invoker is typically provided by a 3rd party application provider who has service agreement with PLMN operator. The API invoker may reside within the same trust domain as the PLMN operator network. The API invoker may be either an application on a server or an application on a UE.

The API invoker supports the following capabilities:

- Triggering API invoker onboarding/offboarding;
- Supporting the authentication by providing the API invoker identity and other information required for authentication of the API invoker;
- Supporting mutual authentication with CAPIF;
- Obtaining the authorization prior to accessing the service API;
- Discovering service APIs information; and
- Invoking the service APIs.

NOTE: The details of the specific service APIs are out of scope of the present document.

6.3.3 CAPIF core function

The CAPIF core function consists of the following capabilities:

- Authenticating the API invoker based on the identity and other information required for authentication of the API invoker;
- Mutual authentication with the API invoker;
- Publishing, storing, retrieving, unpublishing, updating and discovering of service APIs information;
- Controlling the service API access based on PLMN operator configured policies;
- Storing the logs for the service API invocations and providing the service API invocation logs to authorized entities;
- Reporting Charging based events to the charging function (e.g. events of API invoker onboarding);
- Monitoring the service API invocations;
- Onboarding a new API invoker and offboarding an API invoker;
- Storing policy configurations related to CAPIF and service APIs;
- Accessing the logs for auditing (e.g. detecting abuse);
- Publishing, retrieving, unpublishing, updating,and discovering service APIs information with another CAPIF core function in CAPIF interconnection; and
- Slice related API exposure in, e.g., API publish, API discovery, API invoker authorization, API access control.

NOTE: Charging events reported by the CCF are specified in 3GPP TS 32.254 [20].

6.3.4 API exposing function

The API exposing function is the provider of the service APIs and is also the service communication entry point of the service API to the API invokers. The API exposing function consists of the following capabilities:

- Authenticating the API invoker based on the identity and other information required for authentication of the API invoker provided by the CAPIF core function;
- Requesting the revocation of the API invoker authorization;
- Enforcing the authorization provided by the CAPIF core function;
- Supporting authorization revocation;
- Reporting Charging based events to the charging function (e.g. events of service API invocations);
- Logging the service API invocations at the CAPIF core function; and
- Hiding the topology of the PLMN trust domain from API invokers, depending on configured policy.

NOTE: Charging events reported by the AEF (NEF) are specified in 3GPP TS 32.254 [20].

6.3.5 API publishing function

The API publishing function enables the API provider to publish the service APIs information in order to enable the discovery of service APIs by the API invoker. The API publishing function consists of the following capability:

- Publishing the service API information of the API provider to the CAPIF core function; and
- Retrieving service API information from the CAPIF core function.

6.3.6 API management function

The API management function enables the API provider to perform administration of the service APIs. The API management function consists of the following capabilities:

- Auditing the service API invocation logs received from the CAPIF core function;
- Monitoring the events reported by the CAPIF core function;
- Configuring the API provider policies to the CAPIF core function;
- Monitoring the status of the service APIs;
- Onboarding new API invokers and offboarding API invokers; and
- Registering and maintaining registration information of the API provider domain functions on the CAPIF core function.

NOTE: The API invoker onboarding/offboarding in the API management function is out of the scope of the current release.

6.3.7 Authorization function

The authorization function has the following capabilities:

- Acts in an authorization server role as specified in clause 1.1 of IETF RFC 6749 [14],
- Verifies authorization information for service APIs, upon request from AEF (when the AEF does not have information required to authorize the service API invocation),
- Uses the RO authorization information (such as user consent) obtained via ROF, when legally required, as part of the authorization of the API invoker to access the RO resources related to a service API.

NOTE 1: In the current release, the authorization function is an internal entity of the CAPIF core function.

NOTE 2: The persistent storage of the resource owner authorization data is subject to local/regional regulations and is not in scope of this specification.

6.3.8 Resource owner function

The resource owner function is responsible for interactions with the resource owner, when needed for the resource owner network resource consumption during obtain service API authorization procedure, e.g. in a similar way to the resource owner's user agent shown in clause 4.1 of IETF RFC 6749 [14].

The resource owner function enables the following:

- Authorization for resource access; and
- Managing and revoking authorization for resource access.

NOTE: The procedures corresponding to these capabilities of the resource owner function are FFS and out of scope of the current release of the specification.

6.4 Reference points

6.4.1 General

The reference points for CAPIF are described in the following subclauses.

6.4.2 Reference point CAPIF-1 (between the API invoker and the CAPIF core function)

The CAPIF-1 reference point, which exists between the API invoker and the CAPIF core function, is used for the API invoker within the PLMN trust domain to discover service APIs, to authenticate and to get authorization.

The CAPIF-1 reference point supports:

- Onboarding the new API invokers and offboarding API invokers;
- Authenticating the API invoker based on the identity and credentials of the API invoker;
- Mutual authentication between the API invoker and the CAPIF core function;
- Providing authorization for the API invoker prior to accessing the service API;
- Providing authorization for the API invoker based on RNAA; and
- Discovering the service APIs information.

NOTE: The security aspects of CAPIF-1 are specified in subclause 6.2 of 3GPP TS 33.122 [12].

6.4.3 Reference point CAPIF-1e (between the API invoker and the CAPIF core function)

The CAPIF-1e reference point, which exists between the API invoker and the CAPIF core function, is used for the API invoker outside the PLMN trust domain to discover service APIs, to authenticate and to get authorization.

The CAPIF-1e reference point supports all the functions of CAPIF-1.

NOTE: The security aspects of CAPIF-1e are specified in subclause 6.3 of 3GPP TS 33.122 [12].

6.4.4 Reference point CAPIF-2 (between the API invoker and the API exposing function)

The CAPIF-2 reference point, which exists between the API invoker and the API exposing function belonging to the same trust domain, is used for the API invoker to communicate with the service APIs.

The CAPIF-2 reference point supports:

- Authenticating the API invoker based on the identity and credentials of the API invoker;
- Authorization verification for the API invoker upon accessing the service API; and
- Invocation of service APIs.

NOTE 1: The aspects related to the specific service API invocation in reference point CAPIF-2 are out of scope of the present document.

NOTE 2: The security aspects of CAPIF-2 are specified in subclause 6.4 of 3GPP TS 33.122 [12].

6.4.5 Reference point CAPIF-2e (between the API invoker and the API exposing function)

The CAPIF-2e reference point, which exists between the API invoker and the API exposing function belonging to a different trust domain, is used for the API invoker to communicate with the service APIs.

The CAPIF-2e reference point supports all the functions of CAPIF-2.

NOTE: The security aspects of CAPIF-2e are specified in subclause 6.5 of 3GPP TS 33.122 [12].

6.4.6 Reference point CAPIF-3 (between the API exposing function and the CAPIF core function)

The CAPIF-3 reference point, which exists between the API exposing function and the CAPIF core function, is used for exercising access and policy related control for service API communications initiated by the API invoker.

The CAPIF-3 reference point supports:

- Authenticating the API invoker based on the identity and credentials of the API invoker;
- Providing authorization for the API invoker prior to accessing the service API;
- Providing authorization revocation information;
- Authorization verification for the API invoker upon accessing the service API;
- Authorization verification for the API invoker based on RNAA;
- Controlling the service API access based on PLMN operator configured policies; and
- Logging the service API invocations;

NOTE: The security aspects of CAPIF-3 are specified in subclause 6.6 of 3GPP TS 33.122 [12].

6.4.7 Reference point CAPIF-4 (between the API publishing function and the CAPIF core function)

The CAPIF-4 reference point, which exists between the API publishing function and the CAPIF core function, is used for publishing the service API information.

The CAPIF-4 reference point supports:

- Publishing the service APIs information by the API publishing function.

NOTE: The security aspects of CAPIF-4 are specified in subclause 6.6 of 3GPP TS 33.122 [12].

6.4.8 Reference point CAPIF-5 (between the API management function and the CAPIF core function)

The CAPIF-5 reference point, which exists between the API management function and the CAPIF core function, is used for management of service API, API invoker and API provider domain function information.

The CAPIF-5 reference point supports:

- Accessing the service API invocation logs by the API management function;
- Enabling the API management function to monitor the events reported due to the service APIs invocations;
- Onboarding new API invokers by provisioning the API invoker information at the CAPIF core function, requesting explicit grant of new API invokers onboarding and confirming onboarding success;
- Offboarding API invokers;
- Enabling the API management function to configure policies at the CAPIF core function e.g. service API invocation throttling, blocking API invocation for certain duration;
- Enabling the API provider to monitor the status of service APIs (e.g. pilot or live status, start or stop status of service API);
- Registering API provider domain functions on the CAPIF core function; and
- Update of the registration information of API provider domain functions on the CAPIF core function.

NOTE 1: The security aspects of CAPIF-5 are specified in subclause 6.6 of 3GPP TS 33.122 [12].

NOTE 2: The API invoker onboarding/offboarding over CAPIF-5 is out of the scope of the current release.

6.4.9 Reference point CAPIF-3e (between the API exposing function and the CAPIF core function)

The CAPIF-3e reference point, which exists between the API exposing function within the 3rd party trust domain and the CAPIF core function within the PLMN trust domain, is used for exercising access and policy related control for service API communications initiated by the API invoker.

The CAPIF-3e supports all the functions of CAPIF-3.

NOTE: The security aspects of CAPIF-3e are specified in clause 6.10 of 3GPP TS 33.122 [12].

6.4.10 Reference point CAPIF-4e (between the API publishing function and the CAPIF core function)

The CAPIF-4e reference point, which exists between the API publishing function within the 3rd party trust domain and the CAPIF core function within the PLMN trust domain, is used for publishing the service API information.

The CAPIF-4e reference point supports all the functions of CAPIF-4.

NOTE: The security aspects of CAPIF-4e are specified in clause 6.10 of 3GPP TS 33.122 [12].

6.4.11 Reference point CAPIF-5e (between the API management function and the CAPIF core function)

The CAPIF-5e reference point, which exists between the API management function within the 3rd party trust domain and the CAPIF core function within the PLMN trust domain, is used for management of service API, API invoker and API provider domain function information.

The CAPIF-5e reference point supports all the functions of CAPIF-5.

NOTE: The security aspects of CAPIF-5e are specified in clause 6.10 of 3GPP TS 33.122 [12].

6.4.12 Reference point CAPIF-7 (between the API exposing functions)

The CAPIF-7 reference point, which exists between the API exposing functions belonging to the same trust domain, is used for the forwarding or routing or nesting of the API invoker's service API invocation from one API exposing function to the other API exposing function deployed in the PLMN trust domain.

The CAPIF-7 reference point supports all the functions of CAPIF-2.

The CAPIF-7 reference point supports invocation of service APIs originated by the API invoker using CAPIF-2.

NOTE 1: The aspects related to the specific service API invocation in reference point CAPIF-7 are out of scope of the present document.

NOTE 2: The security aspects of CAPIF-7 are specified in clause 6.9 of 3GPP TS 33.122 [12]. Nested API invocations over CAPIF-7 interface are specified in clause 6.14 of 3GPP TS 33.122 [12].

6.4.13 Reference point CAPIF-7e (between the API exposing functions)

The CAPIF-7e reference point, which exists between the API exposing functions belonging to different trust domains, is used for the forwarding or routing of the API invoker's service API invocation from one API exposing function to the other API exposing function between different trust domains.

The CAPIF-7e reference point supports all the functions of CAPIF-2e.

NOTE: The security aspects of CAPIF-7e are specified in clause 6.9 of 3GPP TS 33.122 [12].

6.4.14 Reference point CAPIF-6 (between the CAPIF core functions of the same CAPIF provider)

The CAPIF-6 reference point exists between the CAPIF core functions within the same trust domain of CAPIF provider.

The CAPIF-6 reference point supports:

- Publishing the service APIs information;
- Discovering the service APIs information;
- Obtaining security information for the API invoker upon accessing the service API; and
- Providing authorization for the API invoker prior to accessing the service API.

NOTE: The security aspects of CAPIF-6 are specified in clauses 6.6, 6.10 and 6.13 of 3GPP TS 33.122 [12].

6.4.15 Reference point CAPIF-6e (between the CAPIF core functions of different CAPIF providers)

The CAPIF-6e reference point exists between the CAPIF core function within the 3rd party trust domain and the CAPIF core function within the PLMN trust domain.

The CAPIF-6e reference point supports all the functions of CAPIF-6.

NOTE: The security aspects of CAPIF-6e are specified in 3GPP TS 33.122 [12], clauses 6.6, 6.10 and 6.13.

6.4.16 Reference point CAPIF-8 (between the CAPIF core function and the resource owner function)

The CAPIF-8 reference point exists between the CAPIF core function and the resource owner function.

The CAPIF-8 reference point supports:

- Providing authorization for resource access; and
- Managing and revoking the provided authorization.

NOTE: The functionalities over CAPIF-8 is FFS and out of scope of the current release of the specification.

6.5 Service-based interfaces

The CAPIF architecture contains the following service-based interfaces:

- Cccf: Service-based interface exhibited by CAPIF core function.
- Caef: Service-based interface exhibited by API exposing function.

7 Application of functional model to deployments

7.1 General

The CAPIF deployments in centralized and distributed models are described in clause 7.2 and clause 7.3. The multiple CCFs deployment is described in clause 7.4.

The RNAA deployments are described in clause 7.5.

The CAPIF deployment models shown are not exhaustive.

7.2 Centralized deployment

The CAPIF can be deployed centrally as illustrated in the figure 7.2-1.

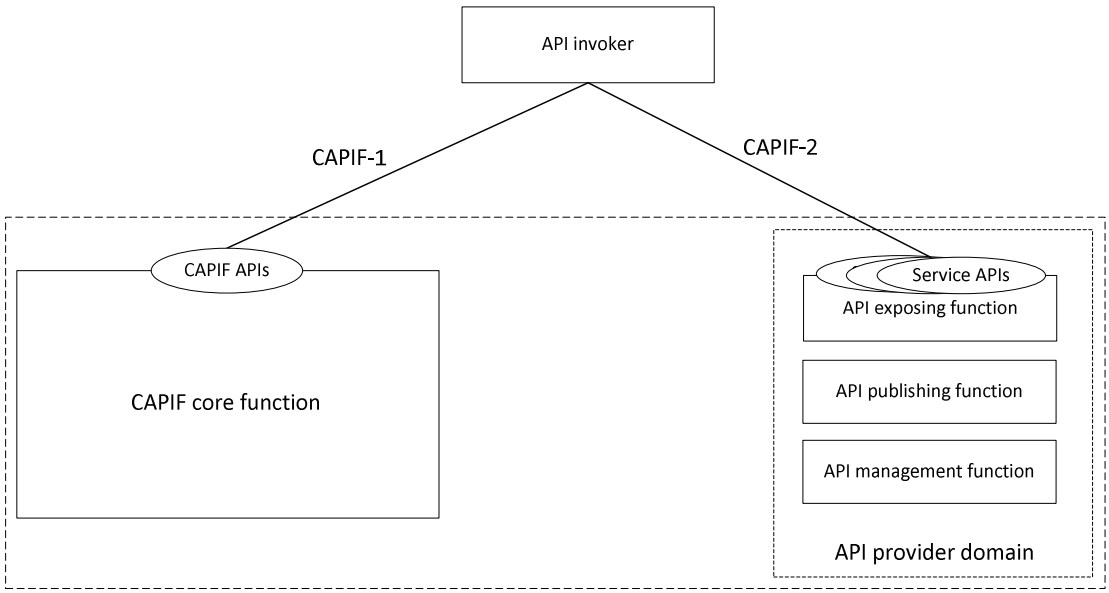


Figure 7.2-1: Centralized deployment of CAPIF

In one centralized deployment, the CAPIF core function and the API provider domain functions are co-located. The API invoker can interact independently with the CAPIF core function and the API exposing function including the service

APIs. The CAPIF appears as a gateway for all API invoker interactions. The API invoker obtains the service API information and its entry point details from the CAPIF core function via CAPIF-1. The service communication point of entry for the service API is the API exposing function which also applies any access control or policy control to the internal interactions between the API invoker and the service API in coordination with the CAPIF core function.

NOTE: The API invoker can be outside the PLMN trust domain and will access the CAPIF via CAPIF-1e and CAPIF-2e instead of CAPIF-1 and CAPIF-2.

Another variation of the centralized deployment is where the CAPIF core function and the API exposing function is co-located where as other API provider domain functions (API publishing function and API management function) are not co-located with the API exposing function. In such deployment scenario, the CAPIF core function interacts with the API publishing function and the API management function via CAPIF-4 and CAPIF-5 reference points respectively.

7.3 Distributed deployment

The CAPIF can be deployed in a distributed manner illustrated in the figure 7.3-1.

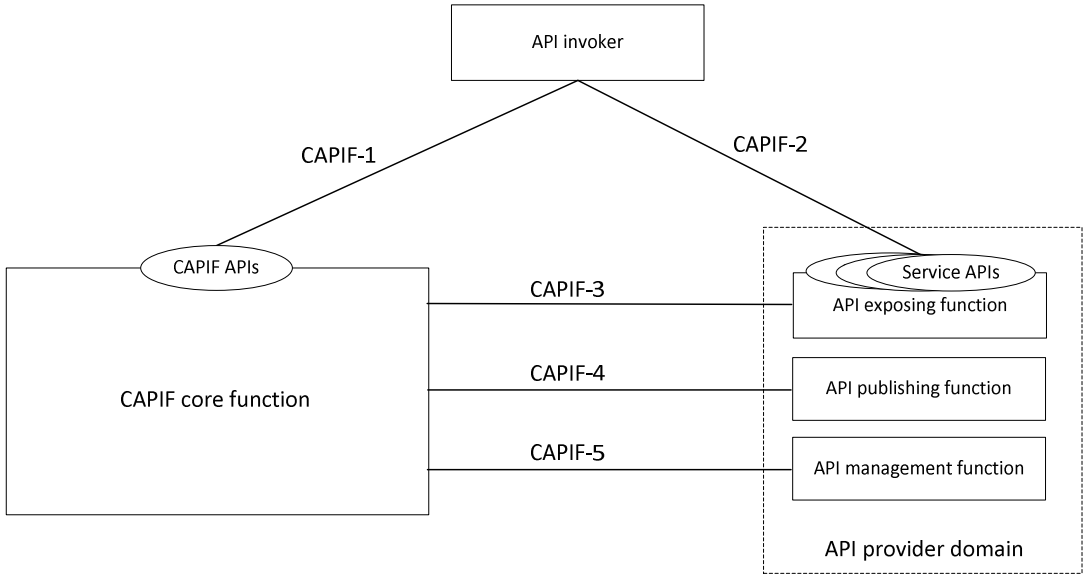


Figure 7.3-1: Distributed deployment of the CAPIF within PLMN trust domain

In distributed deployment, the CAPIF core function and the API provider domain functions are not co-located. The CAPIF core function interacts with the API exposing function, the API publishing function and the API management function via CAPIF-3, CAPIF-4 and CAPIF-5 reference points respectively. The API invoker can interact independently with the CAPIF core function and the API exposing function including the service APIs. In this deployment, the API exposing function appears as an agent for all service API invocations from the API invoker. The API invoker obtains the service API information and its entry point details from the CAPIF core function via CAPIF-1 interface. The first point of entry for the service API is the API exposing function during API invocation. The API exposing function acts as agent for service API applying any access control or policy control to the interactions between the API invoker and the service API in coordination with the CAPIF core function via CAPIF-3 interface.

The CAPIF can be deployed by splitting the functionality of the API exposing function among multiple API exposing function entities, of which one acts as the entry point. However there will be single API publishing function and single API management function in the API provider domain although there could be multiple API exposing function entities. The CAPIF deployment with cascading API exposing functions is as illustrated in the figure 7.3-2.

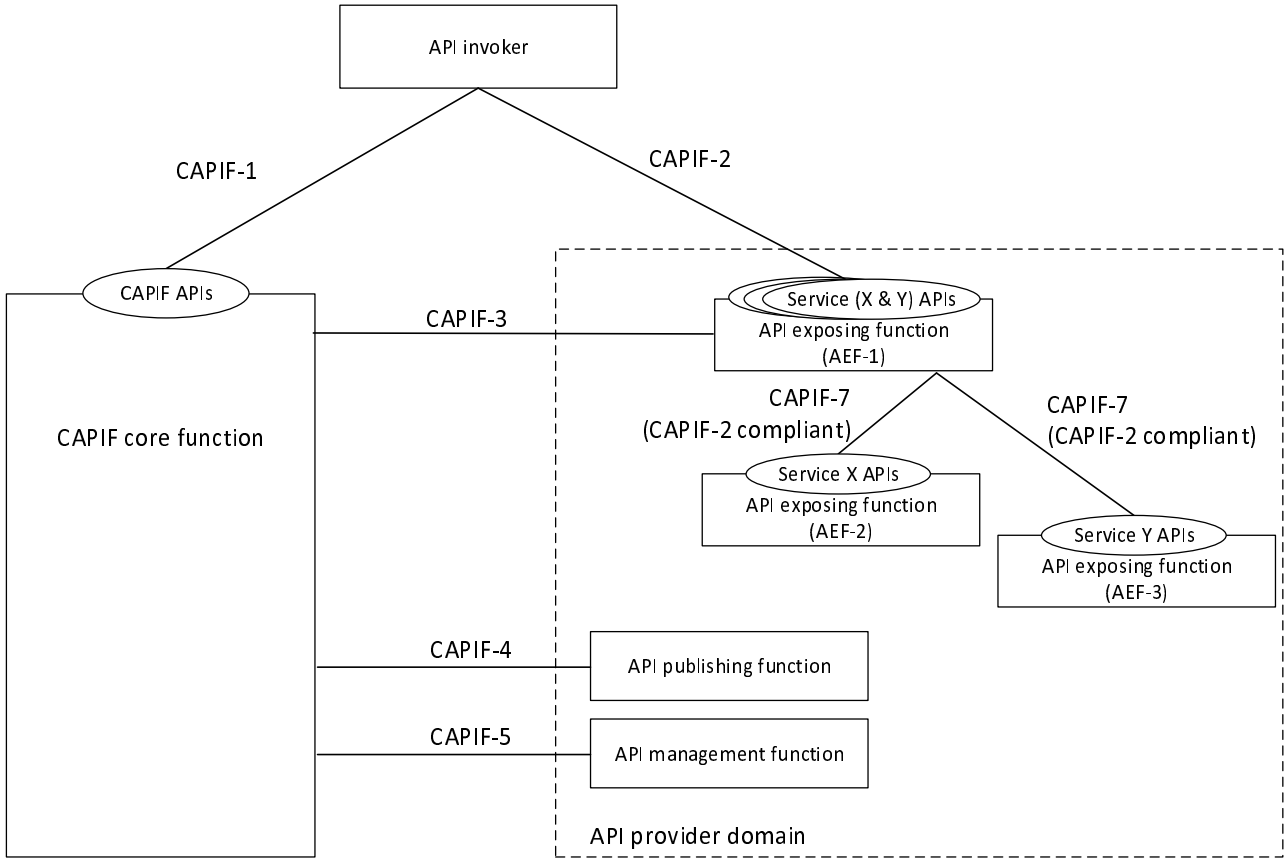


Figure 7.3-2: Distributed deployment of the CAPIF with cascading API exposing functions

In this deployment option, the API exposing function can have several instances like AEF-1, AEF-2 and AEF-3 which can be assigned with different roles. The roles for each API exposing function are decided by the operator. In this illustration, the API exposing functions AEF-2 and AEF-3 provide service APIs for service X and service Y respectively. The API exposing function AEF-1 provides the service communication entry point to the service APIs for service X APIs and service Y APIs. The API exposing function AEF-1 for instance can hide the topology of service X APIs and service Y APIs from the API invoker. The API exposing function AEF-1 also applies any access control or policy control to the interactions between the API invoker and service X APIs and between the API invoker and service Y APIs, in coordination with the CAPIF core function using CAPIF-3.

The API invoker interacts with the CAPIF core function via CAPIF-1. The API invoker interacts with service (X&Y) APIs on the API exposing function AEF-1 via CAPIF-2. The API exposing function AEF-1 forwards the invocation of the service X API or service Y API from the API invoker to the API exposing functions AEF-2 or AEF-3 respectively via CAPIF-2. The API messages are forwarded via CAPIF-7 (in compliance with CAPIF-2 interaction between the API invoker and the AEF-1) in the interactions between API exposing functions. The API invoker cannot directly interact with service X APIs and service Y APIs provided by API exposing functions AEF-2 and AEF-3 respectively.

Different splits of responsibility are possible. In another example illustrated in figure 7.3-3, the API exposing function AEF-1 could provide topology hiding for API exposing functions AEF-2 and AEF-3, plus access control for AEF-3. The API exposing function AEF-2 would provide its own access control, interacting with the CAPIF core function via CAPIF-3.

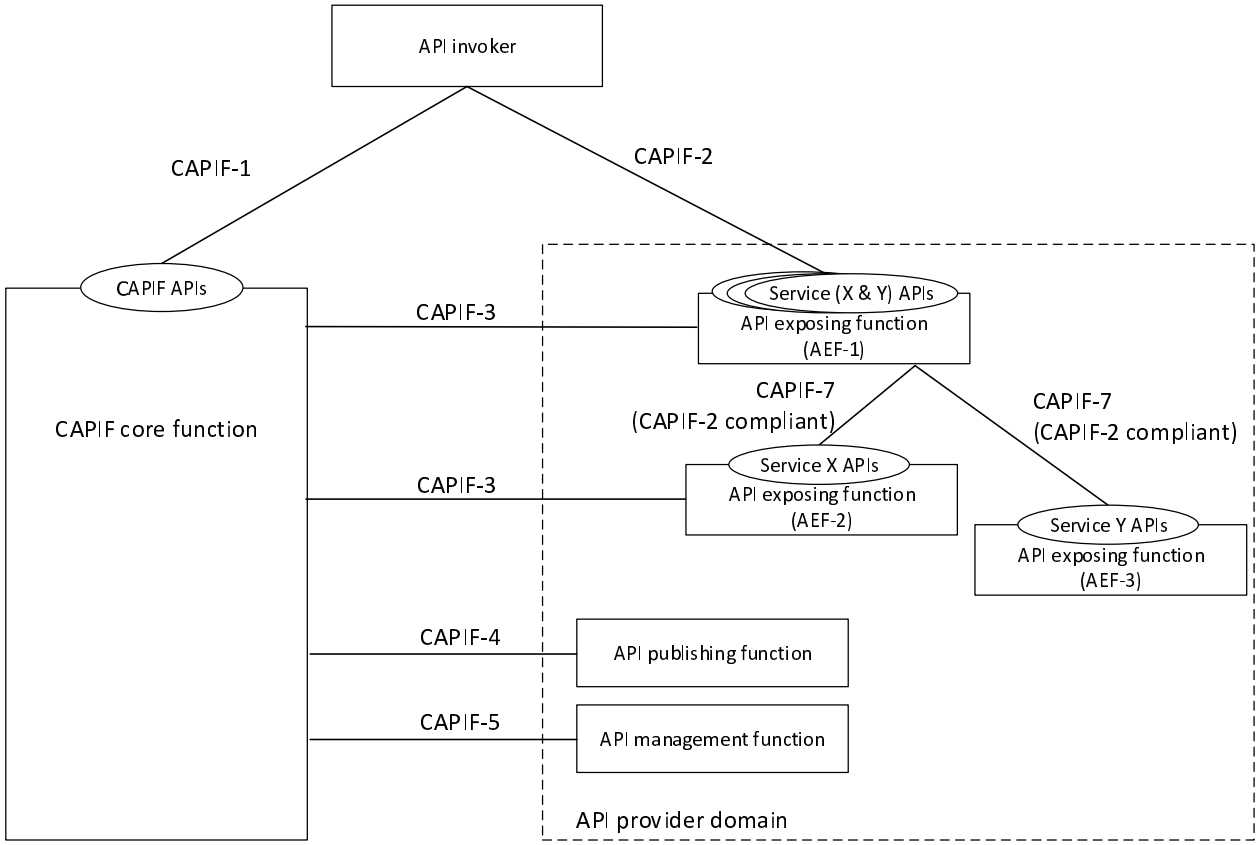


Figure 7.3-3: Another example of distributed deployment of the CAPIF with cascading API exposing functions

NOTE 1: The API invoker can be outside the PLMN trust domain and will access the CAPIF via CAPIF-1e and CAPIF-2e instead of CAPIF-1 and CAPIF-2.

When considering the 3rd party trust domain deployment, the API provider domain belongs to a 3rd party trust domain, the CAPIF core function belongs to PLMN trust domain and the API invoker belongs to PLMN trust domain as illustrated in figure 7.3-4.

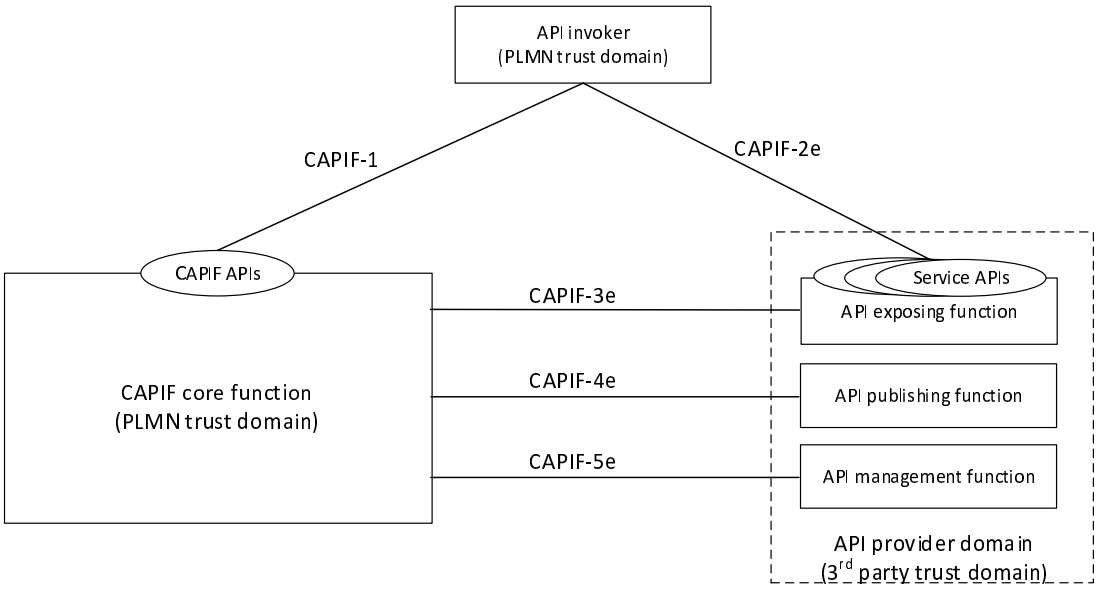


Figure 7.3-4: Distributed deployment of CAPIF considering PLMN trust domain and 3rd party trust domain

The interactions between the AEF and the CAPIF core function is based on CAPIF-3e. The interactions between the API publisher function and the CAPIF core function is based on CAPIF-4e. The interactions between the API management function and the CAPIF core functions are based on CAPIF-5e. The interactions between the API invoker and the AEF are based on CAPIF-2e. The API provider domain functions may be deployed in the PLMN trust domain and the interactions of the API provider domain functions within CAPIF of the PLMN trust domain is not shown in the figure 7.3-4 and is as illustrated in figure 7.3-1.

NOTE 2: For deployments illustrated in figure 7.3-2 and figure 7.3-3, when the API provider domain belongs to the 3rd party trust domain, the interactions between the AEF of the API provider domain and API invoker belonging to the PLMN trust domain are carried over CAPIF-2e reference point and the interactions between the entities of the API provider domain and the CAPIF core function belonging to the PLMN trust domain are carried over CAPIF-3e, CAPIF-4e and CAPIF-5e as illustrated in figure 7.3-4.

7.4 Multiple CCFs deployment

Multiple CAPIF core functions may be deployed within the PLMN trust domain as illustrated in the figure 7.4-1. For simplicity, the API invoker is not shown.

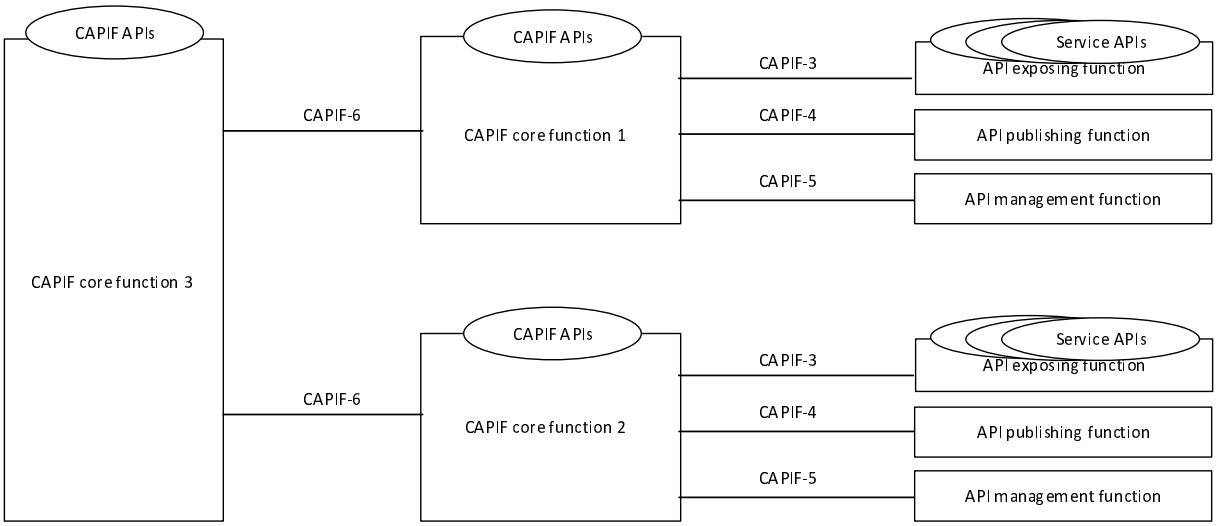


Figure 7.4-1: Multiple CCFs deployment within the PLMN trust domain

In the distributed deployment, the CAPIF core function 1 and the CAPIF core function 2 interact with CAPIF core function 3 via CAPIF-6 reference point. The CAPIF core function 3 assumes the role of a centralized repository of service APIs in the PLMN trust domain.

NOTE: The CAPIF core function 3 can be connected with the API exposing function(s) and API invokers.

The CAPIF core function 1 and the CAPIF core function 2 publishes the service API provided by its connected API exposing function(s) to the CAPIF core function 3, and obtains the service API information provided by other CAPIF core function(s).

An API invoker (not shown in the figure for simplicity) connected to the CAPIF core function 1 is able to discover and invoke the service APIs provided by the API exposing function connected to the CAPIF core function 2.

7.5 RNAA deployments

CAPIF supports RNAA and has enabled API invoker(s) to have authorized access to resources of a resource owner provided by service APIs offered by the AEF. The CCF acts as the Authorization Function and supports the authentication and authorization of API invoker(s). After checking the resource owner authorization information (e.g., user consent) via the ROF, the CCF (Authorization function) provides the access token for a service API access to the API invoker. The API invoker performs service API invocations on the AEF by utilizing the access token.

The API invoker may be deployed in the following ways:

a. API invoker may be deployed as AF on the UE (i.e. 3rd party application).

b. API invoker may be deployed as AF on the UE supporting several other 3rd party applications deployed on the UE.

c. API invoker may be deployed on the network as AF.

The resource owner is connected via a UE and can use a Resource Owner Function on the UE to interact with the CCF acting as the Authorization Function to grant RO permission to the API invoker to access resource(s) of the resource owner provided by the service API. The CCF (Authorization Function) uses the RO authorization information, as part of API invoker authorization process.

NOTE: The details of the protocol for CAPIF supporting RNAA is specified in 3GPP TS 33.122 [12].

8 Procedures and information flows

8.1 Onboarding the API invoker to the CAPIF

8.1.1 General

The procedure in this subclause corresponds to the architectural requirements for onboarding the API invoker to the CAPIF. The CAPIF enables a one time onboarding process that enrolls the API invoker as a recognized user of the CAPIF, which may be triggered by the API invoker via CAPIF-1 or CAPIF-1e, or may be based on provisioning.

8.1.2 Information flows

8.1.2.1 Onboard API invoker request

Table 8.1.2.1-1 describes the information flow onboard API invoker request from the API invoker to the CAPIF core function.

Table 8.1.2.1-1: Onboard API invoker request

Information element	Status	Description
Onboarding information	M	The information of the API invoker including enrolment details, required for onboarding
> Onboarding criteria	O	Indicates that the API Invoker wishes to on board itself to the CCF only if the CCF supports the features mentioned in criteria information.
>> Criteria information	M	This information includes at least one of the criteria listed (supported security methods, supported security methods for given AEFs / service APIs/ Service API categories) for the CCF to onboard the API invoker.
APIs for enrollment	O	List of APIs being enrolled for.
Proposed expiration time	O	Proposed expiration time for the onboarding.

8.1.2.2 Onboard API invoker response

Table 8.1.2.2-1 describes the information flow onboard API invoker response from the CAPIF core function to the API invoker.

Table 8.1.2.2-1: Onboard API invoker response

Information element	Status	Description
Onboarding status	M	The result of onboarding request i.e., success indication is included if the API invoker is granted permission otherwise failure.
Enrolled information	O (see NOTE 1)	Information from the provisioned API invoker profile which includes information to allow the API invoker to be authenticated and to obtain authorization for service APIs, containing the unique API invoker identity associated to the onboarded API invoker profile.
Service API information	O (see NOTE 2)	The service API information as specified in Table 8.7.2.2-1.
Reason	O (see NOTE 3)	This element indicates the reason when onboarding status is failure.
Expiration time	O	Indicates the expiration time of the onboarding. At expiration, CCF cancels the enrollment of the API invoker from CAPIF. If omitted, it indicates the onboarding does not expire.
NOTE 1: Information element shall be present when onboarding status is successful. NOTE 2: Information element may be present when onboarding status is successful. NOTE 3: Information element shall be present when onboarding status is failure.		

8.1.3 Procedure

Figure 8.1.3-1 illustrates the procedure for onboarding the API invoker to the CAPIF. The security aspects of this procedure are specified in subclause 6.1 of 3GPP TS 33.122 [12].

Pre-conditions:

1. The API invoker is not a recognized user of the CAPIF.
2. The API invoker has visibility to APIs information (e.g., API catalogue or dashboard - central place for the API provider to manage which APIs are displayed, giving API invokers the ability to enroll for).

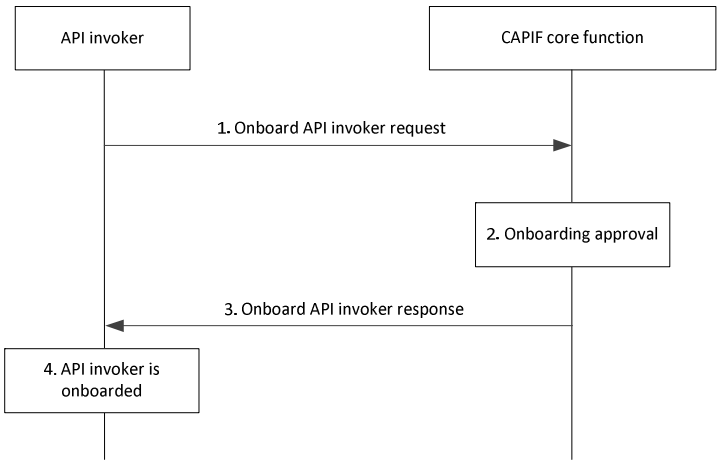


Figure 8.1.3-1: Procedure for onboarding the API invoker to the CAPIF

1. For enrolment of the API invoker to be a recognized user of the CAPIF, the API invoker triggers onboard API invoker request towards the CAPIF core function, providing the information as required for the API management and criteria information as required to onboard the API invoker.
2. The CAPIF core function begins the onboarding process by verifying whether all the necessary information has been provided to onboard the API invoker, and further initiates a grant process. If the onboard API invoker request message includes onboard criteria information, then the CCF verifies whether the CCF can support the onboard criteria information. The CCF further proceeds with API invoker onboarding procedure, only if the CCF supports the onboard criteria information from the API Invoker. Successful onboarding results in provisioning

API invoker profile which includes identity for the API invoker. The authorization information and the list of APIs and the categories of APIs that the API invoker can access subsequent to successful onboarding may also be created, considering the network slice information, if applicable. The CAPIF core function may create access control policy (see Table E-1) for the onboarded API invoker considering the network slice information.

NOTE 1: Completion of onboarding process can require explicit grant by the CAPIF administrator or the API management, which is left out-of-scope of this solution. CAPIF can handle the grant process internally without the need of explicit grant by the CAPIF administrator.

NOTE 2: The API invoker profile consists of at least the identity information for the API invoker, information required for the authentication and authorization by the CAPIF (e.g., application service information such as application service provider and application identifier) and the CAPIF identity information.

3. If the API invoker has triggered the onboard API invoker request and is granted permission, the onboard API invoker response provides success indication including information from the provisioned API invoker profile which may include information to allow the API invoker to be authenticated and to obtain authorization for service APIs.

4. As a result of successful onboarding process, the CAPIF core function is able to authenticate and authorize the API invoker.

8.2 Offboarding the API invoker from the CAPIF

8.2.1 General

This subclause defines the procedure for offboarding the API invoker from the CAPIF. The offboarding process makes the API invoker no longer a recognized user of the CAPIF. The procedure is triggered by the API invoker over CAPIF-1 or CAPIF-1e.

8.2.2 Information flows

This subclause describes the information flows for the API invoker offboarding.

8.2.2.1 Offboard API invoker request

Table 8.2.2.1-1 describes the information flow offboard API invoker request from the API invoker to the CAPIF core function.

Table 8.2.2.1-1: Offboard API invoker request

Information element	Status	Description
API invoker identity information	M	Identity information of the API invoker requesting offboarding
Reason	O	Indicate the reason of offboarding

8.2.2.2 Offboard API invoker response

Table 8.2.2.2-1 describes the information flow offboard API invoker response from the CAPIF core function to the API invoker.

Table 8.2.2.2-1: Offboard API invoker response

Information element	Status	Description
Result	M	Indicates the success or failure of the offboarding operation

8.2.3 Procedure

Figure 8.2.3-1 illustrates the procedure for offboarding the API invoker from the CAPIF, triggered by the API invoker. The security aspects of this procedure are specified in subclause 6.8 of 3GPP TS 33.122 [12].

Pre-conditions:

1. The API invoker has been onboarded as a recognized user of the CAPIF.

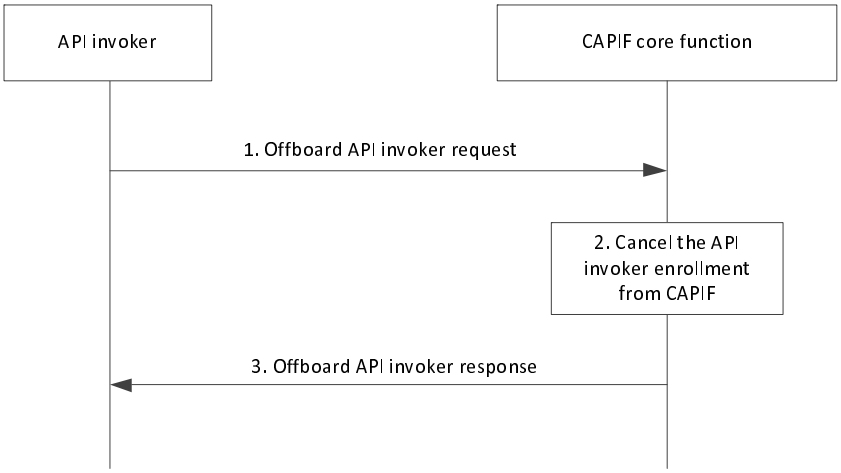


Figure 8.2.3-1: Procedure for offboarding the API invoker from the CAPIF

1. The API invoker triggers offboard API invoker request to the CAPIF core function, providing the information as required for the API management.
2. The CAPIF core function cancels the enrollment of the API invoker from CAPIF. The API invoker ceases to be a recognized user of the CAPIF. All the authorizations corresponding to the API invoker are revoked from CAPIF. Optionally, the information of the API invoker may be retained at the CAPIF core function as per the operator policy.

NOTE: Completion of offboarding process can require explicit notification to the CAPIF administrator or the API management, which is left out-of-scope of this solution. CAPIF can handle the de-provisioning process internally without the need of explicit grant by the CAPIF administrator.

3. The CAPIF core function returns the offboard API invoker response providing successful offboarding indication.

8.3 Publish service APIs

8.3.1 General

The CAPIF supports publishing service APIs by the API provider. The API publishing function can be within PLMN trust domain or within 3rd party trust domain.

8.3.2 Information flows

8.3.2.1 Service API publish request

Table 8.3.2.1-1 describes the information flow service API publish request from the API publishing function to the CAPIF core function.

Table 8.3.2.1-1: Service API publish request

Information element	Status	Description
API publisher information	M	The information of the API publisher may include identity, authentication and authorization information
Service API information	M (see NOTE 2)	The service API information includes the service API name, API provider name (optional), List of public IP ranges of UEs (optional), service API category (e.g. V2X, IoT), service API status (e.g. active, inactive), communication type, description, Serving Area Information (optional), AEF location (optional), interface details (e.g. IP address, port number, URI), protocols, version numbers, data format, Service KPIs (optional), service API operation(s) and resource(s) (optional), and Network Slice Info (optional), security methods.
Shareable information	O (see NOTE 1)	Indicates whether the service API information or the service API category can be published to other CCFs. And if sharing, a list of CAPIF provider domain information where the service API information or the service API category can be published is contained.
NOTE 1: If the shareable information is not present, the service API information is not allowed to be shared. NOTE 2: The active and inactive states of the service APIs of 3GPP AEFs are based on the respective combinations of the management states (administrative, operational) specified in the management domain in TS 28.541 [17], TS 28.532 [18] and TS 28.538 [19].		

The Service KPIs is defined as below:

Table 8.3.2.1-2: Service KPIs

Information element	Status	Description
Maximum Request rate	O	Maximum request rate from the API Invoker supported by the server.
Maximum Response time	O	The maximum response time advertised for the API Invoker's service requests.
Availability	O	Advertised percentage of time the server is available for the API Invoker's use.
Available Compute	O	The maximum compute resource available for the API Invoker.
Available Graphical Compute	O	The maximum graphical compute resource available for the API Invoker.
Available Memory	O	The maximum memory resource available for the API Invoker.
Available Storage	O	The maximum storage resource available for the API Invoker.
Connection Bandwidth	O	The connection bandwidth in Kbit/s advertised for the API Invoker's use.

8.3.2.2 Service API publish response

Table 8.3.2.2-1 describes the information flow service API publish response from the CAPIF core function to the API publishing function.

Table 8.3.2.2-1: Service API publish response

Information element	Status	Description
Result	M	Indicates the success or failure of publishing the service API information
Service API published information reference	O (see NOTE)	The information which can be used for referencing the information (set) about the published service API by the API publishing function.
Service API information	O (see NOTE)	The authorized service API information, which may be a subset or the full set, of the Service API information as specified in Table 8.3.2.1-1.
NOTE: This information element is included when the Result indicates success.		

8.3.3 Procedure

Figure 8.3.3-1 illustrates the procedure for publishing the service APIs. The service API publish mechanism is supported by the CAPIF core function.

Pre-conditions:

1. Authorization details of the APF are available with the CAPIF core function.
2. API invokers may have subscribed with the CAPIF core function to obtain new service API information.

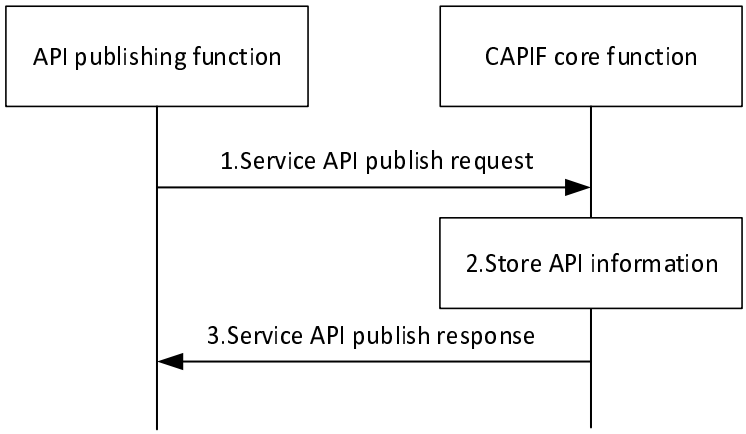


Figure 8.3.3-1: Publish service APIs

1. The API publishing function sends a service API publish request to the CAPIF core function, with the details of the service API. If the service API is to be shared to other CAPIF core functions, the shareable information and the CAPIF provider domain information are included.
2. Upon receiving the service API publish request, the CAPIF core function checks whether the API publishing function is authorized to publish service APIs. If the check is successful, the service API information provided by the API publishing function is stored at the CAPIF core function (API registry).
3. The CAPIF core function provides a service API publish response to the API publishing function indicating success or failure result and triggers notifications to subscribed API invokers as described in subclause 8.8.4.

8.4 Unpublish service APIs

8.4.1 General

The CAPIF supports unpublishing service APIs by the API provider. Once the service API information is unpublished, it is no more available to be discovered by API invokers. The API publishing function can be within PLMN trust domain or within 3rd party trust domain.

8.4.2 Information flows

8.4.2.1 Service API unpublish request

Table 8.4.2.1-1 describes the information flow service API unpublish request from the API publishing function to the CAPIF core function.

Table 8.4.2.1-1: Service API unpublish request

Information element	Status	Description
API publisher information	M	The information of the API publisher may include identity, authentication and authorization information
Service API published information reference	M	The information provided by the CAPIF core function which can be for referencing the information (set) about the published service API by the API publishing function.

8.4.2.2 Service API unpublish response

Table 8.4.2.2-1 describes the information flow service API unpublish response from the CAPIF core function to the API publishing function.

Table 8.4.2.2-1: Service API unpublish response

Information element	Status	Description
Result	M	Indicates the success or failure of unpublishing the service API information

8.4.3 Procedure

Figure 8.4.3-1 illustrates the procedure for unpublishing the service APIs. The service API unpublish mechanism is supported by the CAPIF core function.

Pre-conditions:

1. Authorization details of the APF are available with the CAPIF core function.
2. API invokers may have subscribed with the CAPIF core function to obtain notification regarding service API unpublish.

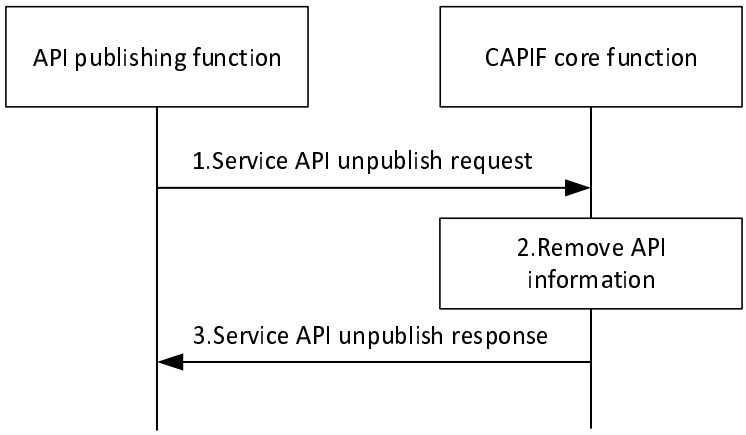


Figure 8.4.3-1: Unpublish service APIs

1. The API publishing function sends a service API unpublish request to the CAPIF core function, with service API published information reference provided by the CAPIF core function when the service API was published.

2. Upon receiving the service API unpublish request, the CAPIF core function checks whether the API publishing function is authorized to unpublish service APIs. If the check is successful, the service API information provided by the API publishing function is removed at the CAPIF core function (API registry).

3. The CAPIF core function provides a service API unpublish response to the API publishing function and triggers notifications to subscribed API invokers as described in subclause 8.8.4.

8.5 Retrieve service APIs

8.5.1 General

The CAPIF supports retrieving the published service APIs information by the API provider. The API publishing function can be within PLMN trust domain or within 3rd party trust domain.

8.5.2 Information flows

8.5.2.1 Service API get request

Table 8.5.2.1-1 describes the information flow service API get request from the API publishing function to the CAPIF core function.

Table 8.5.2.1-1: Service API get request

Information element	Status	Description
API publisher information	M	The information of the API publisher may include identity, authentication and authorization information
Service API published information reference	M	The information provided by the CAPIF core function which can be for referencing the information (set) about the published service API by the API publishing function.

8.5.2.2 Service API get response

Table 8.5.2.2-1 describes the information flow service API get response from the CAPIF core function to the API publishing function.

Table 8.5.2.2-1: Service API get response

Information element	Status	Description
Result	M	Indicates the success or failure of retrieving the service API information
Cause	O	The cause for the request failure.
Service API information	O (see NOTE)	The service API information as specified in Table 8.3.2.1-1.
NOTE: Shall be present if the Result information element indicates that the service API get request is successful. Otherwise service API information shall not be present.		

8.5.3 Procedure

Figure 8.5.3-1 illustrates the procedure for retrieving the service APIs. The service API retrieval mechanism is supported by the CAPIF core function.

Pre-condition:

1. Authorization details of the APF are available with the CAPIF core function.

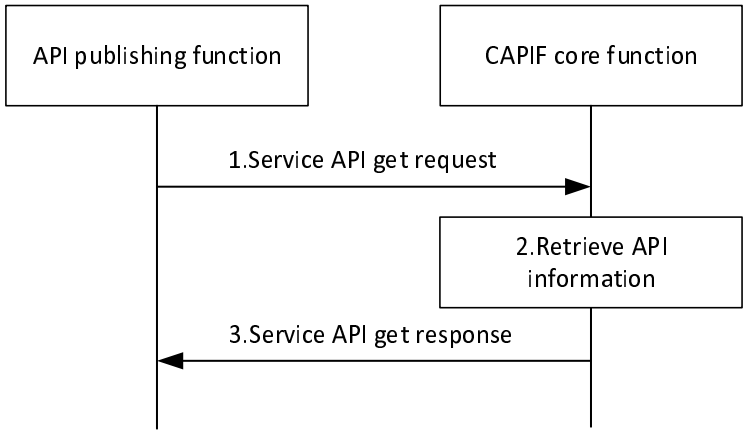


Figure 8.5.3-1: Retrieve service APIs

1. The API publishing function sends a service API get request to the CAPIF core function, with service API published information reference provided by the CAPIF core function when the service API was published.
2. Upon receiving the service API get request, the CAPIF core function checks whether the API publishing function is authorized to get published service APIs information. If the check is successful, the corresponding service API information is retrieved from the CAPIF core function (API registry).
3. The CAPIF core function provides a service API get response to the API publishing function which includes the service API information.

8.6 Update service APIs

8.6.1 General

The CAPIF core function allows the service API provider to update the information related to the published service API, e.g. a change in the characteristics of the service API. This procedure is initiated by the API publishing function to the CAPIF core function. The API publishing function can be within PLMN trust domain or within 3rd party trust domain.

8.6.2 Information flows

8.6.2.1 Service API update request

Table 8.6.2.1-1 describes the information flow service API update request from the API publishing function to the CAPIF core function.

Table 8.6.2.1-1: Service API update request

Information element	Status	Description
API publisher information	M	The information of the API publisher may include identity, authentication and authorization information
Service API published information reference	M	The information (set) provided by the CAPIF core function about the published service API which can be used for reference by the API publishing function.
Service API information	M	The service API information as specified in Table 8.3.2.1-1, as required to replace the existing service API information.
Reason	O	The reason of the update (e.g. change log).

NOTE: How to monitor service API status when the APF is unable to update service API status is not specified in this release.

8.6.2.2 Service API update response

Table 8.6.2.2-1 describes the information flow service API update response from the CAPIF core function to the API publishing function.

Table 8.6.2.2-1: Service API update response

Information element	Status	Description
Result	M	Indicates the success or failure of updating the service API information
Cause	O	The cause for the request failure.
Service API information	O	The authorized service API information during update, applicable when the update result is success. This can be a subset or the full set, of the Service API information as specified in Table 8.3.2.1-1.

8.6.3 Procedure

Figure 8.6.3-1 illustrates the procedure for updating the published service APIs information. The service API update mechanism is supported by the CAPIF core function.

Pre-conditions:

1. Authorization details of the APF are available with the CAPIF core function.
2. API invokers may have subscribed with the CAPIF core function to obtain notification regarding update to service API information.

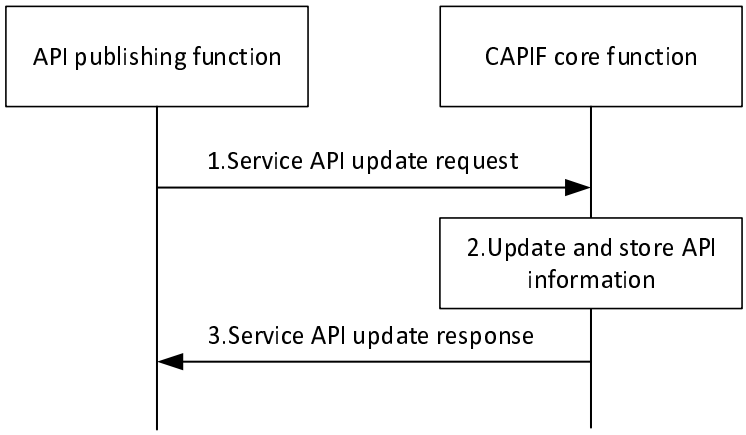


Figure 8.6.3-1: Update service APIs

1. The API publishing function sends a service API update request to the CAPIF core function, which includes the service API published information reference provided by the CAPIF core function when the service API was published and the new service API information which is to be updated.
2. Upon receiving the service API update request, the CAPIF core function checks whether the API publishing function is authorized to update the published service APIs information. If the check is successful, the service API information provided by the API publishing function is updated at the CAPIF core function (API registry).
3. The CAPIF core function provides a service API update response to the API publishing function and triggers notifications to subscribed API invokers as described in subclause 8.8.4.

8.7 Discover service APIs

8.7.1 General

The following procedure in this subclause corresponds to the architectural requirements on discover service APIs. This procedure may be invoked by different types of API invokers (i.e., application management client, hosted application) at different phases (i.e., hosted application development, test phase and hosted application running phase). Different API invokers utlize this procedure for different purpose:

- service API discovery for interested service APIs with service API informaiton including e.g., API name, API provider name, API category.
- service API discovery for the interface information (e.g. IP address, port number, URI) of a particular service API in case of the interface information is not provided in the service API information during hosted application running phase.

8.7.2 Information flows

8.7.2.1 Service API discover request

Table 8.7.2.1-1 describes the information flow service API discover request from the API invoker to the CAPIF core function.

Table 8.7.2.1-1: Service API discover request

Information element	Status	Description
API invoker identity information	M	Identity information of the API invoker discovering service APIs
Query information	O (NOTE 2)	Criteria for discovering matching service APIs (e.g. service API category, Serving Area Information (optional), preferred AEF location (optional), required API provider name (optional), UE IP address (optional), interfaces, protocols, Service KPIs (optional), security methods, service API operation(s) and resource(s) (optional), and Network Slice Info (optional)). (see NOTE 1)
Service API identification	O (NOTE 2)	The identification information of the service API(s) for which the request is targeting
NOTE 1: It should be possible to discover all the service APIs. NOTE 2: Either "Query information" IE or " Service API identification" IE shall be present.		

8.7.2.2 Service API discover response

Table 8.7.2.2-1 describes the information flow service API discover response from the CAPIF core function to the API invoker.

Table 8.7.2.2-1: Service API discover response

Information element	Status	Description
Result	M	Indicates the success or failure of the discovery of the service API information
Service API information (see NOTE 2)	O (see NOTE 1, NOTE 3)	The service API information as specified in Table 8.3.2.1-1, except for the service API status (e.g. active, inactive).
Interface information	O (NOTE 3)	The interface details (e.g. IP address, port number, URI) of the requested service API(s)
CAPIF core function identity information	O (see NOTE 1, NOTE 3)	Indicates the CAPIF core function serving the service API category provided in the query criteria
NOTE 1: The service API information or the CAPIF core function identity information or both shall be present if the Result information element indicates that the service API discover operation is successful. Otherwise, both shall not be present. NOTE 2: If topology hiding is enabled for the service API, the interface details shall be the interface details of AEF acting as service communication entry point for the service API. NOTE 3: "Interface information" IE shall not be present when "Service API information" or "CAPIF core function identity information" is present.		

8.7.3 Procedure

Figure 8.7.3-1 illustrates the procedure for discover service APIs.

The service API discovery mechanism is supported by the CAPIF core function.

Pre-conditions:

1. The API invoker is onboarded and has received an API invoker identity.
2. The CAPIF core function is configured with a discovery policy information (e.g. to restrict discovery to category of APIs) for API invoker(s).

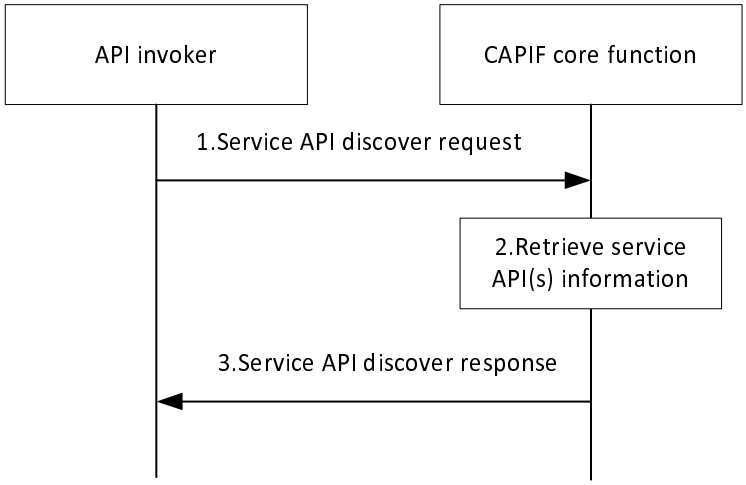


Figure 8.7.3-1: Discover service APIs

1. The API invoker sends a service API discover request to the CAPIF core function. It includes the API invoker identity, either query information or the service API identification.

   For service API discovery for interested service APIs, the query information is included.

   For service API discovery for interface information, service API identification is included.

2. Upon receiving the service API discover request, the CAPIF core function verifies the identity of the API invoker (via authentication).

If the query information is included, the CAPIF core function retrieves the stored service API(s) information from the CAPIF core function (API registry) as per the query information in the service API discover request. Further, the CAPIF core function applies the discovery policy and performs filtering of service APIs information retrieved from the CAPIF core function.

If the service API identification is included, the CAPIF core function retrieves the interface information of the requested service API(s).

3. The CAPIF core function sends a service API discover response to the API invoker.

If the query information is included, the list of service API information is returned.

If the service API identification is included, the interface information is returned.

8.8 Subscription, unsubscription and notifications for the CAPIF events

8.8.1 General

The CAPIF core function enables the subscribing entity (i.e. the API invoker, the API exposing function, the API publishing function, the API management function) to subscribe to and unsubscribe from the CAPIF events such as availability events of service APIs, change in service API information, monitoring service API invocations, API invoker onboarding events, etc. The subscription, unsubscription and notification for the CAPIF events are enabled on the following CAPIF reference points:

- CAPIF-1 or CAPIF-1e: the API invoker can subscribe to and unsubscribe from CAPIF events and receive notifications from the CAPIF core function;
- CAPIF-3 or CAPIF-3e: the AEF can subscribe to and unsubscribe from CAPIF events and receive notifications from the CAPIF core function;
- CAPIF-4 or CAPIF-4e: the API publishing function can subscribe to and unsubscribe from CAPIF events and receive notifications from the CAPIF core function; and
- CAPIF-5 or CAPIF-5e: the API management function can subscribe to and unsubscribe from CAPIF events and receive notifications from the CAPIF core function.

NOTE: Support for subscriptions and notifications can also be part of the actual service APIs. That type of subscriptions and notifications is not covered by the provisions in this clause.

8.8.2 Information flows

8.8.2.1 Event subscription request

Table 8.8.2.1-1 describes the information flow for event subscription request from the subscribing entity to the CAPIF core function.

Table 8.8.2.1-1: Event subscription request

Information element	Status	Description
Identity subscription information	M	The information to determine the identity of the subscribing entity
Event subscription information	M	The event subscription information as defined in Table 8.8.2.1-2
Notification reception information	O	The information of the subscribing entity for receiving the notifications for the event.

Table 8.8.2.1-2: Event subscription information

Information element	Status	Description
Events	M	One or more event identifiers. Table 8.8.6-1 provides a non-exhaustive list of CAPIF events.
Event filter information	O	Contains the matching criteria for the indicated event(s) (e.g., optionally the list of AEF(s) for which the CCF will detect the event service API availability).
Event reporting information	O (NOTE)	Contains, the notification method for the matched event, e.g. periodic, on event detection, etc. The event reporting information may include, per event, a reporting threshold on number of matched events, optionally within a time window, that indicates how many times an event must be matched to trigger a notification.
NOTE: The threshold on number of matched events applies only to the event detection notification method. The time window represents a fixed time duration (e.g., one hour) and may be provided when the reporting threshold on number of matched events is provided. The count of the matched events and the time window are restarted when the event notification is sent or when the time window expires but the threshold to trigger the notification is not reached.		

8.8.2.2 Event subscription response

Table 8.8.2.2-1 describes the information flow for event subscription response from the CAPIF core function to the subscribing entity.

Table 8.8.2.2-1: Event subscription response

Information element	Status	Description
Result	M	Indicates the success or failure of the event subscription operation
Cause	O	The cause for the request failure.
Subscription identifier	O (see NOTE)	The unique identifier for the event subscription.
NOTE: Shall be present if the Result information element indicates that the event subscription operation is successful. Otherwise subscription identifier shall not be present.		

8.8.2.3 Event notification

Table 8.8.2.3-1 describes the information flow for event notification from the CAPIF core function to the subscribing entity. A notification about an event is sent to a subscribing entity if the event subscription information in the related subscription match the corresponding attributes of the event content.

Table 8.8.2.3-1: Event notification

Information element	Status	Description
Subscription identifier	M	The unique identifier of the event subscription
Event identifier	M	The unique identifier for the event. For the list of events, refer subclause 8.8.6
Event related information	M	The event related information (e.g. time at which the event originated, location of event)
Event content	M	The content of the event information.

8.8.2.4 Event notification acknowledgement

Table 8.8.2.4-1 describes the information flow event notification acknowledgement from the subscribing entity to the CAPIF core function.

Table 8.8.2.4-1: Event notification acknowledgement

Information element	Status	Description
Acknowledgement	M	Acknowledgement for the event notification received.

8.8.2.5 Event unsubscription request

Table 8.8.2.5-1 describes the information flow for event unsubscription request from the subscribing entity to the CAPIF core function.

Table 8.8.2.5-1: Event unsubscription request

Information element	Status	Description
Identity information	M	The information to determine the identity of the subscribing entity
Subscription identifier	M	The unique identifier for the event subscription that was provided to the subscribing entity during the CAPIF event subscription operation.

8.8.2.6 Event unsubscription response

Table 8.8.2.6-1 describes the information flow for event unsubscription response from the CAPIF core function to the subscribing entity.

Table 8.8.2.6-1: Event unsubscription response

Information element	Status	Description
Result	M	Indicates the success or failure of the event unsubscription operation
Cause	O	The cause for the request failure.

8.8.2.7 Event subscription update request

Table 8.8.2.7-1 describes the information flow for event subscription update request from the subscribing entity to the CAPIF core function.

Table 8.8.2.7-1: Event subscription update request

Information element	Status	Description
Identity information	M	The information to determine the identity of the subscribing entity
Subscription identifier	M	The unique identifier for the event subscription that was provided to the subscribing entity during the CAPIF event subscription operation.
Event subscription information changes	O (NOTE)	Updates to the event subscription information which is defined in clause 8.8.2.1
Notification reception information changes	O (NOTE)	Updates to the information of the subscribing entity for receiving the notifications for the event
NOTE: At least one of these information elements shall be present.		

8.8.2.8 Event subscription update response

Table 8.8.2.8-1 describes the information flow for event subscription update response from the CAPIF core function to the subscribing entity.

Table 8.8.2.2-1: Event subscription update response

Information element	Status	Description
Result	M	Indicates the success or failure of the event subscription update operation
Cause	O	The cause for the request failure.

8.8.3 Procedure for CAPIF event subscription

Figure 8.8.3-1 illustrates the procedure for CAPIF events subscription.

Pre-conditions:

1. The subscribing entity has the authorization to subscribe for the CAPIF events.

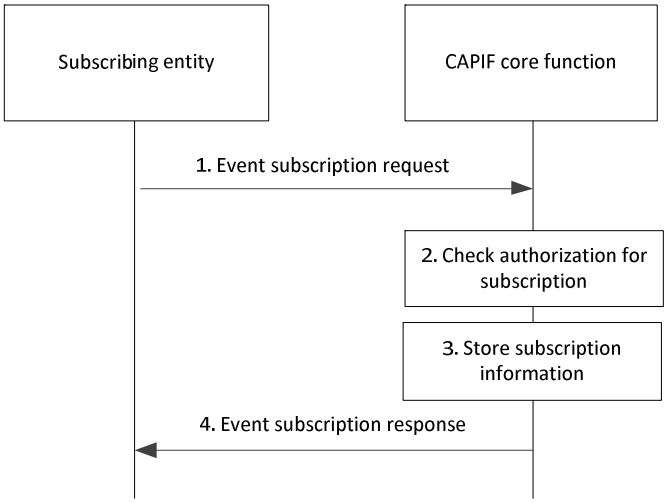


Figure 8.8.3-1: Procedure for CAPIF event subscription

1. The subscribing entity sends an event subscription request to the CAPIF core function in order to receive notification of events.
2. Upon receiving the event subscription request from the subscribing entity, the CAPIF core function checks for the relevant authorization for the event subscription.
3. If the authorization is successful, the CAPIF core function stores the subscription information.
4. The CAPIF core function sends an event subscription response indicating successful operation.

8.8.4 Procedure for CAPIF event notifications

Figure 8.8.4-1 illustrates the procedure for CAPIF event notifications.

Pre-conditions:

1. The subscription procedure as illustrated in figure 8.8.3-1 is performed by the subscribing entity.

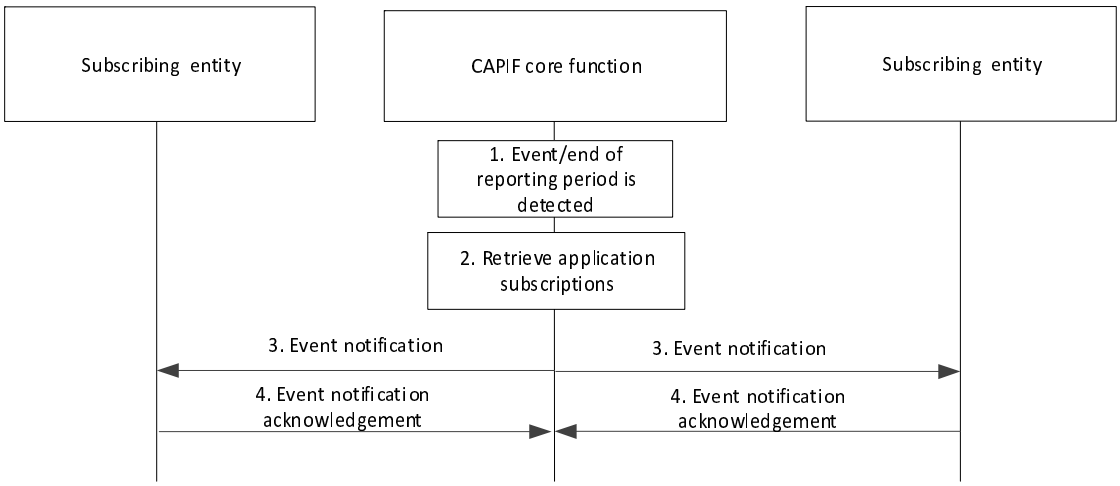


Figure 8.8.4-1: Procedure for CAPIF event notifications

1. The CAPIF core function may detect events to be consumed by the subscribing entity(s) or the end of a reporting period for the subscription of the subscribing entity(s).
2. The CAPIF core function retrieves the list of corresponding subscriptions, and determines the event information to send in the event report.
3. The CAPIF core function sends event notifications to all the subscribing entity(s) that have subscribed for the event when the event matching the criteria (event filter information) and the notification criteria (event reporting information) are fulfilled. If a notification reception information is available as part of the subscribing entity event subscription, then the notification reception information is used by the CAPIF core function to send event notifications to the subscribing entity.
4. The subscribing entity sends an event notification acknowledgement to the CAPIF core function for the event notification received.

8.8.5 Procedure for CAPIF event unsubscription

Figure 8.8.5-1 illustrates the procedure for CAPIF event unsubscription.

Pre-condition:

1. The subscribing entity has subscribed to the CAPIF events.

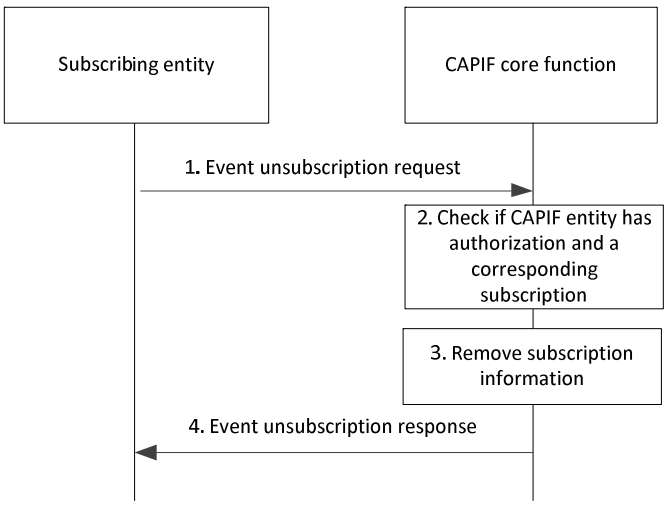


Figure 8.8.5-1: Procedure for CAPIF event unsubscription

1. The subscribing entity sends an event unsubscription request to the CAPIF core function with the information of the subscribed CAPIF event.
2. Upon receiving the event unsubscription request from the subscribing entity, the CAPIF core function checks for the event subscription corresponding to the subscribing entity and further checks if the subscribing entity is authorized to unsubscribe from the CAPIF event.
3. If the event subscription information corresponding to the subscribing entity is available and the subscribing entity is authorized to unsubscribe for the CAPIF event, the CAPIF core function removes the subscription information.
4. The CAPIF core function sends an event unsubscription response indicating successful operation.

8.8.5a Procedure for CAPIF event subscription update

Figure 8.8.5a-1 illustrates the procedure for CAPIF events subscription update.

Pre-conditions:

1. The subscribing entity has the authorization to update subscriptions for CAPIF events.
2. The subscribing entity has created subscriptions for CAPIF events.

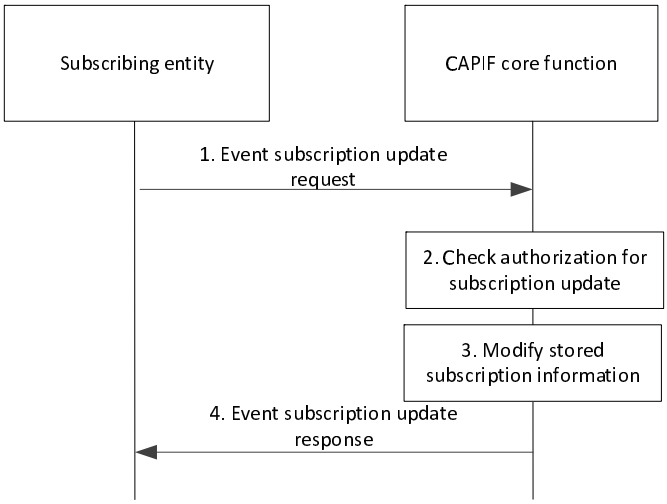


Figure 8.8.5a-1: Procedure for CAPIF event subscription

1. The subscribing entity sends an event subscription update request to the CAPIF core function in order update a previous subscription to receive notification of events.
2. Upon receiving the event subscription update request from the subscribing entity, the CAPIF core function checks for the relevant authorization for the event subscription update.
3. If the authorization is successful, the CAPIF core function updates the subscription information.
4. The CAPIF core function sends an event subscription update response indicating successful operation.

8.8.6 List of CAPIF events

Table 8.8.6-1 provides a non-exhaustive list of CAPIF events.

Table 8.8.6-1: List of CAPIF events

Events	Events Description
Availability of service APIs	Availability events of service APIs (e.g. active, inactive)
Service API updated	Events related to change in service API information
Monitoring service API invocations	Events corresponding to service API invocations
API invoker status	Events related to API invoker status in CAPIF (onboarded, offboarded, onboarding criteria not met status along with the criteria information that is not met.)
API topology hiding status	Events related to API topology hiding status in CAPIF (created, revoked)
System related events	Alarm events providing fault information
Performance related events	Events related to system load conditions
Service API of interest events	Events indicating whether the service API is of interest for the API invoker(s) (e.g., the service API is within the list of service APIs the API Invoker included during onboarding or the service API is within the list of service APIs the API invoker requested during discovery requests and received in responses).

8.9 Revoking subscription of the CAPIF events

8.9.1 General

The CAPIF core function allows to revoke subscription of CAPIF events for the subscribing entity related to the service API changes, such as availability events of service APIs, change in service API information, monitoring service API invocations, API invoker onboarding events, etc. This procedure is initiated by the CAPIF core function.

NOTE: It is optional to trigger notification by the CAPIF core function for revocation of subscription for CAPIF event(s).

8.9.2 Information flows

This subclause describes the information flows for CAPIF event subscription revocation.

8.9.2.1 Subscription revoke notification

Table 8.9.2.1-1 describes the information flow for subscription revoke notification from the CAPIF core function to the subscribing entity.

Table 8.9.2.1-1: Subscription revoke notification

Information element	Status	Description
Identity information	M	The information to determine the identity of the subscribing entity
Subscription identifier	M	The unique identifier for the event subscription that was provided to the subscribing entity during the CAPIF event subscription operation.
Reason	O	Indicate the reason of subscription revocation

8.9.2.2 Subscription revoke notification acknowledgement

Table 8.9.2.2-1 describes the information flow for subscription revoke notification acknowledgement from the subscribing entity to the CAPIF core function.

Table 8.9.2.2-1: Subscription revoke notification acknowledgement

Information element	Status	Description
Acknowledgement	M	The acknowledgement for the received notification.

8.9.3 Procedure

Figure 8.9.3-1 illustrates the procedure for subscription revocation, triggered by the CAPIF core function.

Pre-conditions:

1. The subscribing entity has previously subscribed to CAPIF event(s) to the CAPIF core function.

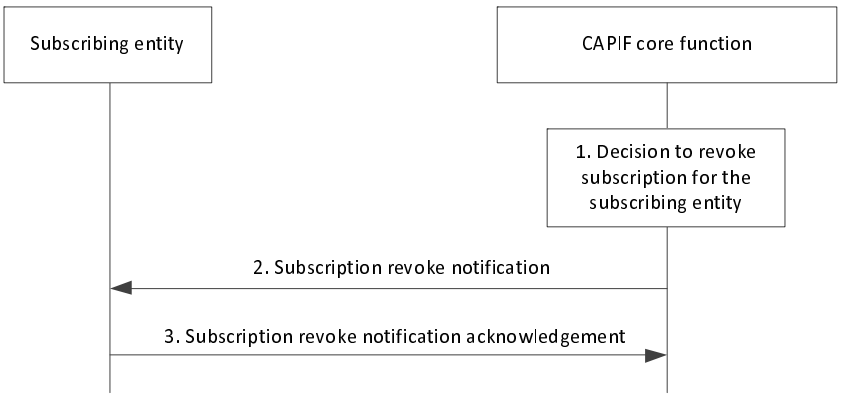


Figure 8.9.3-1: Procedure for revoking subscription of the CAPIF events

1. The CAPIF core function decides to revoke subscription of CAPIF event(s) for the subscribing entity.
2. The CAPIF core function sends subscription revoke notification to the subscribing entity.
3. The subscribing entity provides a subscription revoke notification acknowledgement to the CAPIF core function.

8.10 Authentication between the API invoker and the CAPIF core function

8.10.1 General

The procedure in this subclause corresponds to the architectural requirements for authentication between the API invoker and the CAPIF core function.

8.10.2 Information flows

NOTE: The security aspects of this procedure are specified in subclause 6.2 and subclause 6.3.1 of 3GPP TS 33.122 [12].

8.10.3 Procedure

Figure 8.10.3-1 illustrates the procedure for authentication between the API invoker and the CAPIF core function.

Pre-conditions:

1. The API invoker is onboarded with the CAPIF core function and the API invoker profile is created.

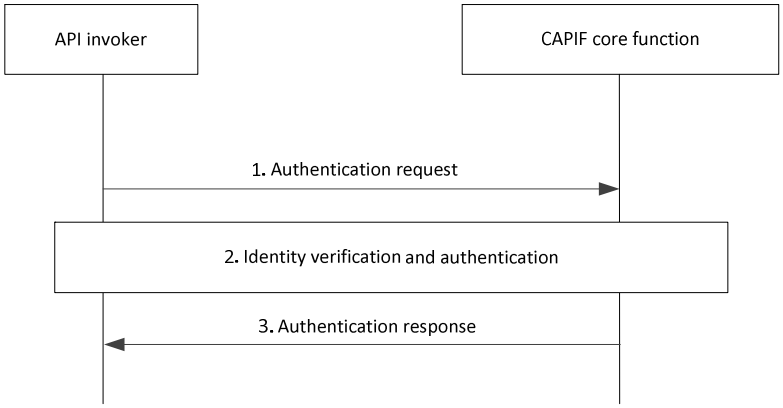


Figure 8.10.3-1: Procedure for authentication between the API invoker and the CAPIF core function

1. The API invoker triggers authentication to the CAPIF core function, including the identity confirmed after successful onboarding.
2. Upon receiving the authentication request, the CAPIF core function verifies the identity with the API invoker profile and authenticates the API invoker.

NOTE: The authentication process is specified in subclause 6.2 and subclause 6.3.1 of 3GPP TS 33.122 [12].

3. The CAPIF core function returns the result of the API invoker identity verification in the authentication response.

8.10A API Invoker obtaining security method to access service API

8.10A.1 General

The API invoker requires to execute this procedure to obtain the security method that shall be used by the API invoker and the API exposing function for CAPIF-2e interface authentication and authorization.

8.10A.2 Information flows

NOTE: The security aspects of this procedure are specified in subclause 6.2 and subclause 6.3.1.2 of 3GPP TS 33.122 [12].

8.10A.3 Procedure

NOTE: This procedure is specified in subclause 6.3.1.2 of 3GPP TS 33.122 [12].

8.11 API invoker obtaining authorization to access service API

8.11.1 General

The API invoker requires to execute this procedure when it needs to obtain or re-obtain (e.g. upon expiry of the authorization information) the authorization to access the service API. Once the API invoker receives the authorization to access the service API, the API invoker can perform one or multiple service API invocations as per the permission limit. This procedure may be performed during the API invoker onboarding process.

8.11.2 Information flows

NOTE: The security aspects of this procedure are specified in subclause 6.4 and subclause 6.5.2 of 3GPP TS 33.122 [12].

8.11.3 Procedure

Figure 8.11.3-1 illustrates the procedure for obtaining authorization to access the service API.

Pre-condition:

1. The API invoker is onboarded and has received an API invoker identity.

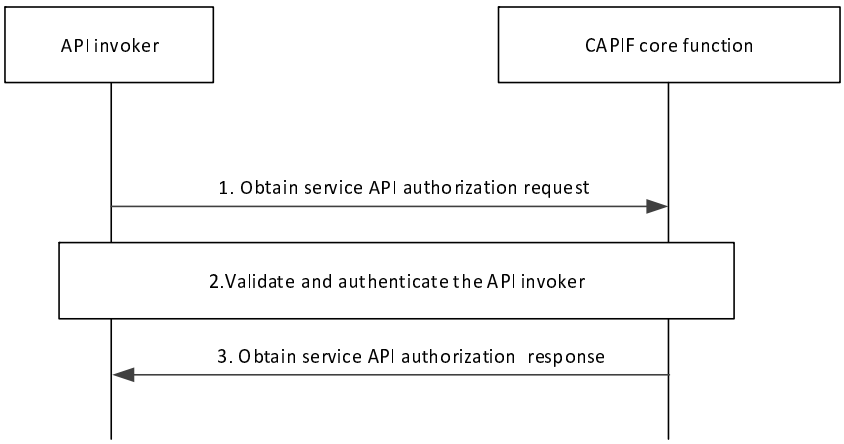


Figure 8.11.3-1: Procedure for the API invoker obtaining authorization for service API access

1. The API invoker sends an obtain service API authorization request to the CAPIF core function for obtaining permission to access the service API by including the API invoker information required for authorization, the information identifying the service API, and any information required for authentication of the API invoker. The request may include desired Network Slice Info of the service API. In addition to the information identifying the service API, the request may include finer granularity level for service API authorization e.g., service API operation and resource.

2. The CAPIF core function validates the authentication of the API invoker (using authentication information) and checks whether the API invoker is permitted to access the requested service API, considering also service API operation(s) and resource(s) restrictions locally configured, if available. The CAPIF core function may additionally verify the Network Slice Info, e.g., check that the authorized Network Slice Info for the API invoker is included in the supported Network Slice Info for the indicated service API.

NOTE 1: The authorization of Network Slice Info is performed by the CAPIF core function within the PLMN trust domain.

NOTE 2: The authentication process is specified in subclause 6.5.2 of 3GPP TS 33.122 [12].

3. Based on the API invoker's subscription information available in the CCF, the authorization information to access the service APIs, which may include access authorization per service operation, resource, optionally including access authorization per service operation and, optionally, resource, is sent to the API invoker in the obtain service API authorization response.

NOTE 3: How the finer service API authorization granularity is indicated in the request from API invoker and is returned to the API invoker in the CCF authorization reply is specified in 3GPP TS 33.122 [12].

NOTE 4: The mechanism for distribution of the authorization information for the API invoker to the API exposing function over CAPIF-3 reference point is specified in subclause 6.5.2 of 3GPP TS 33.122 [12].

8.12 AEF obtaining service API access control policy

8.12.1 General

The CAPIF core function is the central repository of all the policies related to service APIs. The AEF executes this procedure when it needs to obtain the policy to perform access control on the service API invocations (e.g. when policy for performing access control on service API is unavailable at the AEF). The AEF can be within PLMN trust domain or within 3rd party trust domain.

8.12.2 Information flows

8.12.2.1 Obtain access control policy request

Table 8.12.2.1-1 describes the information flow obtain access control policy request from the AEF to the CAPIF core function.

Table 8.12.2.1-1: Obtain access control policy request

Information element	Status	Description
AEF Identity information	M	Identity information of the entity requesting the access control policy.
Service API identification	M	The identification information of the service API for which the access control policy is being requested.

8.12.2.2 Obtain access control policy response

Table 8.12.2.2-1 describes the information flow obtain access control policy response from the CAPIF core function to the AEF.

Table 8.12.2.2-1: Obtain access control policy response

Information element	Status	Description
Result	M	Indicates the success or failure of the obtain access control policy operation
Access control policy information	O (see NOTE)	The access control policy information corresponding to the requested service API. (See Table E-1)
NOTE: Shall be present if the Result information element indicates that the obtain access control policy operation is successful. Otherwise access control policy information shall not be present.		

8.12.3 Procedure

Figure 8.12.3-1 illustrates the procedure for obtaining policy to perform access control on the service API invocations.

Pre-conditions:

1. The AEF is hosting the service API but the policy to perform access control is not available with AEF.
2. The CAPIF core function is configured with the access control policies corresponding to one or more service APIs.
3. Authorization details of the AEF are available with the CAPIF core function.

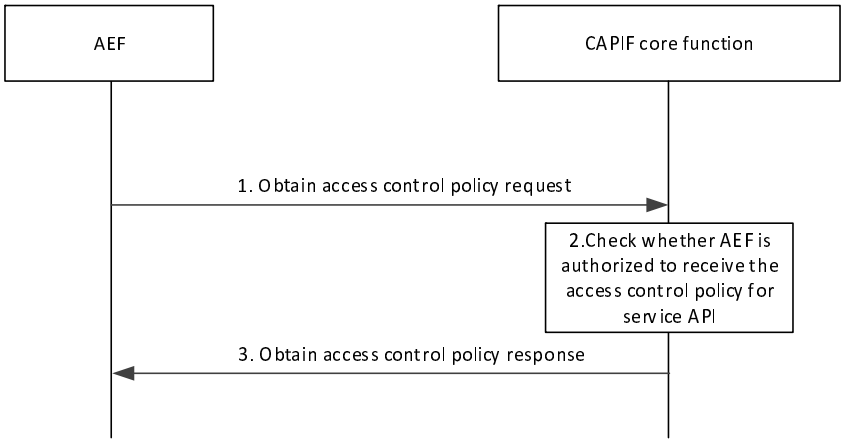


Figure 8.12.3-1: Procedure for the AEF obtaining service API access control policy

1. The AEF sends an obtain access control policy request to the CAPIF core function for obtaining the policy to perform the access control on service API invocations by including the details of the hosted service API.
2. The CAPIF core function checks whether the AEF is authorized to receive the access control policy corresponding to the service APIs requested.
3. If authorization check is successful, the AEF is provided the access control policy for the service API via an obtain access control policy response. If authorization check is not successful, the AEF is provided with a failure indication via a obtain access control policy response. The AEF enforces the access control policy for the service API and API invoker.

NOTE: To maintain synchronization between the AEF and the CAPIF core function for the policy cached at AEF, the AEF can subscribe to the policy update event at CAPIF core function according to the procedure in subclause 8.8.3 and receive notifications about any updated policy at CAPIF core function according to the procedure in subclause 8.8.4.

8.13 Topology hiding

8.13.1 General

The procedure in this subclause corresponds to the architectural requirements for hiding the topology of the PLMN trust domain from the API invokers accessing the service APIs from outside the PLMN trust domain.

8.13.2 Information flows

8.13.2.1 Service API invocation request (API invoker – AEF-1)

The information flow service API invocation request from the API invoker to AEF-1 (AEF acting as service communication entry point) is service API specific and the complete detail of the service API invocation request is out of scope of the present document. Table 8.17.2.1-1 describes the CAPIF related information elements which are included in the service API invocation request.

8.13.2.2 Service API invocation request (AEF-1 – AEF-2)

The information flow service API invocation request from AEF-1 (AEF acting as service communication entry point) to AEF-2 (destination AEF for handling service API) is service API specific and the complete detail of the service API invocation request is out of scope of the present document. Table 8.17.2.1-1 describes the CAPIF related information elements which are included in the service API invocation request.

8.13.2.3 Service API invocation response (AEF-2 – AEF-1)

The information flow service API invocation response from AEF-2 (destination AEF for handling service API) to AEF-1 (AEF acting as service communication entry point) is service API specific and the complete detail of the service API invocation response is out of scope of the present document. Table 8.17.2.2-1 describes the CAPIF related information elements which are included in the service API invocation response.

8.13.2.4 Service API invocation response (AEF-1 – API invoker)

The information flow service API invocation response from AEF-1 (AEF acting as service communication entry point) to the API invoker is service API specific and the complete detail of the service API invocation response is out of scope of the present document. Table 8.17.2.2-1 describes the CAPIF related information elements which are included in the service API invocation response.

8.13.3 Procedure

Figure 8.13.3-1 illustrates the procedure for CAPIF topology hiding.

Pre-conditions:

1. The API invoker has performed the service discovery and received the details of the service API which includes the information about the service communication entry point of the AEF-1 in the CAPIF.
2. The API invoker is authenticated and authorized to use the service API.
3. The AEF-1 in the CAPIF is configured with a policy for topology hiding including the entry point address of the service API (provided via AEF-2).

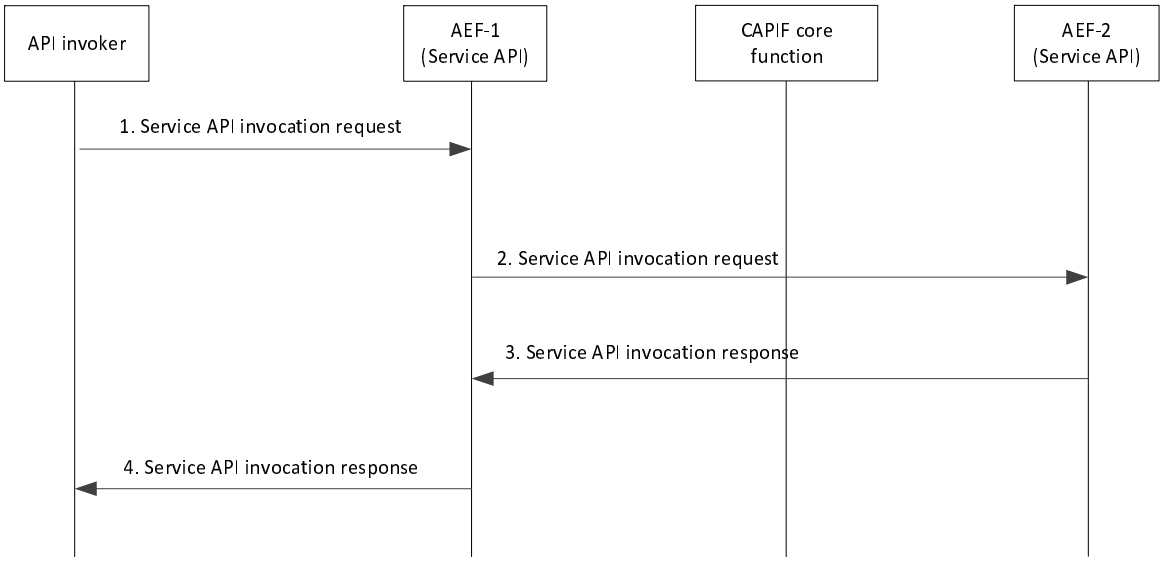


Figure 8.13.3-1: Procedure for CAPIF topology hiding

1. The API invoker performs service API invocation according to the interface of the service API by sending a service API invocation request towards the AEF-1 which exposes the service API towards the API invoker, and acts as topology hiding entity.

NOTE: Steps 2 and 3 are not necessary when the AEF-1 is capable to serve the service API invocation request.

2. The AEF-1 further resolves the actual destination service API address information according to the topology hiding policy and forwards the incoming service API invocation request to the service API of the related AEF-2.
3. The AEF-1 receives a response request for service API invocation from service API provided by AEF-2.
4. The AEF-1 resolves the destination API invoker address and also modifies the source address information of the AEF-2 within the response request as per topology hiding policy and forwards the response request to the API invoker.

8.14 Authentication between the API invoker and the AEF prior to service API invocation

8.14.1 General

The procedure in this subclause corresponds to the architectural requirements for authentication of the API invoker by the AEF.

To reduce latency during API invocation, the API invoker associated authentication information can be made available at the AEF after authentication between the API invoker and the CAPIF core function.

8.14.2 Information flows

NOTE: The security aspects of this procedure are specified in subclause 6.4 and subclause 6.5.2 of 3GPP TS 33.122 [12].

8.14.3 Procedure

Figure 8.14.3-1 illustrates the procedure for authentication between the API invoker and the AEF.

Pre-conditions:

1. Optionally, the CAPIF core function has shared the information required for authentication of the API invoker with the AEF.

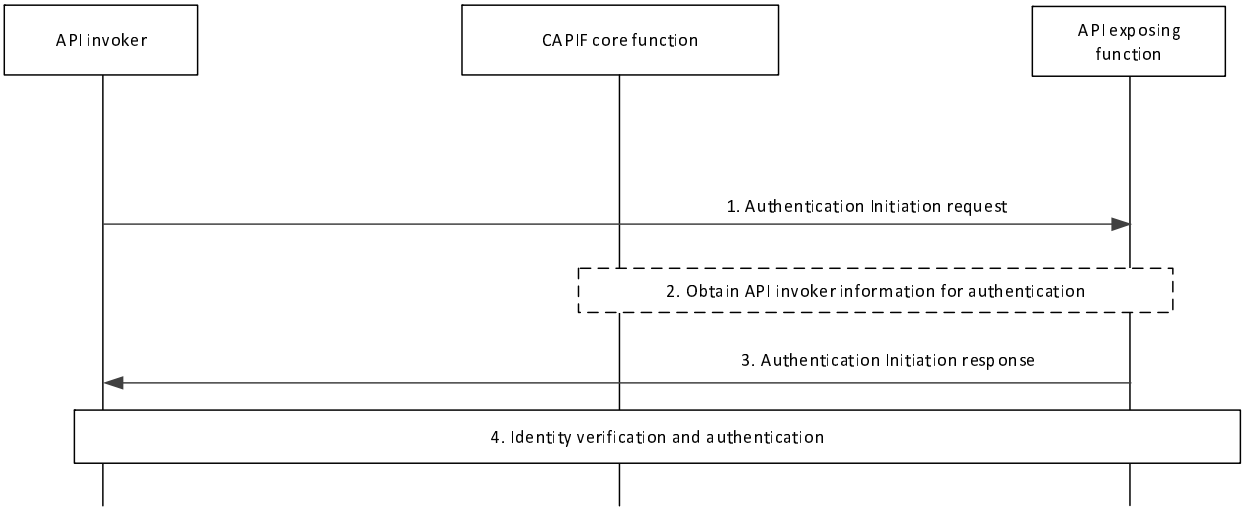


Figure 8.14.3-1: Procedure for authentication between the API invoker and the AEF prior to service API invocation

1. The API invoker triggers authentication initiation to the AEF, including the API invoker identity.
2. The AEF obtains the API invoker information required for authentication by the AEF, if not available.
3. The AEF returns the result of authentication initiation in the authentication initiation response.
4. The AEF verifies the identity of the API invoker and authenticates the API invoker.

NOTE 1: The authentication process is specified in subclause 6.4 and subclause 6.5.2 of 3GPP TS 33.122 [12].

NOTE 2: The authentication is terminated at the AEF acting as the service communication entry point when topology hiding is enabled for the service API.

8.15 Authentication between the API invoker and the AEF upon the service API invocation

8.15.1 General

The procedure in this subclause corresponds to the architectural requirements for authentication of the API invoker by the AEF upon the service API invocation.

To reduce latency during API invocation, the API invoker associated authentication information can be made available at the AEF after authentication between the API invoker and the CAPIF core function.

8.15.2 Information flows

NOTE: The security aspects of this procedure are specified in subclause 6.4 and subclause 6.5.2 of 3GPP TS 33.122 [12].

8.15.2.1 Service API invocation request with authentication information

The information flow service API invocation request with authentication information from the API invoker to the AEF is service API specific and the complete detail of the service API invocation request is out of scope of the present document. Table 8.15.2.1-1 describes only the CAPIF related information elements which are included in the service API invocation request.

Table 8.15.2.1-1: Service API invocation request with authentication information

Information element	Status	Description
API invoker identity information	M	The information that determines the identity of the API invoker
Authentication information	M (see NOTE)	The authentication information obtained before initiating the service API invocation request
Service API identification	M	The identification information of the service API for which invocation is requested. The service API identification is part of the specific service API invocation request.
NOTE: The specific security information of this information element is specified in subclause 6.5.2 of 3GPP TS 33.122 [12].		

8.15.2.2 Service API invocation response

The information flow service API invocation response from the AEF to the API invoker is service API specific and the complete detail of the service API invocation response is out of scope of the present document. Table 8.15.2.2-1 describes only the CAPIF related information elements which are included in the service API invocation response.

Table 8.15.2.2-1: Service API invocation response

Information element	Status	Description
Result	M	Indicates the success or failure of service API invocation.
Cause	O	The cause for the request failure.

8.15.3 Procedure

Figure 8.15.3-1 illustrates the procedure for authentication of the API invoker by the AEF, where the authentication information is carried in the API invocation request.

Pre-conditions:

1. Optionally, the CAPIF core function has shared the information required for authentication of the API invoker with the AEF.

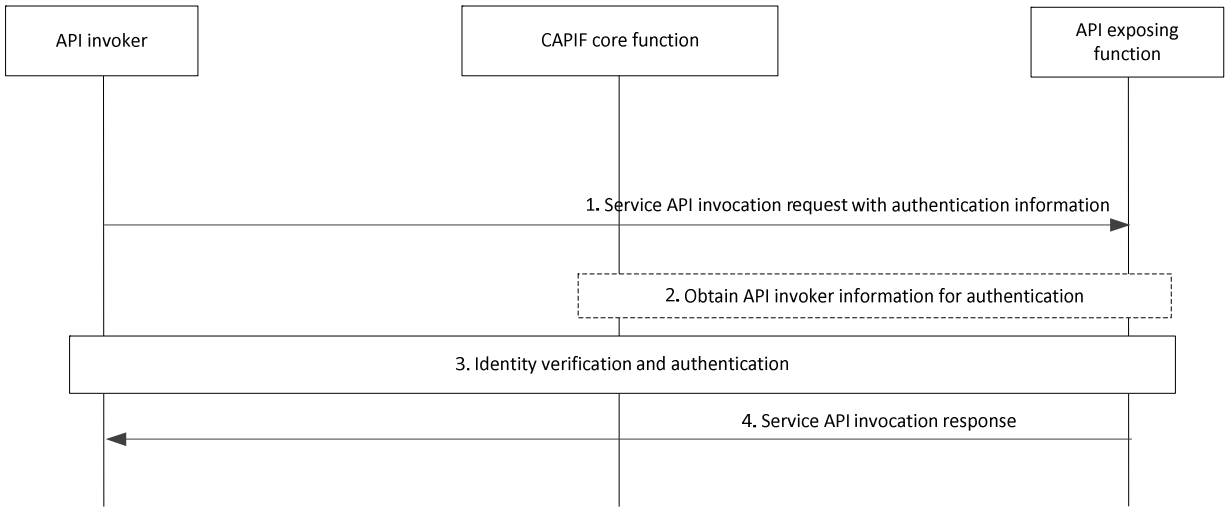


Figure 8.15.3-1: Procedure for authentication between the API invoker and the AEF upon the service API invocation

1. The API invoker invokes a service API invocation request with authentication information to the AEF, and includes in this request authentication information, including the API invoker identity.
2. The AEF obtains the API invoker information required for authentication by the AEF, if not available.
3. The AEF verifies the identity of the API invoker and authenticates the API invoker.

NOTE 1: The authentication process is specified in subclause 6.5.2 of 3GPP TS 33.122 [12].

4. If the verification was successful, the AEF returns the result of the service API invocation in the Service API invocation response.

NOTE 2: The authentication is terminated at the AEF acting as the service communication entry point when topology hiding is enabled for the service API.

8.16 Service API invocation with AEF authorization

8.16.1 General

The procedure in this subclause corresponds to the architectural requirements to validate authorization of API invokers upon the service API invocation.

To reduce latency during API invocation, the API invoker associated authorization information can be made available at the AEF after authentication between the API invoker and the CAPIF core function.

NOTE: The security aspects of service API invocation are specified in TS 33.122 [12] clause 6.4 (CAPIF-2) and 6.5 (CAPIF-2e).

8.16.2 Information flows

8.16.2.1 Service API invocation request

The information flow service API invocation request from the API invoker to the AEF is service API specific and the complete detail of the service API invocation request is out of scope of the present document. Table 8.16.2.1-1 describes only the CAPIF related information elements which are included in the service API invocation request.

Table 8.16.2.1-1: Service API invocation request

Information element	Status	Description
API invoker identity information	M	The information that determines the identity of the API invoker
Authorization information	O (see NOTE)	The authorization information obtained before initiating the service API invocation request
Network Slice Info	O (see NOTE)	The desired network slice information of the service API
Service API identification	M	The identification information of the service API for which invocation is requested. The service API identification is part of the specific service API invocation request.
NOTE: The inclusion of this information element depends on the chosen solution for authorization.		

8.16.2.2 Service API invocation response

The information flow service API invocation response from the AEF to the API invoker is service API specific and the complete detail of the service API invocation response is out of scope of the present document. Table 8.16.2.2-1 describes only the CAPIF related information elements which are included in the service API invocation response.

Table 8.16.2.2-1: Service API invocation response

Information element	Status	Description
Result	M	Indicates the success or failure of service API invocation.
Authorization information with resource owner required	O	This information element may be present when the result is failure. Indicates that the required authorization information with resource owner is required.

8.16.2.3 Obtain API invoker information request

Table 8.16.2.3-1 describes the information flows to request security information of the API invoker using the Obtain API Invoker information request from the AEF to the CAPIF Core Function.

Table 8.16.2.3-1: Obtain API invoker information request

Information element	Status	Description
API invoker identity information	M	The information that determines the identity of the API invoker
Authentication indication	O (see NOTE)	Indicates whether the consumer requests API Invoker authentication information.
Authorization indication	O (see NOTE)	Indicates whether the consumer requests API Invoker authorization information.
NOTE: Either the authentication indication or the authorization indication shall be included.		

8.16.2.4 Obtain API invoker information response

Table 8.16.2.3-1 describes only the information flows returned from the CAPIF Core Function to the AEF with the security method(s) information for the API Invoker.

Table 8.16.2.4-1: Obtain API Invoker information response

Information element	Status	Description
Result	M	Indicates the success or failure of service API invocation.
Security Information	O	This information element shall be present when the result is success. It includes the security information as specified in 3GPP TS 33.122 [12].
Failure Cause	O	This information element may be present when the result is failure. Indicates that the required authorization information with resource owner is required.

8.16.3 Procedure

Figure 8.16.3-1 illustrates the procedure for API invoker authorization to access service APIs.

Pre-conditions:

1. The API invoker has been authenticated.
2. The API invoker associated authorization information is available at AEF.

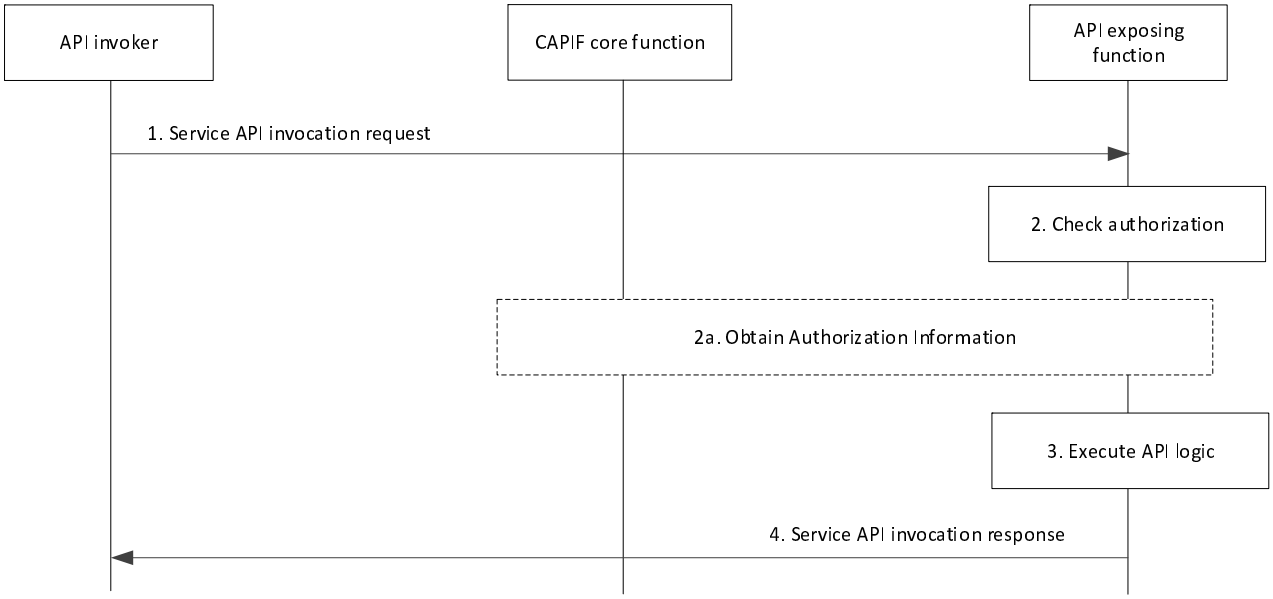


Figure 8.16.3-1: Procedure for API invoker authorization to access service APIs

1. The API invoker triggers service API invocation request to the AEF, including the service API to be invoked.

NOTE 1: Authentication can also be performed if not authenticated previously.

NOTE 2: The API invoker can trigger several service API invocations asynchronously.

2. Upon receiving the service API invocation request, the AEF checks whether the API invoker is authorized to invoke that service API, based on the authorization information.

2a. If the AEF does not have information required to authorize service API invocation, the AEF obtains the authorization information from the CAPIF core function as described in clauses 8.16.2.3 and 8.16.2.4.

3. The AEF executes the service logic for the invoked service API.

4. The API invoker receives the service API invocation response as a result of the service API invocation. If in step 2, the AEF determines that the required resource owner authorization information is not available for the requested service API, then the AEF responds to API Invoker with failure status message indicating that the authorization information with resource owner is not available, as specified in Table 8.16.2.2-1.

8.17 CAPIF access control

8.17.1 General

The CAPIF controls the access of service API by the API invoker based on policy or usage limits.

8.17.2 Information flows

8.17.2.1 Service API invocation request

The information flow service API invocation request from the API invoker to the AEF is service API specific and the complete detail of the service API invocation request is out of scope of the present document. Table 8.17.2.1-1 describes only the CAPIF related information elements which are included in the service API invocation request.

Table 8.17.2.1-1: Service API invocation request

Information element	Status	Description
API invoker identity information	M	The information that determines the identity of the API invoker
Authorization information	O (see NOTE)	The authorization information obtained before initiating the service API invocation request
Service API identification	M	The identification information of the service API for which invocation is requested. The service API identification is part of the specific service API invocation request.
NOTE: The inclusion of this information element depends on the chosen solution for authorization.		

8.17.2.2 Service API invocation response

The information flow service API invocation response from the AEF to the API invoker is service API specific and the complete detail of the service API invocation response is out of scope of the present document. Table 8.17.2.2-1 describes only the CAPIF related information elements which are included in the service API invocation response.

Table 8.17.2.2-1: Service API invocation response

Information element	Status	Description
Result	M	Indicates the success or failure of service API invocation.
Cause	O	The cause for the request failure.

8.17.3 Procedure

Figure 8.17.3-1 illustrates the procedure for service API access control.

Pre-conditions:

1. The API invoker has performed the service API discovery and received the details of the service API which includes the information about the service communication entry point of the AEF in the CAPIF.

2. The API invoker is authenticated and authorized to use the service API.

3. The AEF in the CAPIF is configured with at least one access policy to be applied to the service API invocation corresponding to the API invoker and service API.

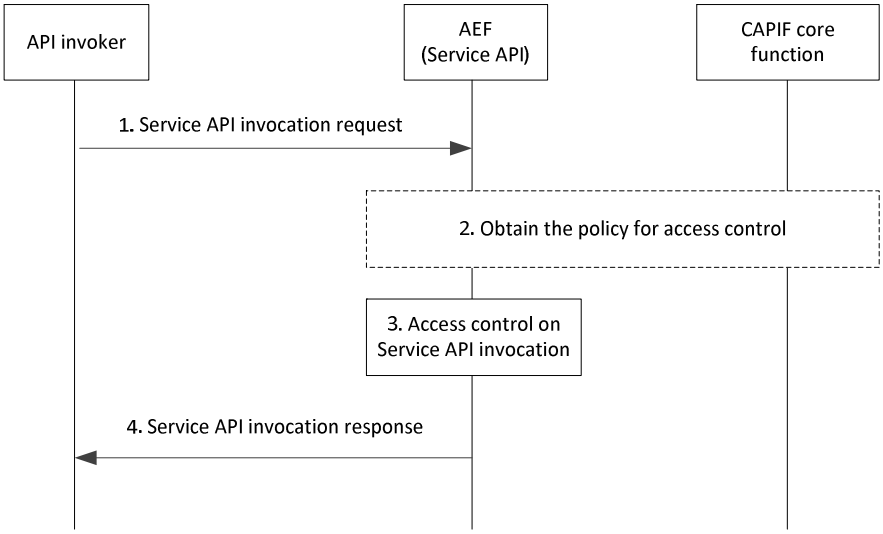


Figure 8.17.3-1: Procedure for service API access control

1. The API invoker performs service API invocation according to the interface of the service API by sending a service API invocation request towards the AEF which exposes the service API towards the API invoker. The AEF acts as an access control entity.
2. If the access control policy is not configured with AEF, then the AEF may obtain the access control policy configuration from the CAPIF core function as described in clause 8.12.3.
3. Upon receiving the service API invocation request from the API invoker, the AEF checks for configuration for access control. As per the configuration for access control, the AEF performs access control on the service API invocation request as per the operator policy.
4. The API invoker receives a service API invocation response for service API invocation from the AEF providing the service API.

8.18 CAPIF access control with cascaded AEFs

8.18.1 General

The procedure in this subclause corresponds to the architectural requirements related to some common access control requirements for service API invocations. It provides access control, based on two cascaded API Exposing Function (AEF) instances. While one AEF instance provides the entry point for the service API and acts as access controller, further AEF instances deliver the functionality of the actual service APIs.

8.18.2 Information flows

8.18.2.1 Service API invocation request

The information flow service API invocation request from the API invoker to the AEF and between AEFs is service API specific and the complete detail of the service API invocation request is out of scope of the present document. Table 8.17.2.1-1 describes the CAPIF related information elements which are included in the service API invocation request.

8.18.2.2 Service API invocation response

The information flow service API invocation response from the AEF to the API invoker and between AEFs is service API specific and the complete detail of the service API invocation response is out of scope of the present document.

Table 8.17.2.2-1 describes the CAPIF related information elements which are included in the service API invocation response.

8.18.3 Procedure

Figure 8.18.3-1 illustrates the procedure for CAPIF access control.

Pre-conditions:

1. The API invoker has performed the service discovery and received the details of the service API which includes the information about the service communication entry point of the AEF-1 in the CAPIF.
2. The API invoker is authenticated and authorized to use the service API.
3. The AEF-1 in the CAPIF is configured with at least one access policy to be applied to the service API invocation corresponding to the API invoker and service API.

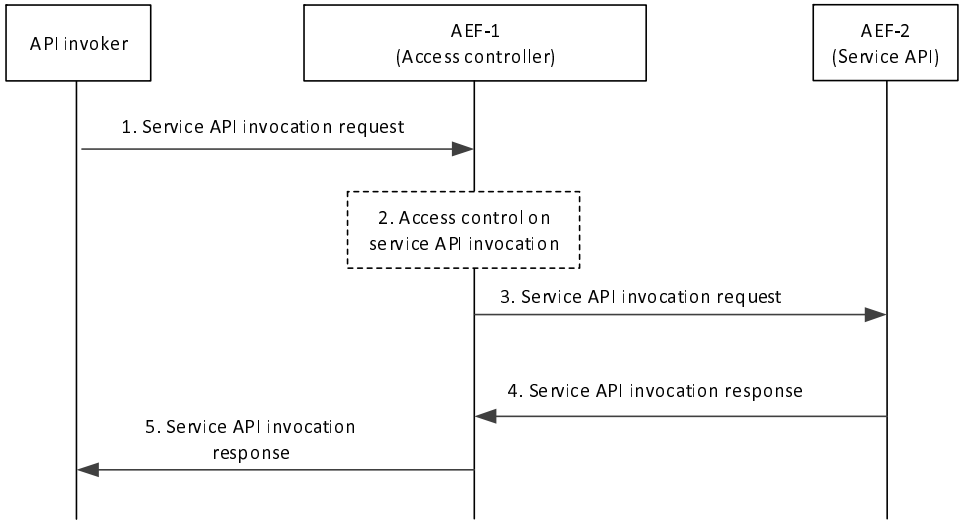


Figure 8.18.3-1: Procedure for CAPIF access control with cascaded AEFs

1. The API invoker performs service API invocation according to the interface of the service API by sending a service API invocation request towards the AEF-1 which exposes the service API towards the API invoker, and acts as access control entity.
2. Upon receiving the service API invocation request from the API invoker, the AEF-1 checks for configuration for access control as described in clause 8.12.3. As per the configuration for access control, the AEF-1 performs access control on the service API invocation as per the operator policy.
3. The AEF-1 forwards the incoming service API invocation request to the service API provided by AEF-2.
4. The AEF-1 receives a service API invocation response for service API invocation from AEF-2.
5. The AEF-1 resolves the destination API invoker address and modifies the source address information of AEF-2 within the service API invocation response and forwards the service API invocation response to the API invoker.

8.19 Logging service API invocations

8.19.1 General

The procedure in this subclause corresponds to the architectural requirements for logging service API invocations at AEF. The AEF can be within PLMN trust domain or within 3rd party trust domain.

8.19.2 Information flows

8.19.2.1 API invocation log request

Table 8.19.2.1-1 describes the information flow API invocation log request from the API exposing function to the CAPIF core function.

Table 8.19.2.1-1: API invocation log request

Information element	Status	Description
API exposing identity information	M	Identity information of the AEF logging service API(s) invocations
API invocation log information	M	API invocation log information such as API invoker's ID, IP address, service API name, version, invoked operation, input parameters, invocation result, time stamp information, Network Slice Info, invocation latency (the time interval between the reception of the API invocation request and the sending of the API invocation response at the AEF)

8.19.2.2 API invocation log response

Table 8.19.2.2-1 describes the information flow API invocation log response from the CAPIF core function to the API exposing function.

Table 8.19.2.2-1: API invocation log response

Information element	Status	Description
Result	M	Indicates the success or failure of API(s) invocation log request
Cause	O	The cause for the request failure.

8.19.3 Procedure

Figure 8.19.3-1 illustrates the procedure for logging service API invocations at AEF.

Pre-conditions:

1. The API invoker(s) has invoked certain service API(s).
2. Authorization details of the AEF are available with the CAPIF core function.

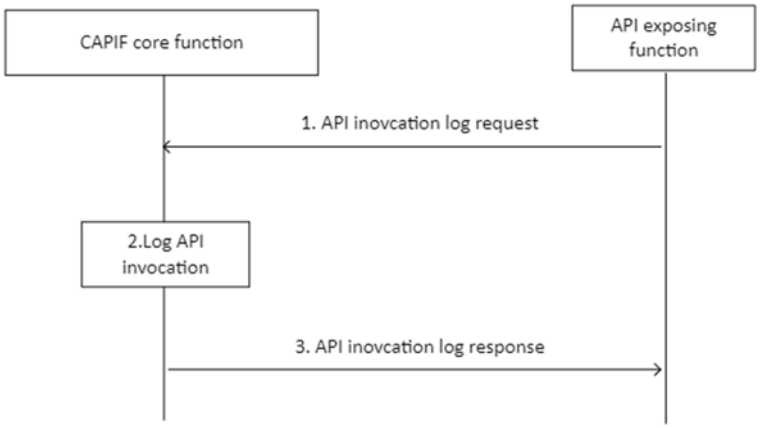


Figure 8.19.3-1: Procedure for logging service API invocations

1. Upon invocation of service API(s) from one more API invokers, the AEF triggers API invocation log request towards the CAPIF core function.

NOTE 1: The AEF can collect the log information associated to several API invocations before triggering API invocation log request asynchronously.

2. The CAPIF core function makes a log entry and stores the information e.g. for monitoring purposes, for access by authorized users and entities.

NOTE 2: API invocation log is stored for a configured duration.

3. AEF receives the API invocation log response from the CAPIF core function.

8.20 Charging the invocation of service APIs

8.20.1 General

The procedure in this subclause corresponds to the architectural requirements for charging the invocation of service APIs. The AEF can be within PLMN trust domain or within 3rd party trust domain. The charging of the invocation of the CAPIF core function APIs and the charging of the invocation of service APIs is specified in 3GPP TS 32.254 [20].

8.20.2 Information flows

NOTE: Charging aspects of CAPIF framework, i.e., charging architecture and information flows, are specified in 3GPP TS 32.254 [20].

8.20.3 Procedure

NOTE: Charging aspects of CAPIF framework, i.e., charging architecture, information flows and deployment options, are specified in 3GPP TS 32.254 [20].

8.21 Monitoring service API invocation

8.21.1 General

The procedure in this subclause corresponds to the architectural requirements for monitoring service API invocation.

8.21.2 Information flows

8.21.2.1 Monitoring service API event notification

The information flow for the monitoring service API event notification from the CAPIF core function to the API management function is same as the event notification from the CAPIF core function to the subscribing entity. Table 8.8.2.3-1 describes the information elements which are included in the monitoring service API event notification.

8.21.2.2 Monitoring service API event notification acknowledgement

The information flow for the monitoring service API event notification acknowledgement from the API management function to the CAPIF core function is same as the event notification acknowledgement from subscribing entity to the CAPIF core function. Table 8.8.2.4-1 describes the information elements which are included in the monitoring service API event notification acknowledgement.

8.21.3 Procedure

Figure 8.21.3-1 illustrates the procedure for monitoring service API invocation.

Pre-conditions:

1. The API management function has subscribed to monitoring event including filters such as invoker's ID and IP address, service API name and version, input parameters, and invocation result.

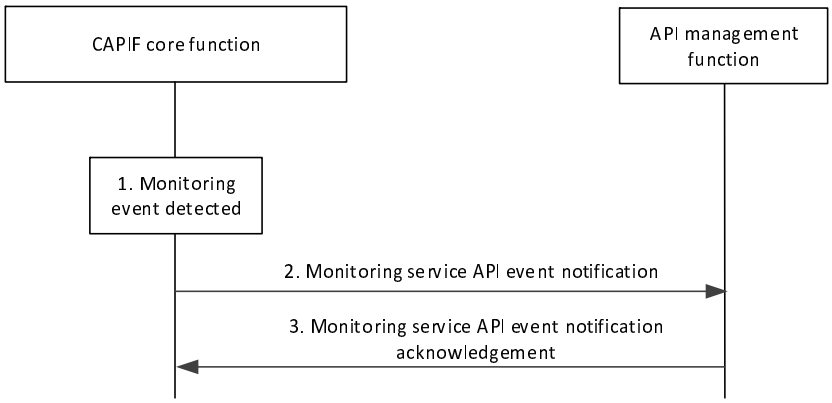


Figure 8.21.3-1: Procedure for monitoring service API invocation

1. The CAPIF core function monitors the service API invocations applying the monitoring filters specified before.
2. Detection of a monitoring event by the CAPIF core function triggers notification to the API management function with the details of the monitored event.

NOTE: API provider action subsequent to monitoring service API notification is out-of-scope of this specification.

3. The API management function sends a monitoring service API event notification acknowledgement to the CAPIF core function for the notification received.

8.22 Auditing service API invocation

8.22.1 General

The procedure in this subclause corresponds to the architectural requirements for auditing service API invocation. This procedure can be used for auditing of other CAPIF interactions i.e. service API invocation events, API invoker

onboarding events and API invoker interactions with the CAPIF (e.g. authentication, authorization, discover service APIs) as well. The API management function can be within PLMN trust domain or within 3rd party trust domain.

8.22.2 Information flows

8.22.2.1 Query service API log request

Table 8.22.2.1-1 describes the information flow query service API log request from the API management function to the CAPIF core function.

Table 8.22.2.1-1: Query service API log request

Information element	Status	Description
Identity information	M	Identity information of the entity querying service API log request
Query information	M	List of query filters such as invoker's ID and IP address, service API name and version, input parameters, and invocation result

8.22.2.2 Query service API log response

Table 8.22.2.2-1 describes the information flow query service API log response from the CAPIF core function to the API management function.

Table 8.22.2.2-1: Query service API log response

Information element	Status	Description
Result	M	Indicates the success or failure of query service API log request
API invocation log information	O (see NOTE)	API invocation log information such as API invoker's ID, IP address, service API name, version, invoked operation, input parameters, invocation result, time stamp information, Network Slice Info
NOTE: Information element shall be present when result indicates success.		

8.22.3 Procedure

Figure 8.22.3-1 illustrates the procedure for auditing service API invocation.

Pre-conditions:

1. Service API invocation logs are available at the CAPIF core function.
2. Authorization details of the AMF are available with the CAPIF core function.

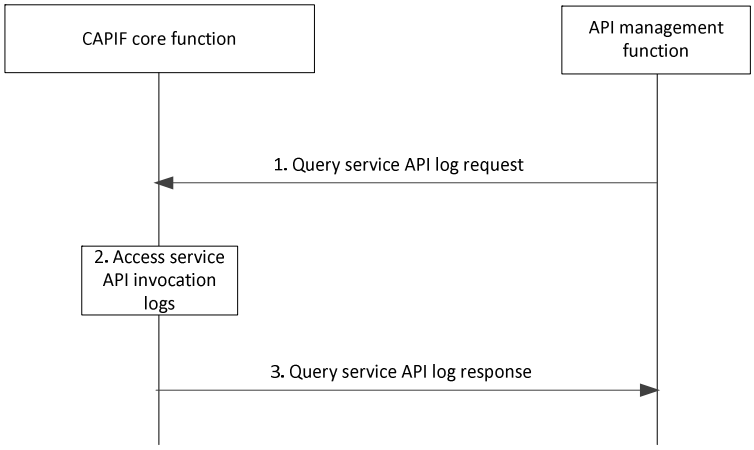


Figure 8.22.3-1: Procedure for auditing service API invocation

1. For auditing service API invocations, the API management function triggers query service API log request to the CAPIF core function.
2. Upon receiving the query service API log request, the CAPIF core function accesses the necessary service API log information for auditing purposes.
3. The CAPIF core function returns the log information to the API management function in the query service API log response.

NOTE: The API management function detecting abuse of the service API invocation and actions, subsequent to query service API log response, are out-of-scope of this specification.

8.23 CAPIF revoking API invoker authorization

8.23.1 General

The CAPIF controls the access of service API by the API invoker based on policy or usage limits. If the usage limits have exceeded, the authorization of the API invoker for accessing the service APIs is revoked. The decision to revoke the API invoker authorization may be triggered by the AEF or the CAPIF core function. The AEF can be within PLMN trust domain or within 3rd party trust domain.

In RNAA scenarios, the decision to revoke the API invoker authorization may be initiated by the CAPIF core function based on triggers at the CAPIF core function.

8.23.2 Information flows

8.23.2.1 Revoke API invoker authorization request

Table 8.23.2.1-1 describes the information flow revoke API invoker authorization request from the API exposing function to the CAPIF core function or from the CAPIF core function to the API exposing function.

Table 8.23.2.1-1: Revoke API invoker authorization request

Information element	Status	Description
API invoker identity information	M	The information that determines the identity of the API invoker
Service API identification list	M	The identification information of the service API(s) for which the authorization is revoked.
Security information	O (NOTE 1, 2)	Security information (as specified in 3GPP TS 33.122 [12]) for authorization revocation.
Cause	M	The cause for revoking the API invoker authorization (e.g., RO authorization revocation).
NOTE 1: The security information may be included in the Revoke API invoker authorization request from the CAPIF core function to the API exposing function (i.e., it is not used in the request from the API exposing function to the CAPIF core function). NOTE 2: If the security information is not included, the revocation applies to all the permissions granted for the service API.		

8.23.2.2 Revoke API invoker authorization response

Table 8.23.2.2-1 describes the information flow revoke API invoker authorization response from the CAPIF core function to the API exposing function or from the API exposing function to the CAPIF core function.

Table 8.23.2.2-1: Revoke API invoker authorization response

Information element	Status	Description
Result	M	Indicates the success or failure of revoke API invoker authorization.
Cause	O	The cause for the request failure.

8.23.2.3 Revoke API invoker authorization notify

Table 8.23.2.3-1 describes the information flow revoke API invoker authorization notify from the CAPIF core function to the API invoker.

Table 8.23.2.3-1: Revoke API invoker authorization notify

Information element	Status	Description
API invoker identity information	M	The information that determines the identity of the API invoker whose authorization has been revoked
Service API identification list	M	The identification information of the service API(s) for which the authorization is revoked.
Security information	O (NOTE)	Security information (as specified in 3GPP TS 33.122 [12]) for authorization revocation.
Cause	M	The cause for revoking the API invoker authorization (e.g., RO authorization revocation).
NOTE: If the security information is not included, the revocation applies to all the permissions granted for the service API(s).		

8.23.2.4 Revoke API invoker authorization notify acknowledgement

Table 8.23.2.4-1 describes the information flow revoke API invoker authorization notify acknowledgement from the API Invoker to the CAPIF core function.

Table 8.23.2.4-1: Revoke API invoker authorization notify acknowledgement

Information element	Status	Description
Acknowledgement	M	Acknowledgement for the revoke API invoker authorization notify received.

8.23.3 Procedure for CAPIF revoking API invoker authorization initiated by AEF

Figure 8.23.3-1 illustrates the procedure for revoking API invoker authorization to access service API initiated by the AEF.

Pre-conditions:

1. The API invoker is authenticated and authorized to use the service API.
2. The AEF in the CAPIF is configured with the access policy to be applied to the service API invocation corresponding to the API invoker and the service API.
3. Authorization details of the AEF are available with the CAPIF core function.

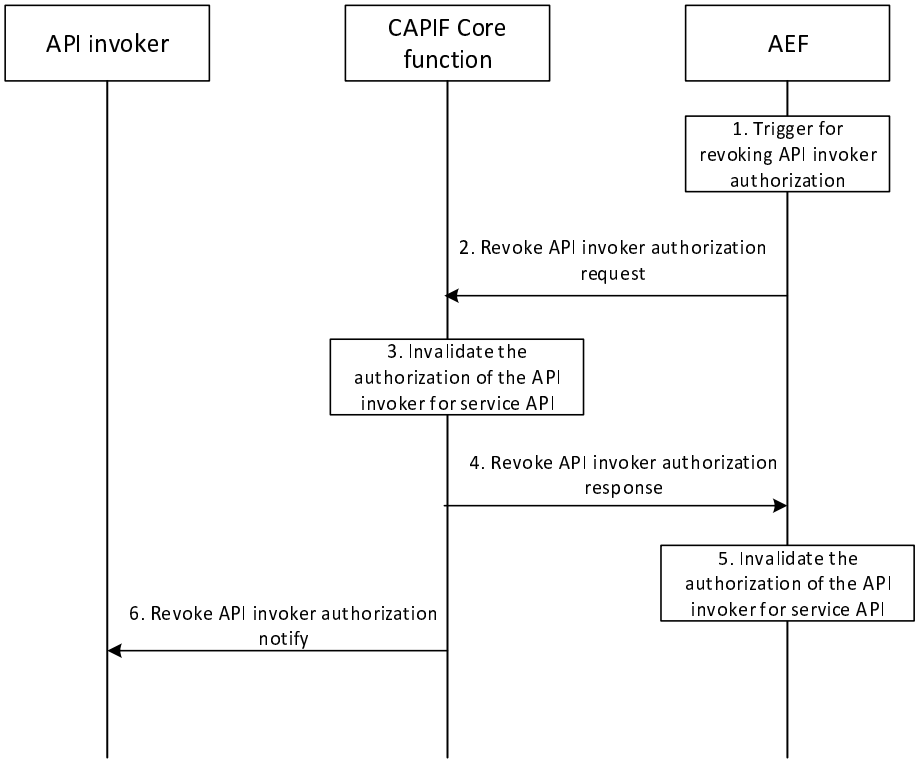


Figure 8.23.3-1: Procedure for revoking API invoker authorization initiated by AEF

1. The AEF triggers the revocation of the API invoker authorization.
2. The AEF sends revoke API invoker authorization request to the CAPIF core function with the details of the API invoker and the service API.
3. Upon receiving the information to revoke the API invoker's authorization for service API invocation, the CAPIF core function invalidates the API invoker authorization corresponding to the service API.
4. The CAPIF core function sends a revoke API invoker authorization response to the AEF.
5. Upon successful revocation of API invoker authorization corresponding to the service API at the CAPIF core function, the AEF invalidates the API invoker authorization corresponding to the service API.

6. The CAPIF core function sends a revoke API invoker authorization notify to the API invoker whose authorization to access the service API has been revoked.

8.23.4 Procedure for CAPIF revoking API invoker authorization initiated by CAPIF core function

Figure 8.23.4-1 illustrates the procedure for revoking API invoker authorization to access service API initiated by the CAPIF core function. This procedure is also used for revoking API invoker authorization supporting RNAA scenarios.

Pre-conditions:

1. The API invoker is authenticated and authorized to use the service API.
2. The AEF in the CAPIF is configured with the access policy to be applied to the service API invocation corresponding to the API invoker and the service API.

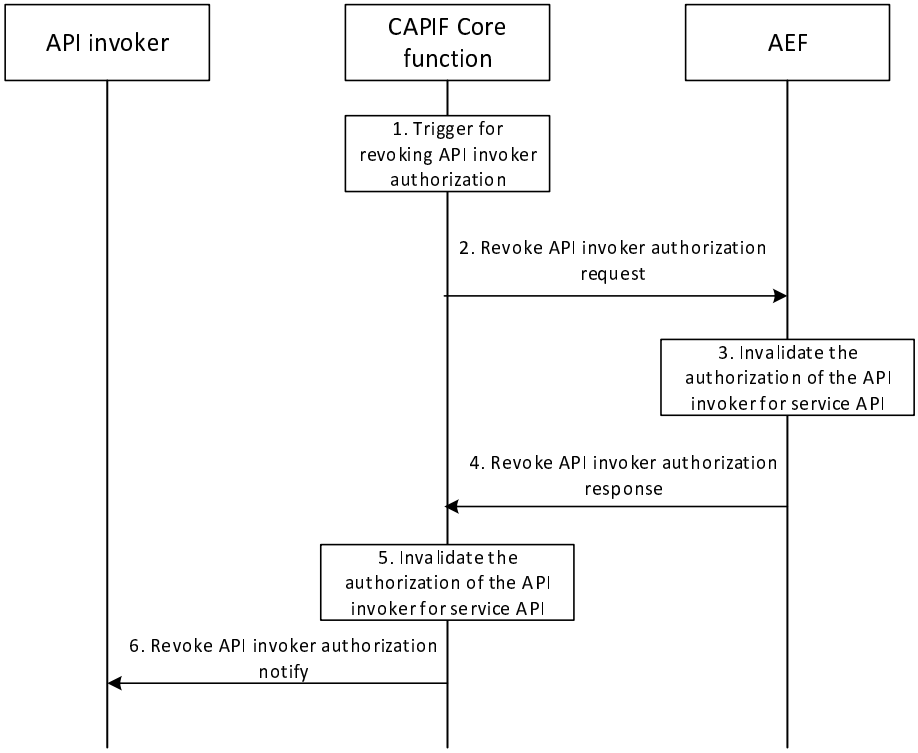


Figure 8.23.4-1: Procedure for revoking API invoker authorization initiated by CAPIF core function

1. The CAPIF core function is triggered to revoke the API invoker authorization.
2. The CAPIF core function sends revoke API invoker authorization request to the AEF with the details of the API invoker and the service API, and optionally, the security information as specified in clause 6.5.3.4 of 3GPP TS 33.122 [12].
3. Upon receiving the information to revoke the API invoker's authorization for service API invocation, the AEF invalidates the API invoker authorization corresponding to the service API or, if provided, the API invoker authorization corresponding to the received security information.
4. The AEF sends a revoke API invoker authorization response to the CAPIF core function.
5. The CAPIF core function invalidates the API invoker authorization corresponding to the service API or, if provided, to the received security information.
6. The CAPIF core function sends a revoke API invoker authorization notify to the API invoker whose authorization to access the service API has been revoked.

8.24 API topology hiding management

8.24.1 General

The following procedure in this subclause corresponds to the architectural requirements on API topology hiding. The procedure in this subclause supports API topology hiding by dynamically configuring the address of the AEF providing the Service API to the AEF entry point providing the topology hiding. The API publishing function and the API exposing function may be in different API Provider domains, e.g. within PLMN trust domain and within 3rd party trust domain, or may be the same API Provider domain, e.g., in the PLMN trust domain or 3rd party trust domain.

8.24.2 Information flows

8.24.2.1 API topology hiding notify

Table 8.24.2.1-1 describes the information flow API topology hiding notify from the CAPIF core function to the API exposing function.

Table 8.24.2.1-1: API topology hiding notify

Information element	Status	Description
Service API identification	M	The identification information of the service API with the API topology hiding
API exposing function(s) information	M	Indicates the one or more AEF(s) which provides the service API to apply the topology hiding including the interface details (e.g. IP address, port number, URI).
Action	M	Indicates the notification action for the API topology hiding (created or revoked).

8.24.3 Procedure

Figure 8.24.3-1 illustrates the procedure for API topology hiding management by API (un)publish function.

Pre-condition:

1. Authorization details of the APF are available with the CAPIF core function.
2. The API exposing function has subscribed to CAPIF event for API topology hiding status.

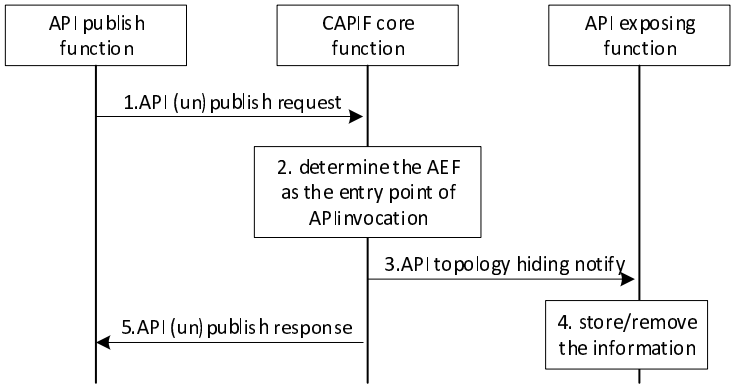


Figure 8.24.3-1: API topology hiding via API (un)publish

1. The API publishing function sends a service API publish request as described in subclause 8.3.2.1 or a service API unpublish request as described in subclause 8.4.2.1 to the CAPIF core function.

2. Upon receiving the service API (un)publish request, the CAPIF core function checks whether the API publishing function is authorized to perform the service API (un)publish. If authorized, based on the service APIs and policy:

   - For service API publish, the CCF applies the topology hiding by selecting an AEF providing the topology hiding as the entry point for service API invocation. The selected AEF information is stored with the service API information received from API publish function at the CAPIF core function (API registry).

   - For service API unpublish, the previously selected AEF as topology hiding entry point and the associated service API information at the CAPIF core function (API registry) are removed.

3. The CCF sends the API topology notify to the AEF selected as the entry point for service API invocation. The service API identification and the AEF(s) information which provides the service API details are included.

4. Upon receiving the notification, the AEF stores the received information for further service API invocation request forwarding if the action in the API topology notify indicates "created" or removes the stored API forwarding information if the action in the API topology notify indicates "revoked".

5. The CCF sends an API (un)publish response to the API publish function.

8.25 Support for CAPIF interconnection

8.25.1 General

The procedures in this subclause corresponds to the architectural requirements on CAPIF interconnection.

8.25.2 Information flows

8.25.2.1 Interconnection API publish request

Table 8.25.2.1-1 describes the information flow interconnection API publish request from CAPIF core function to CAPIF core function.

Table 8.25.2.1-1: Interconnection API publish request

Information element	Status	Description
CCF information	M	The information of the CAPIF core function which publishes APIs, may include identity, authentication and authorization information
Service API information	O (see NOTE 1)	The service API information as specified in Table 8.3.2.1-1.
Service API category	O	Service API category.
Shareable information	O (see NOTE 2)	Indicates whether the service API information or the service API category can be published to other CCFs. And if sharing, a list of CAPIF provider domain information where the service API information or the service API category can be published is contained.
NOTE 1: Either the Service API information or Service API category shall be present. NOTE 2: If the shareable information is not present, the service API information is not allowed to be shared. There is one and only one CAPIF provider domain information sharable via the CAPIF-6e interface.		

8.25.2.2 Interconnection API publish response

Table 8.25.2.2-1 describes the information flow interconnection API publish response from CAPIF core function to CAPIF core function.

Table 8.25.2.2-1: Interconnection API publish response

Information element	Status	Description
Result	M	Indicates the success or failure of publishing the service API information
Cause	O	The cause for the request failure.
Service API published information reference	O (see NOTE)	The information which can be used for referencing the information (set) about the published service API by the CCF which publishes service APIs
NOTE: This information element is included when the Result indicates success.		

8.25.2.3 Interconnection service API discover request

Table 8.25.2.3-1 describes the information flow interconnection service API discover request from one CAPIF core function to another CAPIF core function.

Table 8.25.2.3-1: Interconnection service API discover request

Information element	Status	Description
CAPIF core function identity information	M	Identity information of the CAPIF core function discovering service APIs
Query information	M	Criteria for discovering matching service APIs or CAPIF core function (e.g. service API category, Serving Area Information (optional), UE IP address (optional), preferred AEF location (optional), required API provider name (optional), interfaces, protocols, Service KPIs (optional), and Network Slice Info (optional)) (see NOTE)
NOTE: It should be possible to discover all the service APIs.		

8.25.2.4 Interconnection service API discover response

Table 8.25.2.4-1 describes the information flow interconnection service API discover response from one CAPIF core function to another CAPIF core function.

Table 8.25.2.4-1: Interconnection service API discover response

Information element	Status	Description
Result	M	Indicates the success or failure of the discovery of the service API information
Cause	O	The cause for the request failure.
Service API information	O (see NOTE)	List of service APIs corresponding to the request, including service API information as specified in Table 8.7.2.2-1.
CAPIF core function identity information	O (see NOTE)	Indicates the CAPIF core function matching the query criteria.
NOTE: The service API information or the CAPIF core function identity information or both shall be present, if the Result information element indicates that the interconnection service API discover operation is successful. Otherwise, both shall not be present.		

8.25.2.5 Interconnection API unpublish request

Table 8.25.2.5-1 describes the information flow interconnection API unpublish request from a CAPIF core function to another CAPIF core function.

Table 8.25.2.5-1: Interconnection API unpublish request

Information element	Status	Description
CCF information	M	The information of the CAPIF core function which unpublishes service APIs, may include identity, authentication and authorization information
Service API published information reference (see NOTE)	M	The information which can be used for referencing the information (set) about the published service API by the CCF which publishes service APIs
NOTE: Obtained during the interconnection API publish procedure in clause 8.25.3.1.		

8.25.2.6 Interconnection API unpublish response

Table 8.25.2.6-1 describes the information flow interconnection API unpublish response from a CAPIF core function to another CAPIF core function.

Table 8.25.2.6-1: Interconnection API unpublish response

Information element	Status	Description
Result	M	Indicates the success or failure of unpublishing the service API information
Cause	O	The cause for the request failure.

8.25.2.7 Interconnection get service API request

Table 8.25.2.7-1 describes the information flow interconnection get service API request from a CAPIF core function to another CAPIF core function.

Table 8.25.2.7-1: Interconnection get service API request

Information element	Status	Description
CCF information	M	The information of the CAPIF core function that wants to retrieve service APIs information. It may include identity, authentication and authorization information
Service API published information reference (see NOTE)	M	The information which can be used for referencing the information (set) about the published service API by the CCF which publishes service APIs
NOTE: Obtained during the interconnection API publish procedure in clause 8.25.3.1.		

8.25.2.8 Interconnection get service API response

Table 8.25.2.8-1 describes the information flow interconnection get service API response from a CAPIF core function to another CAPIF core function.

Table 8.25.2.8-1: Interconnection get service API response

Information element	Status	Description
Result	M	Indicates the success or failure of getting the service API information.
Cause	O	The cause for the request failure.
Service API information	O (see NOTE)	The service API information as specified in Table 8.3.2.1-1.
NOTE: Shall be present if the Result information element indicates that the interconnection get service API request is successful. Otherwise, service API information shall not be present.		

8.25.2.9 Interconnection update service API request

Table 8.25.2.9-1 describes the information flow interconnection update service API request from a CAPIF core function to another CAPIF core function.

Table 8.25.2.9-1: Interconnection update service API request

Information element	Status	Description
CCF information	M	The information of the CCF requesting the update operation. It may include identity, authentication and authorization information.
Service API information	O (see NOTE 1)	The service API information as specified in Table 8.3.2.1-1.
Service API category	O (see NOTE 1)	The category of the service APIs to be published, (e.g. V2X, IoT).
Shareable information	O (see NOTE 2)	Indicates whether the service API information or the service API category can be published to other CCFs. And if sharing, a list of CAPIF provider domain information where the service API information or the service API category can be published is contained.
NOTE 1: Either the Service API information or Service API category shall be present. NOTE 2: If the shareable information is not present, the service API information is not allowed to be shared. There is one and only one CAPIF provider domain information sharable via the CAPIF-6e interface.		

8.25.2.10 Interconnection update service API response

Table 8.25.2.10-1 describes the information flow interconnection update service API response from a CAPIF core function to another CAPIF core function.

Table 8.25.2.10-1: Interconnection update service API response

Information element	Status	Description
Result	M	Indicates the success or failure of updating the service API information
Cause	O	The cause for the request failure.
Service API information	O	The authorized service API information during update, applicable when the update result is success. This can be a subset or the full set, of the Service API information as specified in Table 8.3.2.1-1.

8.25.2.11 Interconnection revoke API invoker authorization request

Table 8.25.2.11-1 describes the information flow interconnection revoke API invoker authorization request from one CCF to another CCF.

Table 8.25.2.11-1: Interconnection revoke API invoker authorization request

Information element	Status	Description
CCF information	M	The information of the requesting CAPIF core function, may include identity, authentication and authorization information.
API invoker related information	M	Information related to the API invoker, e.g. information that determines the identity of the API invoker.
Service API identification list	M	The identification information of the service API(s) for which the authorization is revoked.
Security information	O (NOTE 1,2)	Security information (as specified in 3GPP TS 33.122 [12]) for authorization revocation.
Cause	M	The cause for revoking the API invoker authorization
NOTE 1: The security information may be included in the Interconnect Revoke API invoker authorization request from the CAPIF core function to which the API invoker is onboarded to the CAPIF Core Function where the API exposing function is registered (i.e., it is not used in the request from the CAPIF Core Function where the API exposing function is registered). NOTE 2: If the security information is not included, the revocation applies to all the permissions granted for the service API.		

8.25.2.12 Interconnection revoke API invoker authorization response

Table 8.25.2.12-1 describes the information flow interconnection revoke API invoker authorization response from one CCF to another CCF.

Table 8.25.2.12-1: Interconnection revoke API invoker authorization response

Information element	Status	Description
Result	M	Indicates the success or failure of interconnection revoke API invoker authorization.
Cause	O	The cause for the request failure.

8.25.2.13 Interconnection revoke API invoker authorization notify

Table 8.25.2.13-1 describes the information flow interconnection revoke API invoker authorization notify from one CCF to another CCF.

Table 8.25.2.13-1: Interconnection revoke API invoker authorization notify

Information element	Status	Description
CCF information	M	The information of the notifying CAPIF core function, may include identity, authentication and authorization information.
API invoker related information	M	Information related to the API invoker, e.g. information that determines the identity of the API invoker.
Service API identification list	M	The identification information of the service API(s) for which the authorization is revoked.
Security information	O (NOTE)	Security information (as specified in 3GPP TS 33.122 [12]) for authorization revocation.
Cause	M	The cause for revoking the API invoker authorization
NOTE: If the security information is not included, the revocation applies to all the permissions granted for the service API.		

8.25.2.14 Interconnection obtain access control policy request

Table 8.25.2.14-1 describes the information flow interconnection obtain access control policy request from one CCF to another CCF.

Table 8.25.2.14-1: Interconnection obtain access control policy request

Information element	Status	Description
Identity information	M	Identity information of the entity requesting the access control policy
Service API identification	M	The identification information of the service API for which the access control policy is being requested.

8.25.2.15 Interconnection obtain access control policy response

Table 8.25.2.15-1 describes the information flow interconnection obtain access control policy response from one CCF to another CCF.

Table 8.25.2.15-1: Interconnection obtain access control policy response

Information element	Status	Description
Result	M	Indicates the success or failure of the obtain access control policy operation
Cause	O	The cause for the request failure.
Access control policy information	O (see NOTE)	The access control policy information corresponding to the requested service API. (See Table E-1)
NOTE: Shall be present if the Result information element indicates that the obtain access control policy operation is successful. Otherwise access control policy information shall not be present.		

8.25.2.16 Interconnection API invoker obtaining authorization to access service API

NOTE: The security aspects of this procedure are specified in clause 6.13 of 3GPP TS 33.122 [12].

8.25.2.17 Interconnection Authentication and authorization between the API invoker and the AEF

NOTE: The security aspects of this procedure are specified in clause 6.13 of 3GPP TS 33.122 [12].

8.25.3 Procedure

8.25.3.1 Service API publish for CAPIF interconnection

This subclause describes the procedure for service API publish for CAPIF interoperation.

Pre-condition:

1. CCF-A and CCF-B connect to each other, and either belong to the single trust domain of the same CAPIF provider or trust domains of different CAPIF providers.
2. CCF-B is configured as the designated CAPIF core function in the trust domain of CAPIF provider A.
3. When CCF-A and CCF-B belong to trust domains of different CAPIF providers, the two CAPIF providers have business agreement for service API sharing.

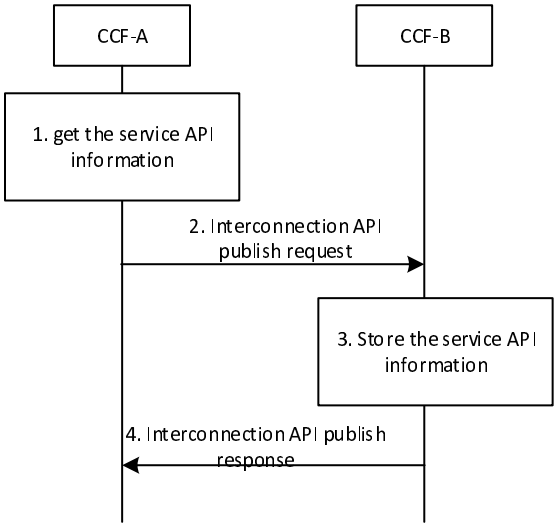


Figure 8.25.3.1-1: Interconnection API publish

1. CCF-A gets the service APIs to be shared with CCF-B from the API publish function which is in the same CAPIF provider domain of CCF-A as described in subclause 8.3.3, or from another CCF as described in this procedure.

2. Based on the shareable information for the service API or the service API category information, the CCF-A determines to publish the service API or the service API category information to the CCF-B. The CCF-A sends the interconnection API publish request to CCF-B with the details of at least one of service APIs or the category information of the service APIs, along with the identity information of CCF-A, shareable information and CAPIF provider domain information if allowed to share. The API topology hiding may be enabled.

3. CCF-B stores the service API information or service API category provided by the CCF-A.

4. CCF-B provides an interconnection API publish response to the CCF-A indicating success or failure result and triggers notifications to subscribed API invokers as described in subclause 8.8.4.

8.25.3.2 Service API discovery involving multiple CCFs

This subclause describes a procedure for service API discovery involving multiple CCFs

Pre-condition:

1. CCF-A and CCF-B connect to each other, and either belong to the single trust domain of the same CAPIF provider or trust domains of different CAPIF providers.

2. When CCF-A and CCF-B belong to trust domains of different CAPIF providers, the two CAPIF providers have business agreement for service API sharing.

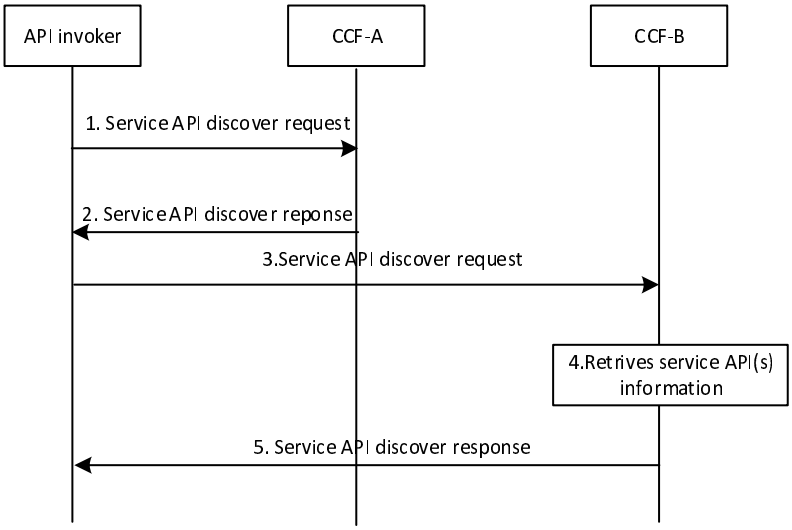


Figure 8.25.3.2-1: Service API discovery involving multiple CCFs

1. The API invoker sends a service API discover request to the CCF-A. It includes the API invoker identity and query information.

2. The CCF-A verifies the identity of the API invoker and retrieves the stored service API(s) information. The information of CCF-B matching the discovery criteria is returned to API invoker in the service API discover response.

NOTE: The remaining steps are only applied when the service API category without detail service API information is included in the interconnection API publish request as described in subclause 8.25.2.1.

3. The API invoker sends an service API discover request to the CCF-B. The identity of API invoker is included. The query information is also provided.

4. Upon receiving the service API discover request, the CCF-B verifies the identity of the API invoker. The CCF-B retrieves the stored service API(s) information as per the query information in the service API discover request. Further, the CCF-B applies the discovery policy and performs filtering of service APIs information which matches the discovery criteria.

5. The CCF-B sends an service API discover response to the API invoker with the list of service API information for which the API invoker has the required authorization.

8.25.3.3 Service API discovery for CAPIF interconnection

This subclause describes a procedure for service API discovery for CAPIF interconnection. The CCF-A and the CCF-B may belong to the same CAPIF provider domain or different CAPIF provider domains. When the CCF-A and the CCF-B belong to different CAPIF provider domains, the two CAPIF providers shall have business agreement for service API discovery.

Pre-conditions:

1. The CCF-A is configured with the CCF-B information.

2. The CCF-B is configured with the CCF-A information.

3. The CCF-A is triggered (e.g. API invoker service API discovery, periodic service API discovery) to perform service API discovery with the CCF-B.

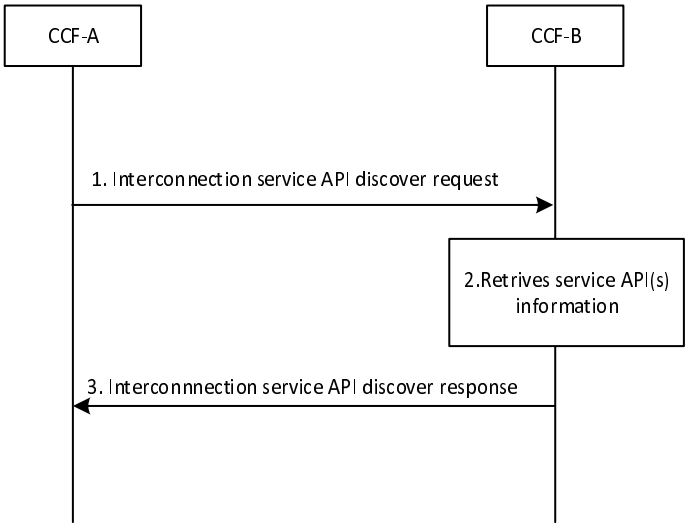


Figure 8.25.3.3-1: Service API discovery for CAPIF interconnection

1. The CCF-A sends the interconnection service API discover request to the CCF-B. The identity of the CCF-A and the query information are included.
2. The CCF-B upon receiving the interconnection service API discover request verifies the identity of the CCF-A. The CCF-B retrieves the stored service API(s) or the CCF(s) information as per the query information in the interconnection service API discover request. Further, the CCF-B applies the discovery policy and performs the filtering of service APIs or the CCF(s) information. The topology hiding policy may be applied to the retrieved list of service API information.
3. The CCF-B sends the interconnection service API discover response to the CCF-A with the list of service API information for which the CCF-A has the required authorization or the CCF(s) information that matches the discovery criteria.

8.25.3.4 Service API unpublish for CAPIF interconnection

This clause describes the procedure for service API unpublish for CAPIF interconnection.

Pre-condition:

1. CCF-A and CCF-B connect to each other, and either belong to the single trust domain of the same CAPIF provider or trust domains of different CAPIF providers. When CCF-A and CCF-B belong to trust domains of different CAPIF providers, the two CAPIF providers have business agreement for service API sharing.
2. CCF-B is configured as the designated CAPIF core function in the trust domain of CAPIF provider A.
3. CCF-A has successfully published service APIs to the designated CCF (i.e., CCF-B) using the procedure in clause 8.25.3.1.

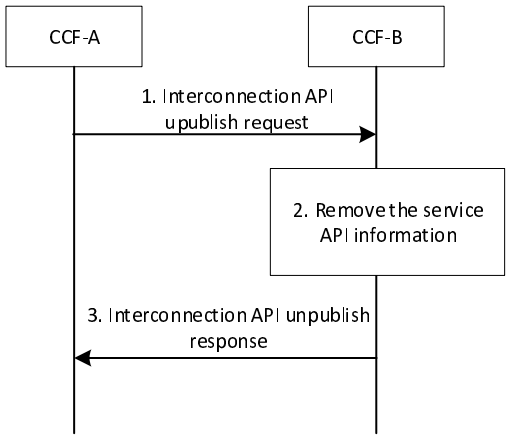


Figure 8.25.3.4-1: Interconnection API unpublish

1. CCF-A sends an interconnection API unpublish request to CCF-B. The request includes a service API published information reference. If the shareable information IE was present in the interconnection API publish request (see clause 8.25.2.1), the API unpublish request should be forwarded to the provided list of CAPIF provider domain information during the publish procedure.

2. Upon receiving the interconnection API unpublish request, the CCF-B checks whether the CCF-A is authorized to unpublish service APIs. If the check is successful, CCF-B will remove the requested service API information from the API registry.

3. CCF-B sends an interconnection API unpublish response to CCF-A containing the operation result and triggers notifications to subscribed API invokers (if any) as described in subclause 8.8.4.

8.25.3.5 Retrieve service APIs for CAPIF interconnection

This clause describes the procedure to retrieve service APIs information in a CAPIF interconnection scenario.

Pre-condition:

1. CCF-A and CCF-B connect to each other, and either belong to the single trust domain of the same CAPIF provider or trust domains of different CAPIF providers. When CCF-A and CCF-B belong to trust domains of different CAPIF providers, the two CAPIF providers have business agreement for service API sharing.

2. CCF-B is configured as the designated CAPIF core function in the trust domain of CAPIF provider A.

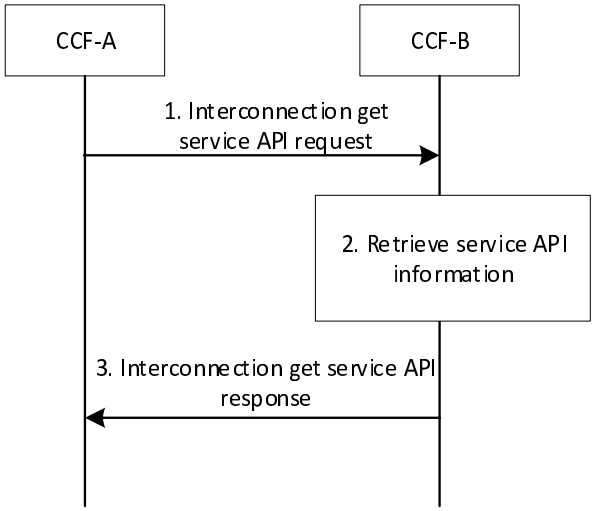


Figure 8.25.3.5-1: Retrieve service APIs for CAPIF interconnection

1. CCF-A sends an interconnection get service API request to CCF-B with service API published information reference provided by the CAPIF core function when the service API was published.
2. Upon receiving interconnection get service API request, CCF-B checks whether CCF-A is authorized to get the published service APIs information. If the check is successful, the corresponding service API information is retrieved from the CAPIF core function (API registry).
3. CCF-B sends an interconnection get service API response to CCF-A which includes the requested service API information if the retrieval operation was successful.

8.25.3.6 Update service APIs for CAPIF interconnection

Figure 8.25.3.6-1 illustrates the procedure for updating the published service APIs information by a CAPIF core function.

Pre-conditions:

1. CCF-A and CCF-B connect to each other, and either belong to the single trust domain of the same CAPIF provider or trust domains of different CAPIF providers. When CCF-A and CCF-B belong to trust domains of different CAPIF providers, the two CAPIF providers have business agreement for service API sharing.
2. CCF-B is configured as the designated CAPIF core function in the trust domain of CAPIF provider A.

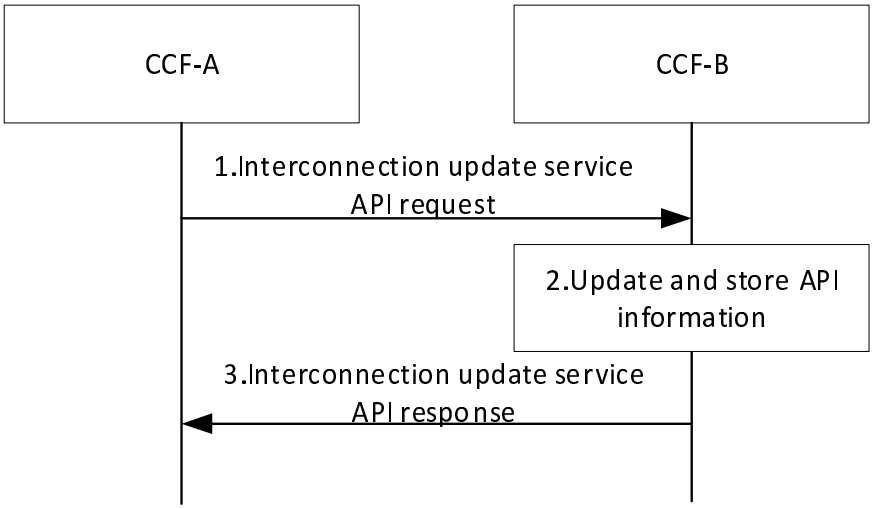


Figure 8.25.3.6-1: Update service APIs for CAPIF interconnection

1. The CCF-A sends an interconnection update service API request to the CCF-B, including the information in table 8.25.2.9-1.
2. Upon receiving the interconnection update service API request, the CCF-B checks whether the CCF-A is authorized to update the published service APIs information. If the check is successful, the service API information provided by the CCF-A is updated at the CCF-B (API registry).
3. The CCF-B provides an interconnection update service API response to the CCF-A containing the result of the update operation and triggers notifications to subscribed API invokers (if any) as described in subclause 8.8.4.

8.25.3.7 API invoker obtaining authorization for service API access in CAPIF interconnection

Pre-condition:

1. The API invoker has discovered service APIs provided by an AEF via procedure defined in step 1 and 2 of clause 8.25.3.2.
2. The API invoker and the CCF-B are in the same trusted domain.
3. The AEF and the CCF-A are in the same trusted domain.
4. The CCF-A and the CCF-B are connected to each other, and they have business agreement for service API authorization.
5. The CCF-A is the authorization function for service API access on the AEF.

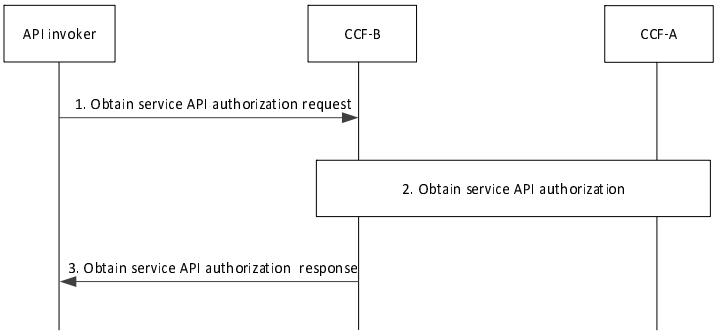


Figure 8.25.3.7-1: Procedure for the API invoker obtaining authorization for service API access

1. The API invoker sends an obtain service API authorization request to the CCF-B for obtaining permission to access the service API by including the API invoker identity information and any information required for authentication of the API invoker.

2. After successful authentication validation of the API invoker, if the CCF-B determines that the service API access authorization cannot be done by itself alone, then the CCF-B obtains service API authorization from the CCF-A, as specified in clause 8.25.2.16.

3. The authorization information to access the service APIs is sent to the API invoker in the obtain service API authorization response.

8.25.3.8 Procedure for CAPIF revoking API invoker authorization in CAPIF interconnection

Pre-conditions:

1. The API invoker is authenticated by CCF-B and authorized by CCF-B and CCF-A to use the service API. The API invoker has obtained authorization to use the service API as specified in clause 8.25.3.7.

2. The CCF-B may have available access control policies corresponding to the service API.

3. The CCF-A is triggered to revoke API invoker authorization for service API access, e.g. AEF requested the revocation of API invoker authorization.

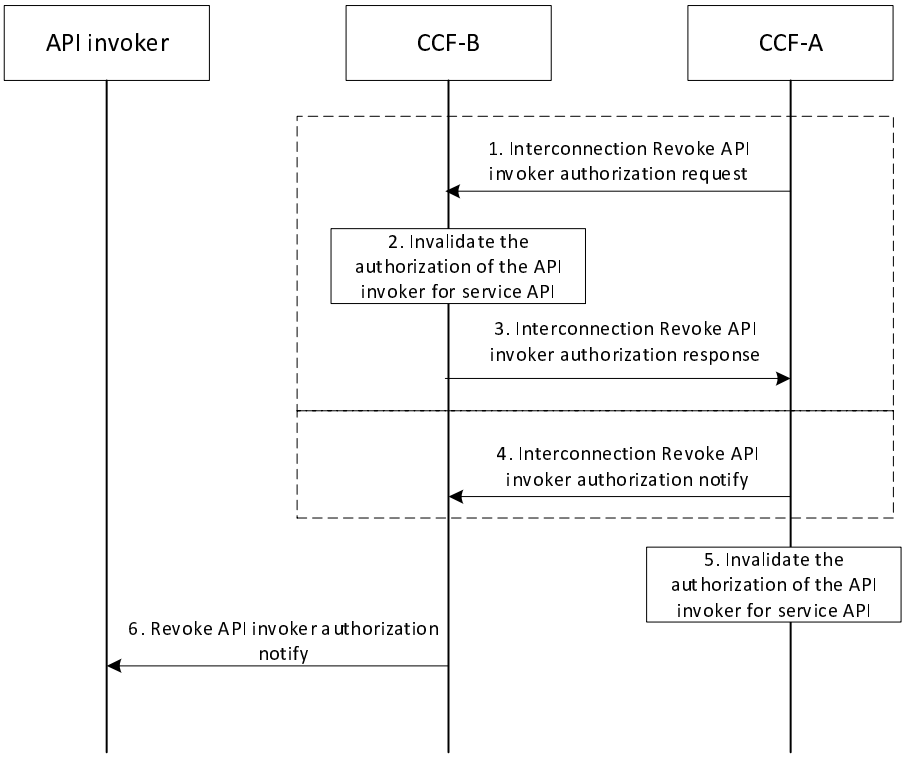


Figure 8.25.3.8-1: Procedure for revoking API invoker authorization in CAPIF interconnection

1. If the CCF-B keeps service API authorization information for the API invoker, and the CCF-A is triggered to revoke API invoker authorization information for service API access, the CCF-A sends an interconnection revoke API invoker authorization request to the CCF-B with the details of the API invoker, the AEF and the service API.

2. Upon receiving the information to revoke the API invoker's authorization for service API invocation, the CCF-B invalidates the API invoker authorization corresponding to the service API.

3. The CCF-B sends an interconnection revoke API invoker authorization response to the CCF-A. The CCF-B sends the Revoke API invoker authorization notify described in step 6.

4. Instead of step 1 to 3, if the CCF-A keeps service API authorization information for the API invoker, and the authorization information is revoked, the CCF-A sends an interconnection revoke API invoker authorization notify to the CCF-B.

5. The CCF-A invalidates the API invoker authorization corresponding to the service API.
6. The CCF-B sends a revoke API invoker authorization notify to the API invoker whose authorization to access the service API has been revoked.

8.25.3.9 Procedure for obtaining access control policy in CAPIF interconnection

Pre-condition:

1. The AEF is hosting the service API but the policy to perform access control is not available with AEF.
2. The CCF-B has available access control policies corresponding to one or more service APIs.
3. The AEF and the CCF-A are in the same trusted domain.

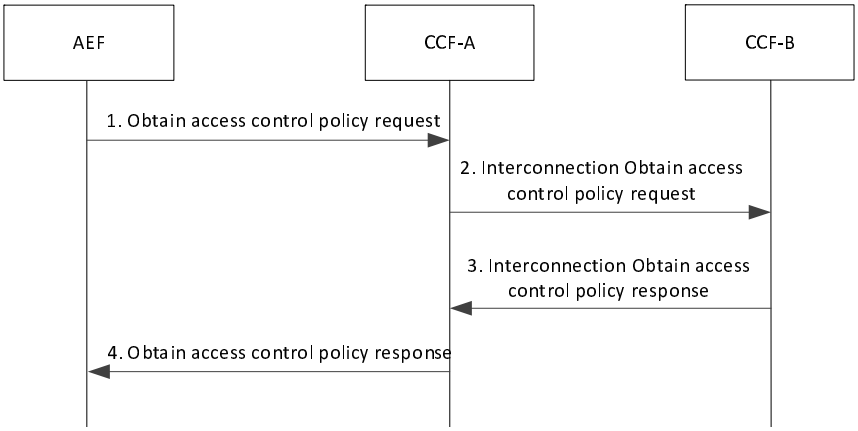


Figure 8.25.3.9-1: Procedure for obtaining access control policy in CAPIF interconnection

1. The AEF sends an obtain access control policy request to the CCF-A for obtaining the policy to perform the access control on service API invocations by including the details of the hosted service API.
2. After successful authentication validation of the AEF, if the CCF-A determines that the service API access control policy authorization cannot be done by itself alone, then the CCF-A sends an interconnection obtain access control policy request to the CCF-B.

NOTE 1: If the CCF-A has enough access control policy available, it executes the procedure defined in clause 8.12.3 without interaction with the CCF-B.

3-4. The CCF-B determines appropriate access control policy for service APIs upon the request sent by the CCF-A. The access control policy information is sent to the AEF (via the CCF-A) in the (interconnection) obtain access control policy response.

NOTE 2: To maintain synchronization between the AEF and the CCF for the policy cached at AEF, the AEF can subscribe to the policy update event at the CCF-A according to the procedure in clause 8.8.3 and receive notifications about any updated policy at the CCF-A according to the procedure in clause 8.8.4. Correspondingly, the CCF-A can subscribe to policy update event at the CCF-B if service API authorization is performed by the CCF-B.

8.25.3.10 Procedure for obtaining security information in CAPIF interconnection

Pre-condition:

1. The AEF has no security information available for authentication and/or authorization.
2. The CCF-B has security information available for authentication and/or authorization corresponding to one or more service APIs.
3. The AEF and the CCF-A are in the same trusted domain.

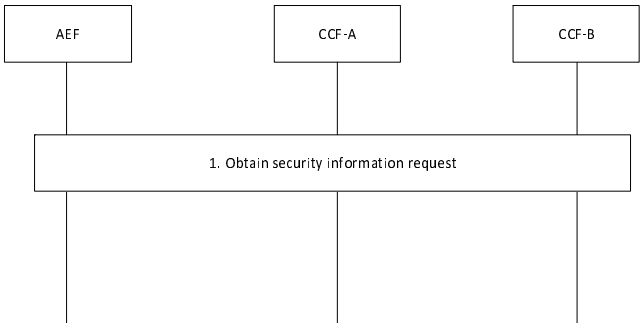


Figure 8.25.3.10-1: Procedure for obtaining security information in CAPIF interconnection

1. The AEF obtains the API invoker information required for authentication and/or authorization by the AEF from the CCF-A by including the details of the API invoker and hosted service API. After successful authentication validation of AEF, if the CCF-A determines that it has no security information, then the CCF-A obtains security information from the CCF-B. Then the security information required for service API invocations is sent to the AEF.

NOTE 1: If the CCF-A has security information available, it executes the procedure defined in clause 8.14.3, clause 8.15.3 or clause 8.16.3 without interaction with the CCF-B.

NOTE 2: The CCF-A can subscribe to API invoker onboarding event at the CCF-B according to the procedure in clause 8.8.3 and receive notifications about any onboarded API invoker and its security related onboarding information at the CCF-B according to the procedure in clause 8.8.4.

8.26 Update API invoker's API list

8.26.1 General

The procedure in this subclause corresponds to the architectural requirements for updating the API invoker's API list on the CAPIF core function. The CAPIF enables API invoker to update its own API list e.g. subsequent to discovering new API(s).

8.26.2 Information flows

8.26.2.1 Update API invoker API list request

Table 8.26.2.1-1 describes the information flow update API invoker API list request from the API invoker to the CAPIF core function.

Table 8.26.2.1-1: Update API invoker API list request

Information element	Status	Description
API invoker identity information	M	Identity information of the API invoker requesting update
APIs for update	M	List of APIs that need update (e.g. enroll new API(s), disenroll API(s)).

8.26.2.2 Update API invoker API list response

Table 8.26.2.2-1 describes the information flow update API invoker API list response from the CAPIF core function to the API invoker.

Table 8.26.2.2-1: Update API invoker API list response

Information element	Status	Description
Result	M	Indicates the success or failure of the update operation
API information	O (see NOTE 1)	List of APIs and the categories of service APIs that the API invoker can access
Reason	O (see NOTE 2)	This element indicates the reason for failure of the operation.
NOTE 1: Information element shall be present when update API invoker API list status is successful. At least one API successfully enrolled is considered as success response. NOTE 2: Information element shall be present when update API invoker API list has failed.		

8.26.3 Procedure

Figure 8.26.3-1 illustrates the procedure for updating the API invoker API list on the CAPIF.

Pre-conditions:

1. The API invoker has been onboarded as a recognized user of the CAPIF and associated API invoker profile is provisioned.

2. The API invoker has visibility to new APIs information (e.g. updates on API catalogue or dashboard, API discovery).

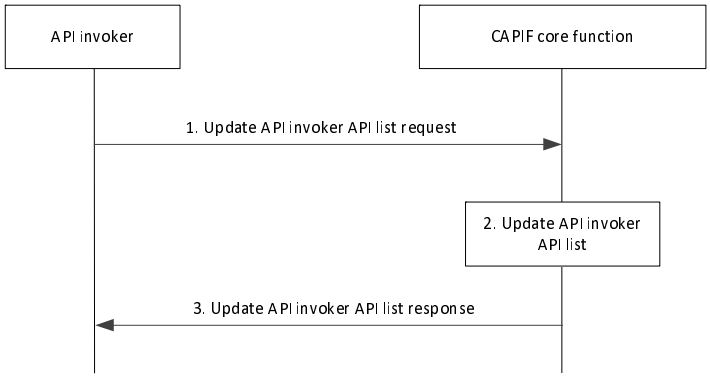


Figure 8.26.3-1: Procedure for updating the API invoker profile on the CAPIF

1. For updating of the API invoker API list on the CAPIF, the API invoker triggers update API invoker API list request towards the CAPIF core function, providing the information to be updated (e.g. enroll new APIs, disenroll APIs).

2. The CAPIF core function updates the API invoker API list of the requesting API invoker, according to the grant from the CAPIF administrator or the API management. For the disenrolled API(s), the authorization corresponding to the API invoker is revoked from CAPIF, if available.

NOTE 1: Completion of updating process can require explicit grant by the CAPIF administrator or the API management, which is left out-of-scope of this solution. CAPIF can handle the grant process internally without the need of explicit grant by the CAPIF administrator.

3. The update API invoker API list response provides a success or failure indication. When not all the APIs included in the request are enrolled, the success result include information about the APIs that the API invoker can access. When the update status is failure, the reason for failure is included.

8.27 Dynamically routing service API invocation

8.27.1 General

The procedure in this subclause corresponds to the architectural requirements for dynamic routing of service API invocation. The CAPIF enables dynamically routing the service API invocation request based on the detailed information of the invocation. The AEF acting as service communication entry point and the destination AEF for handling service API call may be in different API Provider domains, e.g. within the PLMN trust domain and within 3rd party trust domain, or may be the same API Provider domain, e.g., in the PLMN trust domain or 3rd party trust domain.

8.27.2 Information flows

8.27.2.1 Obtain routing information request

Table 8.27.2.1-1 describes the information flow dynamic routing information request from the API exposing function to the CAPIF core function.

Table 8.27.2.1-1: Obtain routing information request

Information element	Status	Description
Service API identification	M	The identification information of the service API for which invocation is requested. The service API identification is part of the specific service API invocation request.
AEF identity information	M	Identity information of the entity requesting the routing information

8.27.2.2 Obtain routing information response

Table 8.27.2.2-1 describes the information flow dynamic routing information response from the CAPIF core function to the API exposing function.

Table 8.27.2.2-1: Obtain routing information response

Information element	Status	Description
Service API identification information	M	The identification information of the service API for which invocation is requested.
Routing rule(s) information for service API invocation	M	Indicates the routing rule(s) for service API invocation, e.g., mapping of IP address range and AEF identity, or mapping of area serving the service API and AEF information.

8.27.3 Procedure

Figure 8.27.3-1 illustrates the procedure for dynamically routing the service API invocation from the AEF acting as service communication entry point to the destination AEF for handling service API.

Pre-conditions:

1. The API invoker has performed the service discovery and received the details of the service API which includes the information about the service communication entry point of the AEF-1 in the CAPIF.
2. The API invoker is authenticated and authorized to use the service API.
3. The AEF-1 is the AEF acting as service communication entry point for the service API, and AEF-2 is one of the multiple destination AEF which provides the service API.

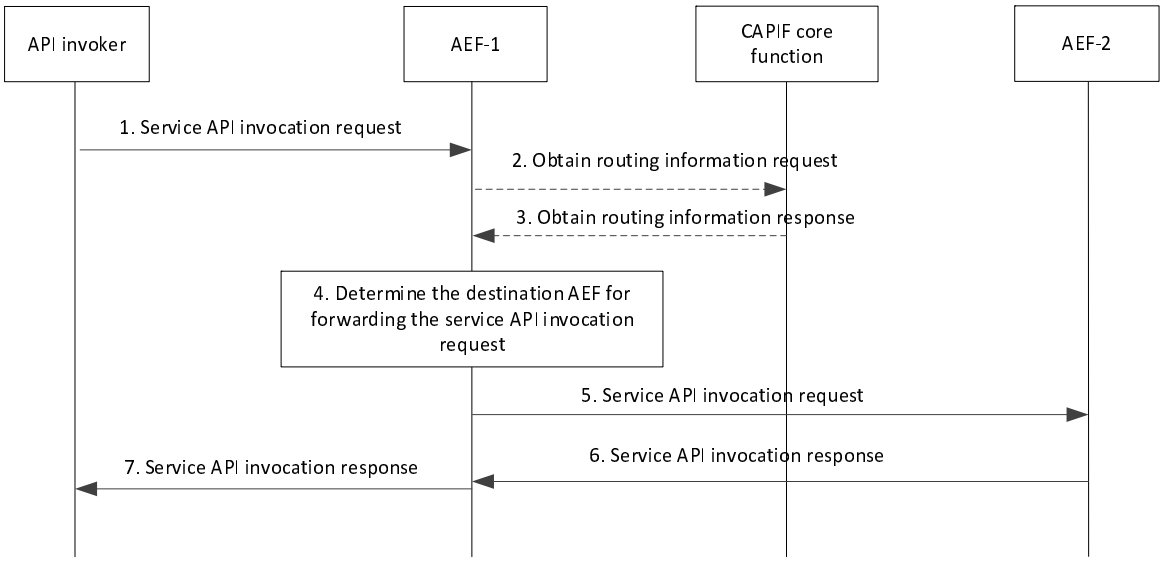


Figure 8.27.3-1: Procedure for dynamic routing of service API invocation

1. The API invoker performs service API invocation according to the interface of the service API by sending a service API invocation request towards the AEF-1 which exposes the service API towards the API invoker, and acts as topology hiding entity.
2. If the routing rule information for the service API invocation is not available, the AEF-1 sends obtain routing information request to the CAPIF core function.
3. The CAPIF core function creates routing rule information for the service API and sends obtain routing information response with the routing rule information.

NOTE: Steps 2 and 3 can be performed before step 1and after receiving the API topology hiding notify as described in subclause 8.24.3.

4. The AEF-1 further resolves the actual destination of the service API address information (AEF-2) according to the routing rule information and the invocation parameters in service API invocation request.
5. The AEF-1 forwards the incoming service API invocation request to AEF-2.
6. The AEF-2 returns the service API invocation response to AEF-1.
7. The AEF-1 sends the service API invocation response to the API invoker.

8.28 Registering the API provider domain functions on the CAPIF

8.28.1 General

The procedure in this subclause corresponds to the architectural requirements for registering the API provider domain functions on the CAPIF. This procedure registers the API provider domain functions as authorized users of the CAPIF functionalities.

NOTE: The security aspects of this procedure are specified in 3GPP TS 33.122 [12] clauses 6.6 and 6.10.

8.28.2 Information flows

8.28.2.1 Registration request

Table 8.28.2.1-1 describes the information flow, registration request, from the API management function to the CAPIF core function.

Table 8.28.2.1-1: Registration request

Information element	Status	Description
List of API provider domain functions	M	List of API provider domain functions including role (e.g. AEF, APF, AMF) and, if required, specific security information.
API provider name	O	The API provider name uniquely identifies an API provider (e.g. Internet Service Provider).
Security information	M	Information for CAPIF core function to validate the registration request

8.28.2.2 Registration response

Table 8.28.2.2-1 describes the information flow, registration response, from the CAPIF core function to the API management function.

Table 8.28.2.2-1: Registration response

Information element	Status	Description
List of identities	M (see NOTE 1)	List of identities, for each successfully registered API provider domain function and any specific security information.
Security information	O	Information to be used by the API provider domain function in subsequent CAPIF API invocations. Provided when registration is successful.
Reason	O (see NOTE 2)	Information related to registration result specific to individual API provider domain functions. Provided when the registration fails.
NOTE 1: Information element shall be present when at least one registration request is successful. NOTE 2: Information element may be present when at least one registration requests fail.		

8.28.3 Procedure

Figure 8.28.3-1 illustrates the procedure for registering API provider domain functions on the CAPIF core function.

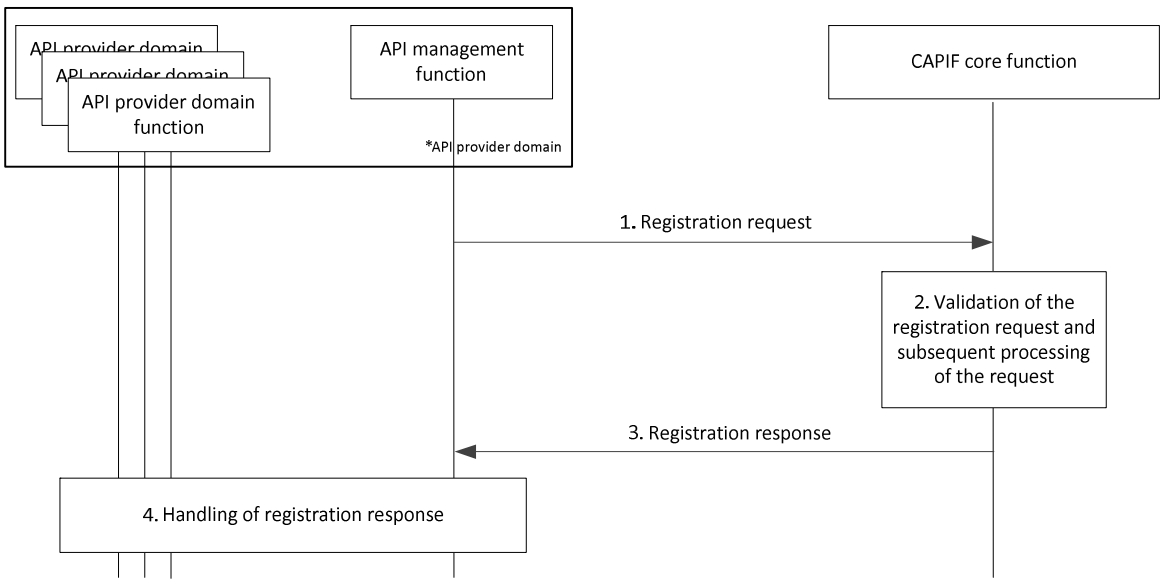


Figure 8.28.3-1: Procedure for registration of API provider domain functions on CAPIF

1. For registration of API provider domain functions on the CAPIF core function, the API management function sends a registration request to the CAPIF core function. The registration request contains a list of information about all the API provider domain functions, which require registration on the CAPIF core function.
2. The CAPIF core function validates the received request and generates the identity and other security related information for all the API provider domain functions listed in the registration request.
3. The CAPIF core function sends the generated information in the registration response message to the API management function.
4. The API management function configures the received information to the individual API provider domain functions.

8.29 Update registration information of the API provider domain functions on the CAPIF

8.29.1 General

The procedure in this subclause corresponds to the architectural requirements for update of the registration information of the API provider domain functions on the CAPIF.

NOTE: The security aspects of this procedure are specified in 3GPP TS 33.122 [12] clauses 6.6 and 6.10.

8.29.2 Information flows

8.29.2.1 Registration update request

Table 8.29.2.1-1 describes the information flow, registration update request, from the API management function to the CAPIF core function.

Table 8.29.2.1-1: Registration update request

Information element	Status	Description
List of API provider domain functions requiring update	M	List of API provider domain functions requiring updates, including role (e.g. AEF, APF, AMF) and, if required, specific security information.
Security information	M	Information for CAPIF core function to validate the registration request

8.29.2.2 Registration update response

Table 8.29.2.2-1 describes the information flow, registration update response, from the CAPIF core function to the API management function.

Table 8.29.2.2-1: Registration update response

Information element	Status	Description
List of identities	M (see NOTE 1)	List of identities, for each successfully updated API provider domain function and any specific security information.
Security information	O	Information to be used by the API provider domain function in subsequent CAPIF API invocations. Provided when registration update is successful.
Reason	O (see NOTE 2)	Information related to registration update result specific to individual API provider domain functions. Provided when the registration fails.
NOTE 1: Information element shall be present when at least one registration update request is successful. NOTE 2: Information element may be present when at least one registration update requests fail.		

8.29.3 Procedure

Figure 8.29.3-1 illustrates the procedure for updating the registration information of the API provider domain functions on the CAPIF core function.

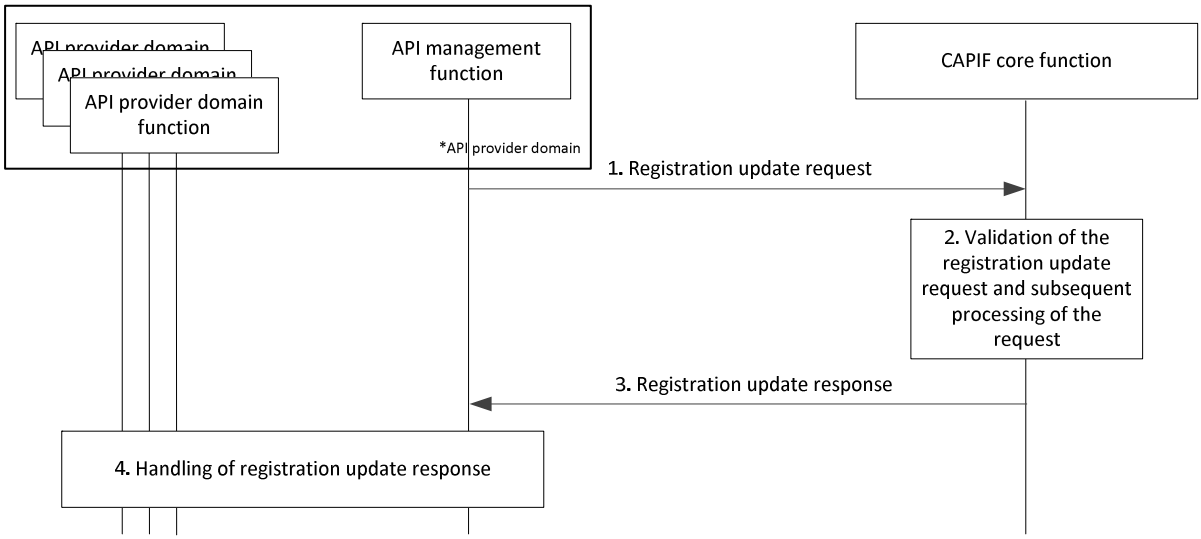


Figure 8.29.3-1: Procedure for update of registration information of API provider domain functions on CAPIF

1. For updating the registration information of API provider domain functions on the CAPIF core function, the API management function sends a registration update request to the CAPIF core function. The registration update request contains a list of information about all the API provider domain functions, which require registration update on the CAPIF core function.

2. The CAPIF core function validates the received request and updates the identity and other security related information for all the API provider domain functions listed in the registration request.

3. The CAPIF core function sends the updated information in the registration update response message to the API management function.

4. The API management function configures the received information to the individual API provider domain functions.

8.30 Deregistering the API provider domain functions on the CAPIF

8.30.1 General

The procedure in this subclause corresponds to the architectural requirements for deregistering the API provider domain functions on the CAPIF. This procedure deregisters the API provider domain functions as authorized users of the CAPIF functionalities.

NOTE: The security aspects of this procedure are specified in 3GPP TS 33.122 [12] clauses 6.6 and 6.10.

8.30.2 Information flows

8.30.2.1 Deregistration request

Table 8.30.2.1-1 describes the information flow, deregistration request, from the API management function to the CAPIF core function.

Table 8.30.2.1-1: Deregistration request

Information element	Status	Description
Security information	M	Information for CAPIF core function to validate the deregistration request

8.30.2.2 Deregistration response

Table 8.30.2.2-1 describes the information flow, deregistration response, from the CAPIF core function to the API management function.

Table 8.30.2.2-1: Deregistration response

Information element	Status	Description
Result	M	Indicates the success or failure of the deregistration operation

8.30.3 Procedure

Figure 8.30.3-1 illustrates the procedure for deregistering API provider domain functions on the CAPIF core function.

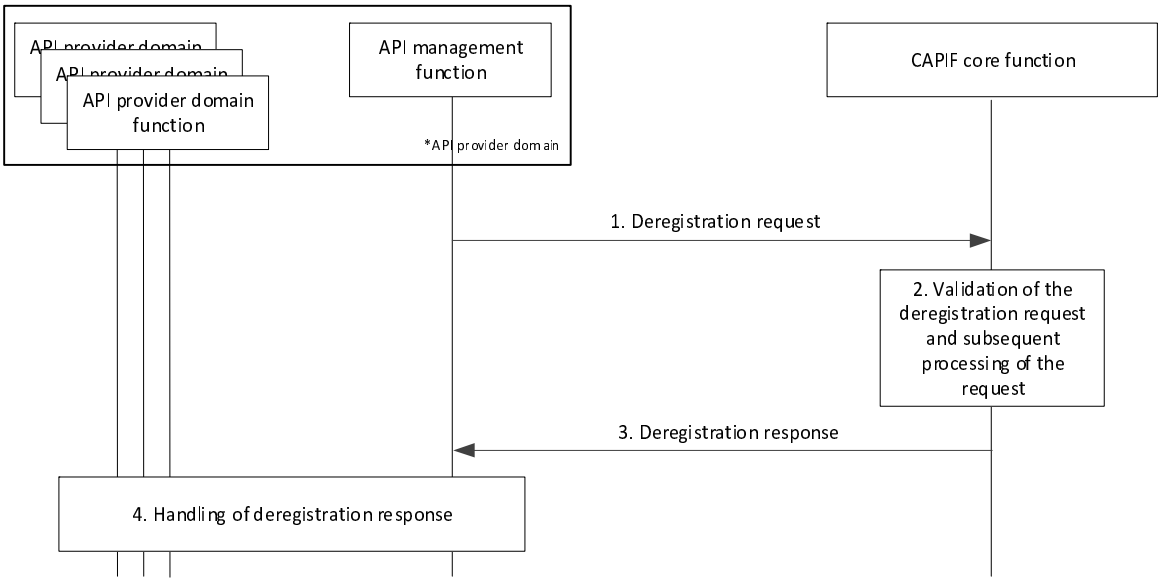


Figure 8.30.3-1: Procedure for deregistration of API provider domain functions on CAPIF

1. For deregistration of API provider domain functions on the CAPIF core function, the API management function sends a deregistration request to the CAPIF core function.
2. The CAPIF core function validates the received request and processes the deregistration request.
3. The CAPIF core function sends a deregistration response message to the API management function.
4. The API management function processes the deregistration to the individual API provider domain functions.

8.31 API invoker obtaining authorization from resource owner

8.31.1 General

CAPIF may authorize the API invoker to invoke the service API based on the authorization information from the resource owner given before the API invocation.

Clause 8.31.3 shows the procedure for obtaining the authorization information.

8.31.2 Information flows

NOTE: The security aspects of this procedure are specified in TS 33.122 [12].

8.31.3 Procedure

Figure 8.31.3-1 illustrates the procedure for API invoker obtaining authorization from resource owner.

Pre-conditions:

1. The resource owner function (ROF) can communicate with the CCF (Authorization function).
2. Access to an API exposing function offered service API requires obtaining authorization from a resource owner (RO).

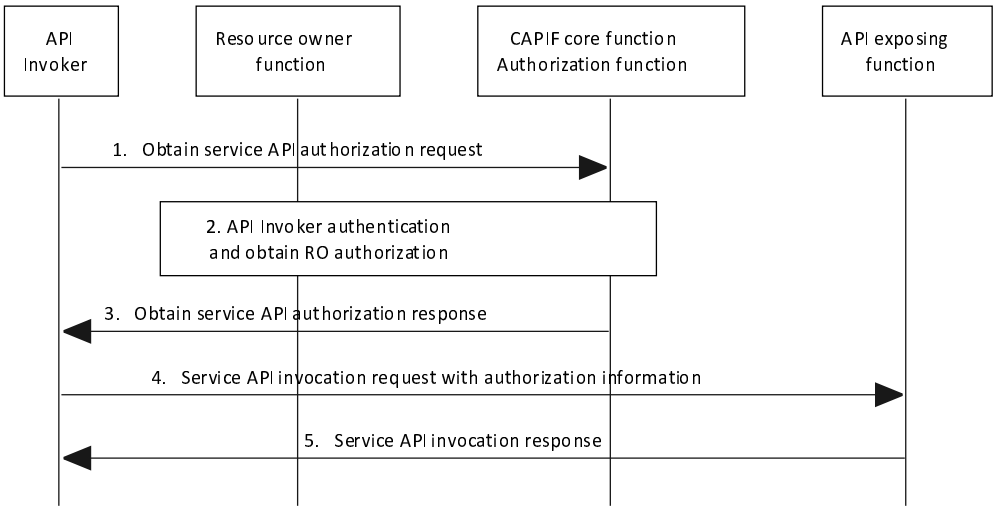


Figure 8.31.3-1: Procedure for API invoker obtaining authorization from resource owner

1. The API invoker requests to obtain authorization information to invoke the service API exposed by the API exposing function (AEF) and to access information owned by the resource owner at the AEF through the invocation of an obtain service API authorization request to the CCF. The request contains API invoker information required for authorization, the information identifying the service API, the purpose for data processing and any information required for authentication of the API invoker. The request may include finer level service API access requirements (e.g., access per service API operation or access per API service resource) and resource owner-related information required to obtain resource owner authorization information.

NOTE 1: The details of the information provided by the API invoker for obtaining service API access are specified in 3GPP TS 33.122 [12].

2. The CCF performs the authentication of the API invoker (using authentication information). Then the Authorization function determines the authorization by checking the authorization information available in the CCF and by checking the authorization information provided by the RO via the ROF. The request to the ROF contains application service information (e.g. the application service provider and application identifier), the purpose for data processing and the resource owner data information for which the API invoker requests access grants.

NOTE 2: The detailed procedure to obtain the RO's authorization information is specified in 3GPP TS 33.122 [12].

NOTE 3: Step 1 can occur when the API invoker receives a failure in Service API invocation response indicating that authorization information from resource owner(s) is required.

3. Based on the RO authorization information obtained via the ROF, the CCF sends an obtain service API authorization response to the API invoker.

4. The API invoker sends service API invocation request to the API exposing function with the authorization information received from CCF (Authorization function) in step 3.

5. The API invoker receives the service API invocation response resulting from the service API invocation once the API exposing function has checked whether the API invoker is authorized to invoke that service API based on the authorization information received from CCF (Authorization function).

8.32 Reducing authorization information inquiry in a nested API invocation

8.32.1 General

The nested API invocation scenario is a scenario where an API invocation towards a first API exposing function triggers that API exposing function to request an API invocation towards a second API exposing function, which is in the same API provider domain as the first API exposing function, which can be the PLMN trust domain or a 3rd party trust domain. This scenario addresses the situation in which a service API may require the services of other service APIs. For example, if the API invoker invokes SEAL SS_LocationInfoRetrieval API (clause 9.4.4 of TS 23.434 [15]), the location management server (acting as an API exposing function for the API invoker and as an API invoker for the NEF) may invoke NEF API to retrieve UE location information from 5GC. In this scenario, the CAPIF may reduce the authorization information inquiries for a nested API invocation using procedure described in clause 8.32.3.

8.32.2 Information flows

NOTE: The security aspects of this procedure are specified in TS 33.122 [12].

8.32.3 Procedure

Figure 8.32.3-1 illustrates the procedure to obtain authorization information in a nested API invocation, in which an API exposing function receiving the service API invocation request interacts with another API exposing function to provide the service.

Pre-conditions:

1. The resource owner function can communicate with the CCF (Authorization function).
2. The API exposing functions 1 and 2 are in the same trust domain.

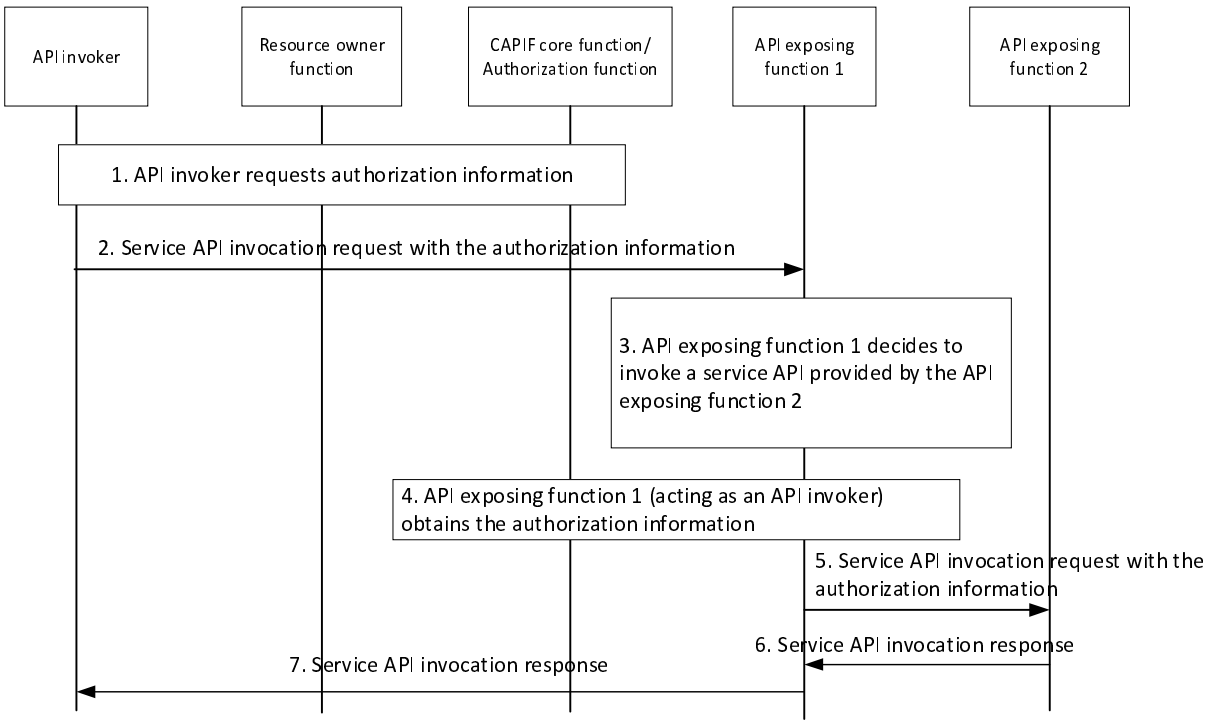


Figure 8.32.3-1: Procedure for obtaining authorization information in a nested API invocation

1. The API invoker requests authorization information to invoke the service API exposed by API exposing function 1.

NOTE 1: This step use either the procedure to obtain authorization to access service API specified in clause 8.11 or the procedure to obtain authorization to access service API that involves the resource owner function as specified in clause 8.31.

2. The API invoker sends a service API invocation request to API exposing function 1 with the authorization information received in step 1. The API exposing function 1 verifies the API invoker request.

3. Based on the service API invocation request, API exposing function 1 decides to invoke another service API exposed by API exposing function 2.

4. API exposing function 1, acting as an API invoker, before interacting with API exposing function 2, shall obtain from the CCF the authorization information to access the service API exposed by API exposing function 2.

NOTE 2: To obtain the authorization information, the API exposing function 1 exchanges the authorization information provided in the API invoker's request (in step 2) with authorization information for accessing the service API exposed by API exposing function 2, e.g. to avoid further interaction with the API invoker. The security mechanism involved in this exchange are out of scope of the present document.

5. API exposing function 1, acting as an API invoker sends a service API invocation request to API exposing function 2 with the authorization information received in step 4.

6. The API exposing function 1 receives the service API invocation response resulting from the service API invocation once API exposing function 2 has checked whether the API exposing function 1, acting as API invoker, is authorized to invoke that service API based on the authorization information.

7. The API invoker receives the service API invocation response resulting from the service API invocation.

8.33 Obtaining collective authorization from a resource owner

8.33.1 General

Using the procedure described in clause 8.31 the CAPIF may authorize an API invoker to invoke a service API based on the authorization information from the resource owner given before the API invocation. This clause describes an extension to that procedure to enable the CAPIF to obtain authorization information from the resource owner relating to multiple API invokers and service APIs in a single interaction.

Clause 8.33.3 shows the procedure for obtaining collective authorization information from a resource owner via their resource owner function.

8.33.2 Information flows

NOTE: The security aspects of this procedure are specified in clause 6.4 and 6.5.3 of 3GPP TS 33.122 [12].

8.33.3 Procedure

Figure 8.33.3-1 illustrates the procedure for obtaining collective authorization from resource owner.

The mechanism to obtain collective authorization from a resource owner is supported by the CAPIF core function.

Pre-conditions:

1. The resource owner function can communicate with the CCF (Authorization function).

2. Access to an API exposing function offered service API requires obtaining authorization from a resource owner.

3. The API invoker is successfully onboarded as described in clause 8.1.

4. The CCF is configured for handling collective authorization, see table 8.33.3-1.

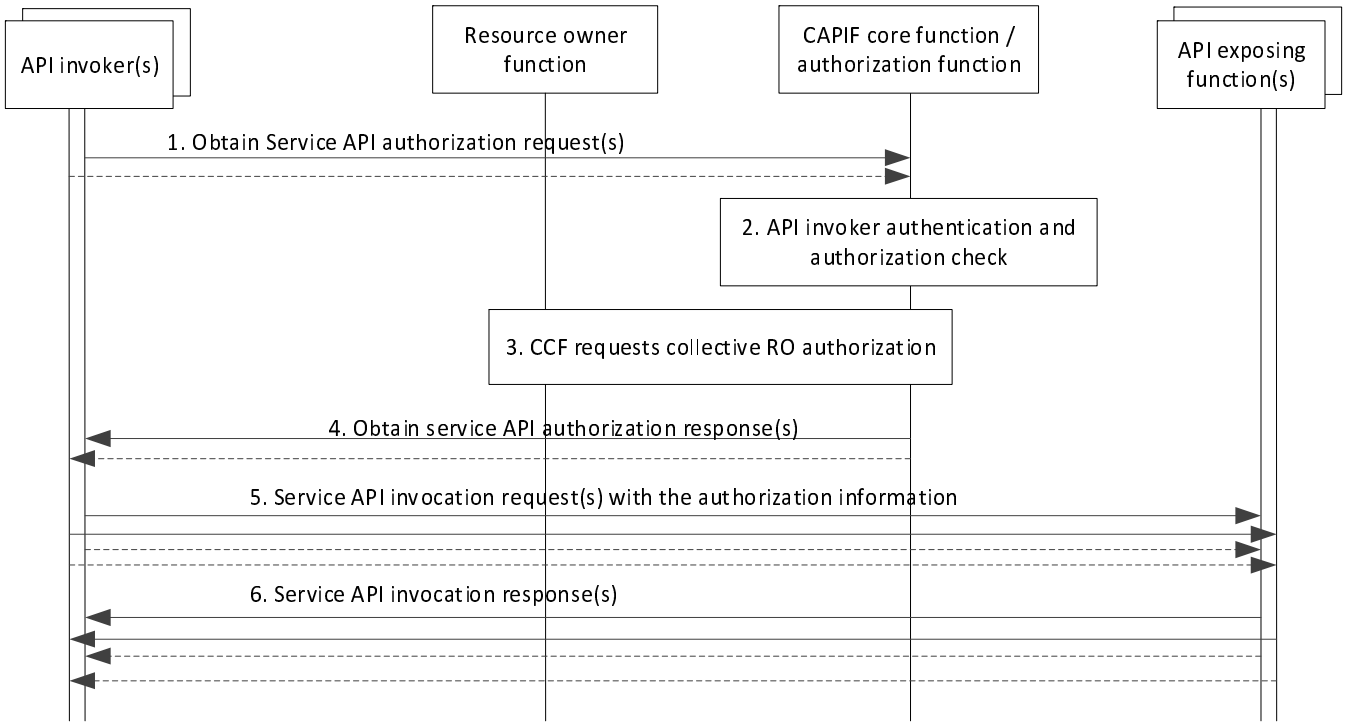


Figure 8.33.3-1: Procedure for obtaining collective authorization from resource owner

1. An API invoker sends obtain service API authorization request to the CAPIF core function for obtaining permission to access a service API by including its API invoker identity information and any information required for authentication of the API invoker. The API invoker includes in the request information about the purpose and the targeted resources and operations to be performed on those resources.

2. The CCF performs the authentication of the API invoker (using authentication information) and checks whether it is permitted to access the requested service API by checking whether a resource owner authorization information is already available for that API invoker. If the authorization for accessing the requested resources for the signalled purpose is already granted, the procedure continues with step 4.

3. If authorization information is not available regarding whether the API invoker is permitted to access the requested service API (i.e., there is no applicable resource owner authorization information available) the collective resource owner authorization configuration (table 8.33.3-1) is assessed.

   If authorization trigger criteria have been specified as part of the collective resource owner authorization configuration, interactions with the resource owner are only initiated once the trigger conditions are met.

   If trigger criteria have not been set or once the trigger conditions are met, the authorization function determines the authorization by interacting with the resource owner via the ROF deployed on a UE. Based on the configured collective resource owner authorization configuration at the CCF and any cached “obtain service API authorization requests”, the scope of the request towards the ROF can be expanded to include additional API invokers and or service APIs based on the assumption that there is an association between the requesting API invoker and the API exposing functions and API invokers listed in the CCF configuration.

NOTE 2: The details of the interaction between RO and ROF for capturing the resource owner authorization are considered out-of-scope of the present specification.

NOTE 3: The detailed procedure to obtain the resource owner's authorization information for multiple API invokers in a single request, including whether such authorization can be limited to the API invoker initiating the request and only for the specific service being requested, is out of scope of this release of the specification.

4. Based on the RO authorization information obtained via the ROF for each API invoker and corresponding service API(s), the CCF sends an API invoker specific service API authorization response (as described in clause 8.31) to the API invoker that initially triggered the collective resource owner authorization information requested and to each API invoker for which a obtain service API authorization request was received (for the

same service API) whilst the collective resource owner authorization configuration duration timer was running, i.e., the cached requests.

5. An API invoker that receives the service API authorization response can send service API invocation requests to the API exposing function with their API invoker specific resource owner authorization information as described in clause 8.31.

6. The API exposing function will send a service API invocation response once it has successfully checked that the API invoker is authorized to invoke that service API based on the authorization information.

Table 8.33.3-1: Collective resource owner authorization configuration

Information element	Status	Description
List of API invoker associations (NOTE 1)	O	Lists of API invokers with their associated service APIs
> API invoker identity information list	M	List of API invoker identities
> Service API identification list	M	List of AEF service API identities.
List of trigger criteria (NOTE 1)	O	Criteria for triggering requesting of resource owner authorization
> Duration (NOTE 2)	O	Initiate request towards RO via ROF after the given duration (in seconds) after receiving a obtain service authorization request from an API invoker
> Timeperiod (NOTE 2)	O	Time period during which notifications may be sent towards RO via ROF.
> Resource owner identifier	O	Can only be included if a duration or timeslot is included. Indicates that the resource owner authorization trigger criteria is to be applied to the specific resource owner identified by their GPSI, i.e., no other resource owner authorization trigger criteria shall be applied for that resource owner.
NOTE 1: At least one of these attributes shall be present NOTE 2: Only one of these attributes shall be present		

8.34 UE-deployed API invoker accessing other UEs' resources of a group

8.34.1 General

This clause specifies the procedures for API invoker on a UE to be able to access the services exposing resources of other UEs of a group. The UEs belong to the same group and the same VAL service provider (IoT case) or the same user and all of them belong to the same CAPIF provider domain.

The authorization relationship between UE2 (the API invoker is in UE2) and UE1, given by the membership to the same group, is within the CAPIF provider domain. The difference of this scenario compared with the procedure of an API invoker obtaining authorization from resource owner described in clause 8.31 is that:

a. the API invoker is not deployed in the UE whose resources are being accessed,

b. the API invoker needs to provide in the request, in addition to the identity of the UE whose resources are being accessed, the Group Identifier and the identity of the UE2; and

c. the group membership needs to be managed by the CAPIF provider as described in Annex E.

NOTE: Other possible scenarios, e.g. where the authorization relationship given by the membership to the same group is within the Application Service Provider, are not specified in this release of the specification.

8.34.2 Information flows

NOTE: The security aspects of this procedure are specified in 3GPP TS 33.122 [12].

8.34.3 Procedure

Figure 8.34.3-1 presents the procedure for enabling a UE-hosted API invoker accessing network-hosted resources owned by other UEs that belong to the same group.

Pre-condition:

1. The network hosts individual resources for a group of UEs which can be accessed via AEF provided the necessary security conditions are met.
2. UE1 and UE2 belong to the same group and the same VAL service provider (IoT case) or the same user.
3. CCF is provisioned with the information of a Group Resource Owner (GRO) which indicates UE 1 as the authorized RO (e.g. by business relationship between the group of UEs) to provide resource owner authorization for the resources belonging to the UEs in this group.

NOTE: In this release of the specification group context information is provisioned by the CAPIF administrator as specified in Annex E and the other mechanisms are out of scope of this release.

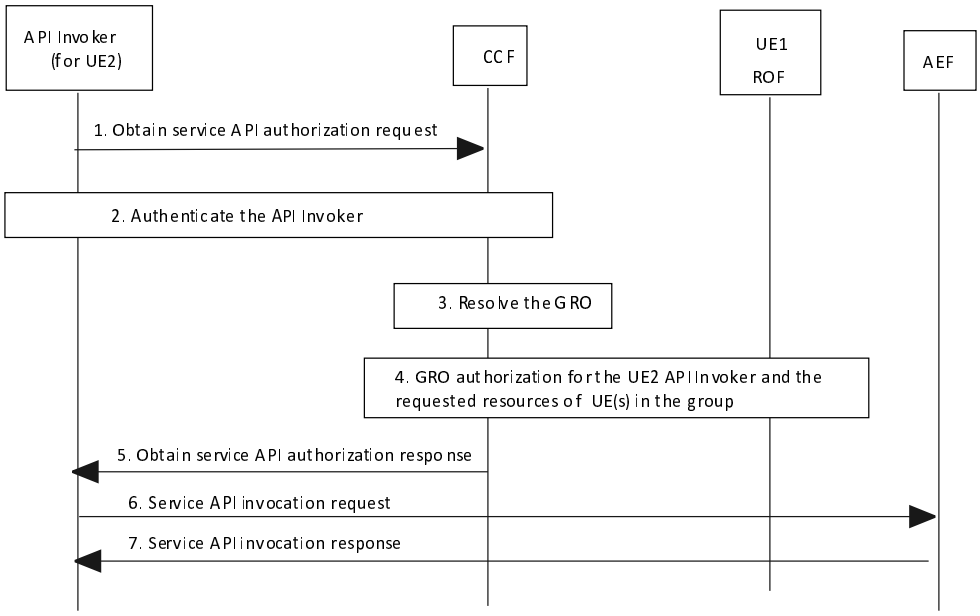


Figure 8.34.3-1: UE-deployed API invoker accessing other UEs' resources of a group

1. The API invoker (e.g., in UE 2) sends an Obtain service API authorization request to the CCF for obtaining permission to access the service API for other UE's resources hosted in the network (e.g., location) by including the API invoker identity information and any information required for authentication of the API invoker. The request includes API invoker information, the group identifier, and the UE in a group whose resources are to be accessed, and the identity of UE2. Additionally, the API invoker may include in the request information about the purpose and operations (scope) to be performed on the targeted resources.
2. CCF performs authentication of the API invoker (using authentication information) as specified in 3GPP TS 33.122 [12].
3. The CCF, based on the group identifier determines that RO authorization is provided by a GRO. The CCF then resolves the identity of the GRO responsible for the group of UEs based on the provisioned group context information related to the GRO, as specified in Annex E. The CCF shall additionally check whether UE2 belongs to the group.
4. If step 3 is successful, the CCF performs the resource owner authorization check using the GRO as the RO for the requested resources of other UE(s) belonging to the group.

NOTE: The details of the procedure to obtain resource owner's authorization information from GRO are out the scope of this release of the specification.

5. Based on the group resource owner authorization received in step 4, the CCF sends an Obtain service API authorization response to the API invoker. The authorization response indicates that the access to UE (resource owner) network resource(s) is authorized for an indicated scope.

6. The API invoker sends service API invocation request to the API exposing function with the RO authorization information received in step 5.

7. The API invoker receives the service API invocation response resulting from the service API invocation once the API exposing function has checked whether the API invoker is authorized to invoke that service API based on the RO authorization information.

8.35 Revoking resource owner authorization

8.35.1 General

The procedure in this subclause corresponds to the architectural requirements for revoking resource owner authorization.

8.35.2 Information flows

The information flows for the interactions between the CAPIF core function, the API Invoker and the AEF are described in clause 8.23.2.

NOTE: The security aspects of this procedure are specified in clause 6.5.3.4 of 3GPP TS 33.122 [12].

8.35.3 Procedure

Figure 8.35.3-1 illustrates the procedure for revoking resource owner authorization to access service API initiated by the resource owner function.

Pre-conditions:

1. The API invoker is authenticated and authorized to use the service API.

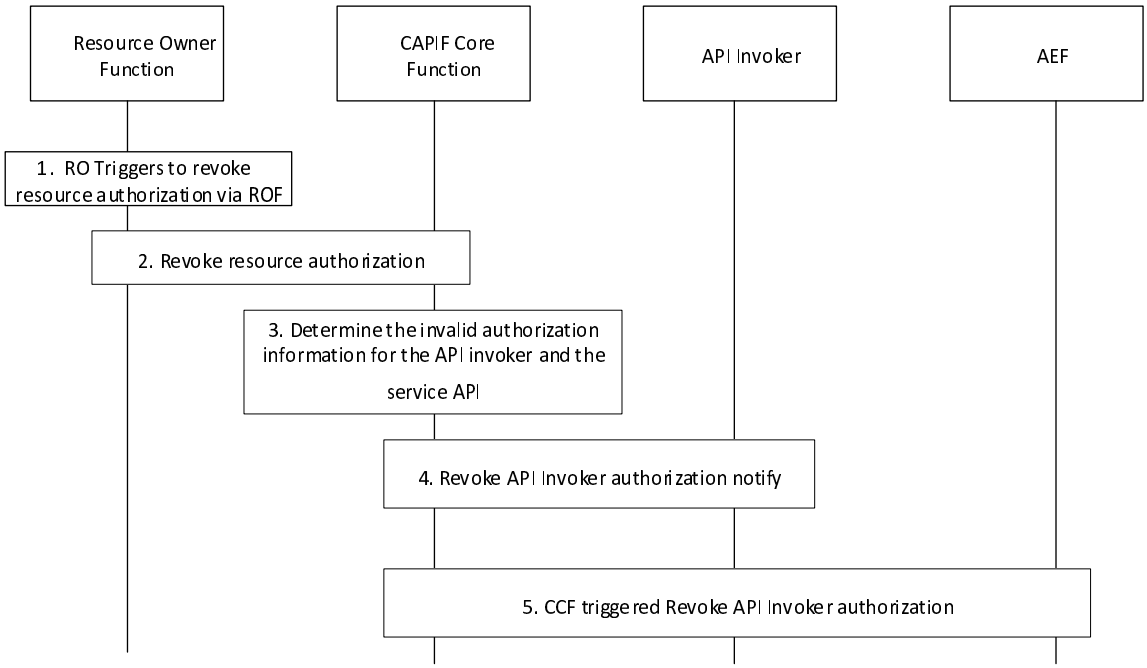


Figure 8.35.3-1: Procedure for revoking resource owner authorization

1. The RO triggers the revocation of the resource authorization via ROF.
2. The ROF requests resource authorization revocation to the CAPIF core function.

NOTE: The detailed procedure of resource authorization revocation is to be specified in TS 33.122 [12].

3. The CAPIF core function determines the revoked authorization information for the API Invoker and Service API as specified in clause 6.5.3.4 of TS 33.122 [12].
4. The CAPIF core function sends a revoke resource authorization notification to the API invoker according to the procedure in clause 8.23.4.
5. The CAPIF core function sends a revoke resource authorization notification to the API exposing function according to the procedure in clause 8.23.4.

8.36 CCF Obtaining Resource Owner Authorization

8.36.1 General

The procedure in this subclause corresponds to the architectural requirements for CCF obtaining resource owner authorization.

8.36.2 Information flows

The information flows for the interactions to obtain RO authorization information from the RO function are part of the Obtain service API authorization procedure described in clause 8.31.2.

NOTE: The security aspects of this procedure are specified in clause 6.5.3.B of 3GPP TS 33.122 [12].

8.36.3 Procedure

Figure 8.36.3-1 illustrates the procedure for CCF obtaining resource owner authorization.

Pre-condition:

1. ROF and CCF have successfully completed mutual authentication.

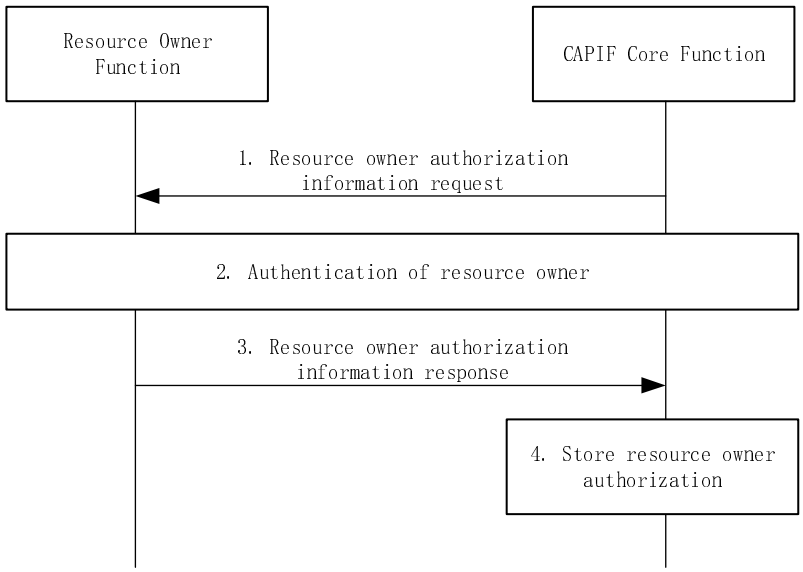


Figure 8.36.3-1: Procedure for obtaining resource owner authorization

1. The CAPIF core function sends the resource owner authorization information request along with the API invoker information received in the obtaining service API authorization procedure as described in clause 8.31.3.
2. Authentication of the resource owner is performed as specified in clause 6.5.3.1 of 3GPP TS 33.122 [12].
3. The resource owner sends the resource owner authorization information response via ROF to the CAPIF core function along with the resource authorization information as specified in clause 8.31.3.
4. The CAPIF core function stores the resource owner authorization information.

8.37 UE-deployed API invoker accessing other UEs' resources

8.37.1 General

This clause specifies the procedures for API invoker on a UE to be able to access the services exposing resources of other UEs.

The authorization relationship between UE2 (the API invoker is in UE2) and UE1 (the resource owner of the accessed network resources is at UE1) is within the Application Service Provider. The difference of this scenario compared with the procedure of an API invoker obtaining authorization from resource owner described in clause 8.31 is that the API invoker is not deployed in the resource owner's UE.

NOTE: Other possible scenarios, e.g. where the authorization relationship between UE(s) is in the CAPIF domain, are not specified in this release of the specification.

8.37.2 Information flows

NOTE: The security aspects of this procedure are specified in 3GPP TS 33.122 [12].

8.37.3 Procedure

Figure 8.37.3-1 presents the procedure for enabling a UE-hosted API invoker accessing network-hosted resources owned by other UEs for the scenarios introduced in clause 8.37.1.

Pre-condition:

1. The network hosts individual resource(s) for a UE which can be accessed via AEF provided the necessary security conditions are met.
2. UE 1 and UE 2 belong to the same CAPIF provider domain. The API invoker knows the RO identifier for UE1.

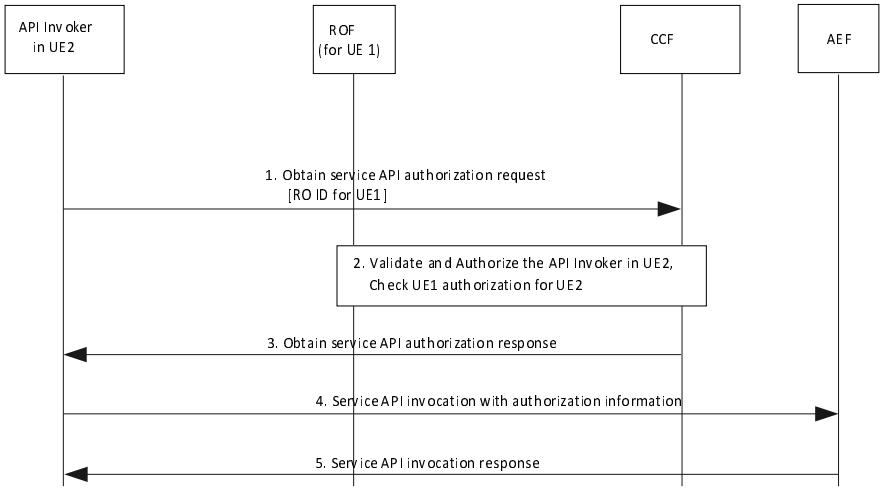


Figure 8.37.3-1: UE-deployed API invoker accessing other UEs' resources

1. The API invoker in UE2 sends an Obtain service API authorization request to the CCF for obtaining permission to access the service API for other UE's (UE1) resources hosted in the network by including the API invoker identity information and any information required for authentication of the API invoker.

   The request shall include the RO identity for UE1, whose resources are trying to be accessed, the API invoker information that is requesting the access, and the information about the purpose and scope to be performed on the targeted resources.

2. The CCF validates and authorizes the API invoker in UE2 as specified in 3GPP TS 33.122 [3]. If the API invoker validation is successful, the CCF checks the Resource Owner authorization information of UE1. The CCF obtains whether the RO (UE1) authorized to the API invoker the access to the network resource(s) for the specified purpose and scope. If the RO authorization information is not available, the CCF may interact with UE1 via ROF to obtain it.

3. Based on the UE1 resource owner authorization obtained in step 2, the CCF sends an Obtain service API authorization response to the API invoker. The response indicates that UE1, as resource owner, authorizes the service API access for an indicated scope.

4. The API invoker sends service API invocation request to the API exposing function with the RO authorization information received in step 2.

5. The API invoker receives the service API invocation response resulting from the service API invocation once the API exposing function has checked whether the API invoker is authorized to invoke that service API based on the RO authorization information.

8.38 Open Discover service APIs

8.38.1 General

The following procedure in this subclause corresponds to supporting open discover service APIs. Such open discovery operation allows requestor to discover service API information about the available set of APIs offered by CCF before onboarding.

8.38.2 Information flows

8.38.2.1 Open Service API discover request

Table 8.38.2.1-1 describes the information flow open service API discover request from the requestor to the CAPIF core function.

Table 8.38.2.1-1: Open Service API discover request

Information element	Status	Description
Query information	O (see NOTE)	Criteria for discovering matching service APIs. It may contain Service API information described in table 8.3.2.1-1, except service API interface details.
NOTE: It should be possible to discover all the service APIs.		

8.38.2.2 Open Service API discover response

Table 8.38.2.2-1 describes the information flow open service API discover response from the CAPIF core function to the requestor.

Table 8.38.2.2-1: Open Service API discover response

Information element	Status	Description
Result	M	Indicates the success or failure of the open discovery of the service API information
Service API information	O (see NOTE)	Filtered service APIs information that can be shared to the requestor not a recognized user of the CAPIF. It may contain Service API information described in table 8.3.2.1-1, except service API interface details.
NOTE: The service API information shall be present if the Result information element indicates that the open service API discover operation is successful. Otherwise, it shall not be present.		

8.38.3 Procedure

Figure 8.38.3-1 illustrates the procedure for open discover service APIs.

The open service API discovery mechanism is supported by the CAPIF core function.

Pre-conditions:

1. The requestor is not a recognized user of the CAPIF.
2. The CAPIF core function is aware of the Service API information that can be disseminated according to the requestor who is not a recognized user of the CAPIF.

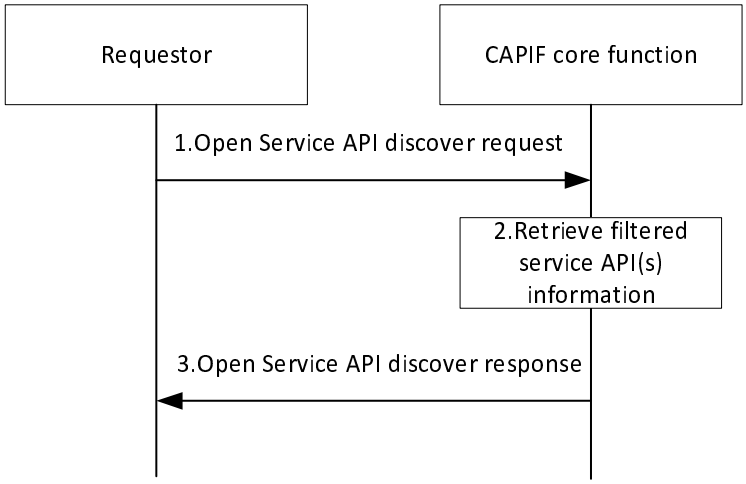


Figure 8.38.3-1: Open Discover service APIs

1. The requestor sends an open service API discover request to the CAPIF core function. It may include the query information from the requestor.

NOTE 1: The security aspects of this procedure are unspecified in this release of the specification.

2. Upon receiving the open service API discover request, if query information is included, the CAPIF core function retrieves the stored service API(s) information as per the query information in the open service API discover request, otherwise all stored service API(s) information is retrieved. Further, since the requestor is not a recognized user of CAPIF, the CAPIF core function may perform the filtering of service APIs information.

NOTE 2: The CAPIF core function could filter the service APIs information to return in the response based on implementation specific means and it is unspecified in this release of the specification.

3. The CAPIF core function sends an open service API discover response to the requestor with the filtered list of service API information.

9 API consistency guidelines

9.1 General

This clause specifies the API consistency guidelines for all northbound APIs utilizing CAPIF architecture. The guidelines are categorized as follows:

- fundamental API guidelines, applicable to all northbound APIs utilizing CAPIF; and
- architecture design considerations, applicable to all northbound APIs utilizing CAPIF.

The API guidelines are also applicable for CAPIF APIs specified in the current specification.

9.2 Fundamental API Guidelines

The specification of each northbound API utilizing the common API framework should define:

1. the function of the API;
2. the resource(s) or endpoints involved;
3. the list of supported operations and their usage;
4. the list of input and output parameters along with applicable schemas, as required;
5. the list of supported response codes;

6. the behaviour of the network entity exposing the APIs (e.g. the CAPIF core function or the API exposing function) for each supported operation;
7. the list of applicable data types; and
8. the list of applicable protocols and data serialization formats.

In order to facilitate the consistency of the northbound APIs utilizing the common API framework it is recommended to adopt the guidelines which define the following:

1. consistent nomenclature for the operations, data structures and resources/endpoints;
2. design principles for the use of operations for common tasks; and
3. a template for the consistent documentation of APIs.

The northbound APIs utilizing the common API framework should support the following properties:

1. be extensible, such that it is possible to accommodate future requirements, including vendor-specific needs;

NOTE: The extension does not replace any existing function in Northbound APIs.

2. support access control mechanisms;
3. support charging, if applicable; and
4. be backward and forward compatible with different versions of the same API.

The guidelines above are generic with regard to the API architecture. They are valid for network APIs that follow the RESTful architectural style and that expose resources towards the API invoker, as well as for network APIs of other architectures that expose general network endpoints towards the API invoker. A network endpoint represents one end of a communication channel through which the API consumer communicates with the API producer, using messages of a protocol defined by the API architecture. A resource is identified, and the corresponding endpoint is addressed, by a resource identifier (such as a URI).

9.3 Architecture design considerations

Northbound APIs utilizing common API framework should adhere to RESTful architecture, whenever possible. Service operations can use custom API operations (RPC-style interaction), when it is seen a better fit for the style of interaction to model, e.g. non-CRUD service operations.

NOTE: The selection of a particular API style is specific to each API implementation, and subject to Stage 3 scope.

The API design:

1. should have a uniform interface that conveys the resource/data model of the API to its client developers and:
   a. the implementation of the resource(s)/operations involved in the APIs should be hidden from the client, and adequate operations should be designed to operate on the resource(s)/data;
   b. any single API should be atomic;
   c. all resources/operations involved in APIs should be accessible through a common approach, and resources/data should be similarly modified using a consistent approach;
2. should allow the client (such as the API invoker) and the server (such as the CAPIF core function or the API exposing function) to evolve independently, i.e. the client should not have to be aware of the execution aspects of the APIs on the server;
3. should be stateless such that each request from the client (such as the API invoker) to the server (such as the CAPIF core function or the API exposing function) contains all of the information necessary for the server to understand the request;

4. should define the usage of standard operations, such as Create, Read, Update and Delete, consistently along with the applicable response codes;
5. should allow to label responses as cacheable or non-cacheable, to improve network efficiency by supporting caching in the client (such as the API invoker), if applicable in the API architecture;
6. should prevent unwanted modification of the resources/data during invocation of APIs; and
7. should support version control.

10 CAPIF core function APIs

10.1 General

Table 10.1-1 illustrates the CAPIF core function APIs.

Table 10.1-1: List of CAPIF core function APIs

API Name	API Operations	Known Consumer(s)	Communication Type
CAPIF_Open_Discover_Service_API	Open_Discover_Service_API	Open	Request/Response
CAPIF_Discover_Service_API	Discover_Service_API	API Invoker, CAPIF core function	Request/ Response
	Subscribe_Event	API Invoker	Request/ Response
	Update_Event_Subscription	API Invoker	Request/ Response
	Notify_Event	API Invoker	Notify
	Unsubscribe_Event	API Invoker	Request/ Response
CAPIF_Publish_Service_API	Publish_Service_API	API Publishing Function, CAPIF core function	Request/ Response
	Unpublish_Service_API	API Publishing Function, CAPIF core function	Request/ Response
	Update_Service_API	API Publishing Function, CAPIF core function	Request/ Response
	Get_Service_API	API Publishing Function, CAPIF core function	Request/ Response
	Subscribe_Event	API Publishing Function	Request/ Response
	Update_Event_Subscription	API Publishing Functionr	Request/ Response
	Notify_Event	API Publishing Function	Notify
	Unsubscribe_Event	API Publishing Function	Request/ Response
CAPIF_Events API	Subscribe_Event	API Invoker, API Publishing Function, API Management Function, API Exposing Function	Request/ Response
	Update_Event_Subscription	API Invoker, API Publishing Function, API Management Function, API Exposing Function	Request/ Response
	Notify_Event	API Invoker, API Publishing Function, API Management Function, API Exposing Function	Notify
	Unsubscribe_Event	API Invoker, API Publishing Function, API Management Function, API Exposing Function	Request/ Response
CAPIF_API_Invoker_management API	Onboard_API_Invoker	API Invoker	Request/ Response
	Offboard_API_Invoker	API Invoker	Request/ Response
	Update_API_Invoker_Details	API Invoker	Request/ Response
	Subscribe_Event	API Management Function	Request/ Response
	Update_Event_Subscription	API Management Function	Request/ Response
	Notify_Event	API Management Function	Notify
	Unsubscribe_Event	API Management Function	Request/ Response
CAPIF_API_Provider_Management_API	Register_API_Provider	API Management Function	Request/Response
	Update_API_Provider	API Management Function	Request/Response
	Deregister_API_Provider	API Management Function	Request/Response

CAPIF_Security API	Obtain_Security_Method	API Invoker	Request/ Response
	Obtain_Authorization	API Invoker	Request/ Response
	Obtain_API_Invoker_Info	API Exposing Function	Request/ Response
	Revoke_Authorization	API Exposing Function	Request/ Response
	Revoke_Authorization_Notify	API Invoker	Notify
CAPIF_Monitoring API	Subscribe_Event	API Management Function	Request/ Response
	Update_Event_Subscription	API Management Function	Request/ Response
	Notify_Monitoring_Service_Event	API Management Function	Notify
	Unsubscribe_Event	API Management Function	Request/ Response
CAPIF_Logging_API_Invocation API	Log_API_Invocation	API exposing function	Request/ Response
CAPIF_Auditing API	Query_API_Invocation_Log	API management function	Request/ Response
CAPIF_Access_Control_Policy API	Obtain_Access_Control_Policy	API exposing function	Request/Response
CAPIF_Routing_Info API	Obtain_Routing_Info	API exposing function	Request/Response

10.2 CAPIF_Discover_Service_API API

10.2.1 General

API description: This API enables the API invoker to communicate with the CAPIF core function to discover the published service API information over CAPIF-1 or CAPIF-1e.

10.2.2 Discover_Service_API operation

API operation name: Discover_Service_API

Description: Provides the published service APIs information.

Known Consumers: API invoker.

Inputs: Refer subclause 8.7.2.1.

Outputs: Refer subclause 8.7.2.2.

See subclause 8.7.3 for the details of usage of this API operation.

10.2.3 Subscribe_Event operation

API operation name: Subscribe_Event

Description: Provides subscription to the CAPIF related event information.

Known Consumers: API invoker.

Inputs: Refer subclause 8.8.2.1.

Outputs: Refer subclause 8.8.2.2.

See subclause 8.8.3 for the details of usage of this API operation.

10.2.4 Notify_Event operation

API operation name: Notify_Event

Description: Provides the relevant CAPIF event information to the subscribed entities.

Known Consumers: API invoker.

Inputs: Refer subclause 8.8.2.3.

Outputs: Refer subclause 8.8.2.4.

See subclause 8.8.4 for the details of usage of this API operation.

10.2.5 Unsubscribe_Event operation

API operation name: Unsubscribe_Event

Description: Unsubscription to the CAPIF event information.

Known Consumers: API invoker.

Inputs: Refer subclause 8.8.2.5.

Outputs: Refer subclause 8.8.2.6.

See subclause 8.8.5 for the details of usage of this API operation.

10.2.6 Update_Event_Subscription operation

API operation name: Update_Event_Subscription

Description: Updates a subscription to CAPIF related event information.

Known Consumers: API invoker.

Inputs: Refer subclause 8.8.2.7.

Outputs: Refer subclause 8.8.2.8.

See subclause 8.8.5a for the details of usage of this API operation.

10.3 CAPIF_Publish_Service_API API

10.3.1 General

API description: This API enables the API publishing function to communicate with the CAPIF core function to publish the service API information and manage the published service API information over CAPIF-4.

NOTE: Stage 3 can decide whether the API for CAPIF_Publish_Service_API can be enabled over CAPIF-4.

10.3.2 Publish_Service_API operation

API operation name: Publish_Service_API

Description: Publish the service API information.

Known Consumers: API publishing function, CAPIF core function.

Inputs: Refer subclause 8.3.2.1, 8.25.2.1.

Outputs: Refer subclause 8.3.2.2, 8.25.2.2.

See subclause 8.3.3 and 8.25.3.1 for the details of usage of this API operation.

10.3.3 Unpublish_Service_API operation

API operation name: Unpublish_Service_API

Description: Remove the published service API information.

Known Consumers: API publishing function, CAPIF core function.

Inputs: Refer subclause 8.4.2.1, 8.25.2.5.

Outputs: Refer subclause 8.4.2.2, 8.25.2.6.

See subclause 8.4.3 and 8.25.3.4 for the details of usage of this API operation.

10.3.4 Update_Service_API operation

API operation name: Update_Service_API

Description: Update the published service API information.

Known Consumers: API publishing function, CAPIF core function.

Inputs: Refer subclause 8.6.2.1, 8.25.2.9.

Outputs: Refer subclause 8.6.2.2, 8.25.2.10.

See subclause 8.6.3 and 8.25.3.6 for the details of usage of this API operation.

10.3.5 Get_Service_API operation

API operation name: Get_Service_API

Description: Retrieve the published service API information.

Known Consumers: API publishing function, CAPIF core function.

Inputs: Refer subclause 8.5.2.1, 8.25.2.7.

Outputs: Refer subclause 8.5.2.2, 8.25.2.8.

See subclause 8.5.3 and 8.25.3.5 for the details of usage of this API operation.

10.3.6 Subscribe_Event operation

API operation name: Subscribe_Event

Description: Provides subscription to the CAPIF related event information.

Known Consumers: API publishing function.

Inputs: Refer subclause 8.8.2.1.

Outputs: Refer subclause 8.8.2.2.

See subclause 8.8.3 for the details of usage of this API operation.

10.3.7 Notify_Event operation

API operation name: Notify_Event

Description: Provides the relevant CAPIF event information to the subscribed entities.

Known Consumers: API publishing function.

Inputs: Refer subclause 8.8.2.3.

Outputs: Refer subclause 8.8.2.4.

See subclause 8.8.4 for the details of usage of this API operation.

10.3.8 Unsubscribe_Event operation

API operation name: Unsubscribe_Event

Description: Unsubscription to the CAPIF event information.

Known Consumers: API publishing function.

Inputs: Refer subclause 8.8.2.5.

Outputs: Refer subclause 8.8.2.6.

See subclause 8.8.5 for the details of usage of this API operation.

10.3.9 Update_Event_Subscription operation

API operation name: Update_Event_Subscription

Description: Updates a subscription to CAPIF related event information.

Known Consumers: API publishing function.

Inputs: Refer subclause 8.8.2.7.

Outputs: Refer subclause 8.8.2.8.

See subclause 8.8.5a for the details of usage of this API operation.

10.4 CAPIF_Events API

10.4.1 General

API description: This API enables the API subscribing entity to communicate with the CAPIF core function to subscribe to and unsubscribe from CAPIF events and receive subsequent notification of CAPIF events. This API is used for the subscription to and notifications of those CAPIF events that are not bound to any of the other CAPIF core function APIs. The following are the key functionalities:

- API invoker subscribes to CAPIF events over CAPIF-1 or CAPIF-1e.
- API invoker receives notifications for subscribed CAPIF events over CAPIF-1 or CAPIF-1e.
- API invoker unsubscribes from CAPIF events over CAPIF-1 or CAPIF-1e.
- API invoker updates subscriptions for CAPIF events over CAPIF-1 or CAPIF-1e.
- API exposing function subscribes to CAPIF events over CAPIF-3.
- API exposing function receives notifications for subscribed CAPIF events over CAPIF-3.
- API exposing function unsubscribes from CAPIF events over CAPIF-3.
- API exposing function updates subscriptions for CAPIF events over CAPIF-3.
- API publishing function subscribes to CAPIF events over CAPIF-4.
- API publishing function receives notifications for subscribed CAPIF events over CAPIF-4.
- API publishing function unsubscribes from CAPIF events over CAPIF-4.
- API publishing function updates subscriptions for CAPIF events over CAPIF-4.
- API management function subscribes to CAPIF events over CAPIF-5.
- API management function receives notifications for subscribed CAPIF events over CAPIF-5.

- API management function unsubscribes from CAPIF events over CAPIF-5.
- API management function updates subscriptions for CAPIF events over CAPIF-5.

NOTE: Stage 3 can further decide if CAPIF_Events API can be further fine grained into more APIs.

10.4.2 Subscribe_Event operation

API operation name: Subscribe_Event

Description: Provides subscription to the CAPIF related event information.

Known Consumers: API invoker, API publishing function, API management function, API exposing function.

Inputs: Refer subclause 8.8.2.1.

Outputs: Refer subclause 8.8.2.2.

See subclause 8.8.3 for the details of usage of this API operation.

10.4.3 Notify_Event operation

API operation name: Notify_Event

Description: Provides the relevant CAPIF event information to the subscribed entities.

Known Consumers: API invoker, API publishing function, API management function, API exposing function.

Inputs: Refer subclause 8.8.2.3.

Outputs: Refer subclause 8.8.2.4.

See subclause 8.8.4 for the details of usage of this API operation.

10.4.4 Unsubscribe_Event operation

API operation name: Unsubscribe_Event

Description: Unsubscription to the CAPIF event information.

Known Consumers: API invoker, API publishing function, API management function, API exposing function.

Inputs: Refer subclause 8.8.2.5.

Outputs: Refer subclause 8.8.2.6.

See subclause 8.8.5 for the details of usage of this API operation.

10.4.5 Update_Event_Subscription operation

API operation name: Update_Event_Subscription

Description: Updates a subscription to CAPIF related event information.

Known Consumers: API invoker, API publishing function, API management function, API exposing function.

Inputs: Refer subclause 8.8.2.7.

Outputs: Refer subclause 8.8.2.8.

See subclause 8.8.5a for the details of usage of this API operation.

10.5 CAPIF_API_invoker_management API

10.5.1 General

API description: This API enables the API invoker to communicate with the CAPIF core function to enroll as a registered user of CAPIF and manage the enrollment information over CAPIF-1 or CAPIF-1e.

10.5.2 Onboard_API_Invoker operation

API operation name: Onboard_API_Invoker

Description: Enrolls the API invoker as a recognized user of the CAPIF.

Known Consumers: API invoker.

Inputs: Refer subclause 8.1.2.1.

Outputs: Refer subclause 8.1.2.2.

See subclause 8.1.3 for the details of usage of this API operation.

10.5.2a Update_API_Invoker_Details operation

API operation name: Update_API_Invoker_Details

Description: Enrolls/disenrolls service APIs from the API invoker profile.

Known Consumers: API invoker.

Inputs: Refer subclause 8.26.2.1.

Outputs: Refer subclause 8.26.2.2.

See subclause 8.26.3 for the details of usage of this API operation.

10.5.3 Offboard_API_Invoker operation

API operation name: Offboard_API_Invoker

Description: Cancels enrollment of the API invoker as a recognized user of the CAPIF.

Known Consumers: API invoker.

Inputs: Refer subclause 8.2.2.1.

Outputs: Refer subclause 8.2.2.2.

See subclause 8.2.3 for the details of usage of this API operation.

10.5.4 Subscribe_Event operation

API operation name: Subscribe_Event

Description: Provides subscription to the CAPIF related event information.

Known Consumers: API management function.

Inputs: Refer subclause 8.8.2.1.

Outputs: Refer subclause 8.8.2.2.

See subclause 8.8.3 for the details of usage of this API operation.

10.5.5 Notify_Event operation

API operation name: Notify_Event

Description: Provides the relevant CAPIF event information to the subscribed entities.

Known Consumers: API management function.

Inputs: Refer subclause 8.8.2.3.

Outputs: Refer subclause 8.8.2.4.

See subclause 8.8.4 for the details of usage of this API operation.

10.5.6 Unsubscribe_Event operation

API operation name: Unsubscribe_Event

Description: Unsubscription to the CAPIF event information.

Known Consumers: API management function.

Inputs: Refer subclause 8.8.2.5.

Outputs: Refer subclause 8.8.2.6.

See subclause 8.8.5 for the details of usage of this API operation.

10.5.7 Update_Event_Subscription operation

API operation name: Update_Event_Subscription

Description: Updates a subscription to CAPIF related event information.

Known Consumers: API management function.

Inputs: Refer subclause 8.8.2.7.

Outputs: Refer subclause 8.8.2.8.

See subclause 8.8.5a for the details of usage of this API operation.

10.6 CAPIF_Security API

10.6.1 General

API description: This API enables the API invoker to communicate with the CAPIF core function to authenticate and obtain authorization to access service APIs over CAPIF-1 or CAPIF-1e. This API also enables the API exposing function (AEF) to obtain API invoker information and revoke API invoker authorization over CAPIF-3.

10.6.2 Obtain_Security_Method operation

API operation name: Obtain_Security_Method

Description: Obtain information about service API security method with CAPIF core function for service API invocations.

Known Consumers: API invoker.

Inputs: Refer subclause 8.10A.2.

Outputs: Refer subclause 8.10A.2.

See subclause 8.10A.3 for the details of usage of this API operation.

10.6.3 Obtain_Authorization operation

API operation name: Obtain_Authorization

Description: Provides the authorization information to access relevant service API.

Known Consumers: API invoker.

Inputs: Refer subclause 8.11.2.

Outputs: Refer subclause 8.11.2.

See subclause 8.11.3 for the details of usage of this API operation.

10.6.4 Obtain_API_Invoker_Info operation

API operation name: Obtain_API_Invoker_Info

Description: Obtains the API invoker information.

Known Consumers: API exposing function.

Inputs: Refer subclause 8.16.2.3.

Outputs: Refer subclause 8.16.2.4.

See subclause 8.16.3 for the details of usage of this API operation.

10.6.5 Revoke_Authorization operation

API operation name: Revoke_Authorization

Description: Revokes API invoker authorization to access service API.

Known Consumers: API exposing function.

Inputs: Refer subclause 8.23.2.

Outputs: Refer subclause 8.23.2.

See subclause 8.23.3 for the details of usage of this API operation.

10.6.6 Revoke_Authorization_Notify operation

API operation name: Revoke_Authorization_Notify

Description: Provides to the API invoker authorization revocation information to access service API.

Known Consumers: API invoker.

Inputs: Refer subclause 8.23.2.3.

Outputs: Refer subclause 8.23.2.4.

See clauses 8.23.3 and 8.23.4 for the details of usage of this API operation.

10.7 CAPIF_Monitoring API

10.7.1 General

API description: This API enables the API management function to communicate with the CAPIF core function to subscribe to and unsubscribe from CAPIF events related to monitoring and receive subsequent notification of CAPIF monitoring events over CAPIF-5.

NOTE: Stage 3 can decide whether the API for CAPIF_Monitoring can be enabled over CAPIF-5.

10.7.2 Subscribe_Event operation

API operation name: Subscribe_Event

Description: Provides subscription to the CAPIF related event information.

Known Consumers: API management function.

Inputs: Refer subclause 8.8.2.1.

Outputs: Refer subclause 8.8.2.2.

See subclause 8.8.3 for the details of usage of this API operation.

10.7.3 Notify_Monitoring_Service_Event operation

API operation name: Notify_Monitoring_Service_Event

Description: Provides the notification of the events related to monitoring service API invocations to the subscribed API management function.

Known Consumers: API management function.

Inputs: Refer subclause 8.21.2.1.

Outputs: Refer subclause 8.21.2.2.

See subclause 8.21.3 for the details of usage of this API operation.

10.7.4 Unsubscribe_Event operation

API operation name: Unsubscribe_Event

Description: Unsubscription to the CAPIF event information.

Known Consumers: API management function.

Inputs: Refer subclause 8.8.2.5.

Outputs: Refer subclause 8.8.2.6.

See subclause 8.8.5 for the details of usage of this API operation.

10.7.5 Update_Event_Subscription operation

API operation name: Update_Event_Subscription

Description: Updates a subscription to CAPIF related event information.

Known Consumers: API management function.

Inputs: Refer subclause 8.8.2.7.

Outputs: Refer subclause 8.8.2.8.

See subclause 8.8.5a for the details of usage of this API operation.

10.8 CAPIF_Logging_API_Invocation API

10.8.1 General

API description: This API enables the API exposing function to communicate with the CAPIF core function to log the information related to service API invocation over CAPIF-3.

NOTE: Stage 3 can decide whether the API for CAPIF_Logging_API_Invocation can be enabled over CAPIF-3.

10.8.2 Log_API_Invocation operation

API operation name: Log_API_Invocation

Description: Enables to log API invocation information.

Known Consumers: API exposing function.

Inputs: Refer subclause 8.19.2.1.

Outputs: Refer subclause 8.19.2.2.

See subclause 8.19.3 for the details of usage of this API operation.

10.9 CAPIF_Auditing API

10.9.1 General

API description: This API enables the API management function to communicate with the CAPIF core function to retrieve the log information related to service API invocation over CAPIF-5.

NOTE: Stage 3 can decide whether the API for CAPIF_Auditing can be enabled over CAPIF-5.

10.9.2 Query_ API_Invocation_Log operation

API operation name: Query_API_Invocation_Log

Description: Query the API invocation log information.

Known Consumers: API management function.

Inputs: Refer subclause 8.22.2.1.

Outputs: Refer subclause 8.22.2.2.

See subclause 8.22.3 for the details of usage of this API operation.

10.10 CAPIF_Access_Control_Policy API

10.10.1 General

API description: This API enables the API exposing function to obtain the policy to perform access control on the service API invocations.

10.10.2 Obtain_Access_Control_Policy operation

API operation name: Obtain_Access_Control_Policy

Description: Allows obtaining the policy to perform access control on the service API invocations.

Known Consumers: API exposing function.

Inputs: Refer subclause 8.12.2.1.

Outputs: Refer subclause 8.12.2.2.

See subclause 8.12.3 for the details of usage of this API operation.

10.11 CAPIF_Routing_Info API

10.11.1 General

API description: This API enables the API exposing function to obtain the routing information to forward the API invocation to another API exposing function.

10.11.2 Obtain_Routing_Info operation

API operation name: Obtain_Routing_Info

Description: Allows obtaining the API routing information.

Known Consumers: API exposing function.

Inputs: Refer subclause 8.27.2.1.

Outputs: Refer subclause 8.27.2.2.

See subclause 8.27.3 for the details of usage of this API operation.

10.12 CAPIF_API_provider_management API

10.12.1 General

API description: This API enables the API Management Function to communicate with the CAPIF core function to register the API provider domain functions as authorized users of the CAPIF functionalities.

10.12.2 Register_API_Provider operation

API operation name: Register_API_Provider

Description: Registers the API provider domain functions as authorized users of the CAPIF.

Known Consumers: API Management Function.

Inputs: Refer subclause 8.28.2.1.

Outputs: Refer subclause 8.28.2.2.

See subclause 8.28.3 for the details of usage of this API operation.

10.12.3 Update_API_Provider operation

API operation name: Update_API_Provider

Description: Updates registration information of the API provider domain functions.

Known Consumers: API Management Function.

Inputs: Refer subclause 8.29.2.1.

Outputs: Refer subclause 8.29.2.2.

See subclause 8.29.3 for the details of usage of this API operation.

10.12.4 Deregister_API_Provider operation

API operation name: Deregister_API_Provider

Description: De-registers the API provider domain functions on the CAPIF core function.

Known Consumers: API Management Function.

Inputs: Refer subclause 8.30.2.1.

Outputs: Refer subclause 8.30.2.2.

See subclause 8.30.3 for the details of usage of this API operation.

10.13 CAPIF_Open_Discover_Service_API API

10.13.1 General

API description: This API enables the API invoker to discover service API information about the available set of APIs offered by CCF before onboarding

10.13.2 Open_Discover_Service_API operation

API operation name: Open_Discover_Service_API

Description: Provides the published service APIs information before onboarding.

Known Consumers: Open.

Inputs: Refer subclause 8.38.2.1.

Outputs: Refer subclause 8.38.2.2.

See subclause 8.38.3 for the details of usage of this API operation.

11 API exposing function APIs

11.1 General

Table 11.1-1 illustrates the API exposing function APIs.

Table 11.1-1: List of API exposing function APIs

API Name	API Operations	Known Consumer(s)	Communication Type
AEF_Security API	Revoke_Authorization	CAPIF Core Function	Request/ Response
	Initiate_Authentication	API Invoker	Request/ Response

11.2 AEF_Security API

11.2.1 General

API description: This API allows CAPIF core function to revoke access to service APIs and API invokers to request the authentication parameters necessary for authentication of the API invoker available with the API exposing function.

11.2.2 Revoke_Authorization operation

API operation name: Revoke_Authorization

Description: Revokes API invoker authorization to access service API.

Known Consumers: CAPIF core function.

Inputs: Refer subclause 8.23.2.

Outputs: Refer subclause 8.23.2.

See subclause 8.23.4 for the details of usage of this API operation.

11.2.3 Initiate_Authentication operation

API operation name: Initiate_Authentication

Description: Authentication between the API invoker and the AEF prior to service API invocation.

Known Consumers: API Invoker.

Inputs: Refer subclause 8.14.2.

Outputs: Refer subclause 8.14.2.

See subclause 8.14.3 for the details of usage of this API operation.

Annex A (informative): Overview of CAPIF operations

Depicted in figure A-1 is the overview of CAPIF operations. CAPIF operations occur between different actors involving the API invoker, the CAPIF core function, the API exposing function, the API publishing function and the API management function, and optionally the resource owner function for RNAA. High level CAPIF interactions between the actors are shown in figure A-1. This figure is only provided for illustration purposes, and does not represent the order of operations.

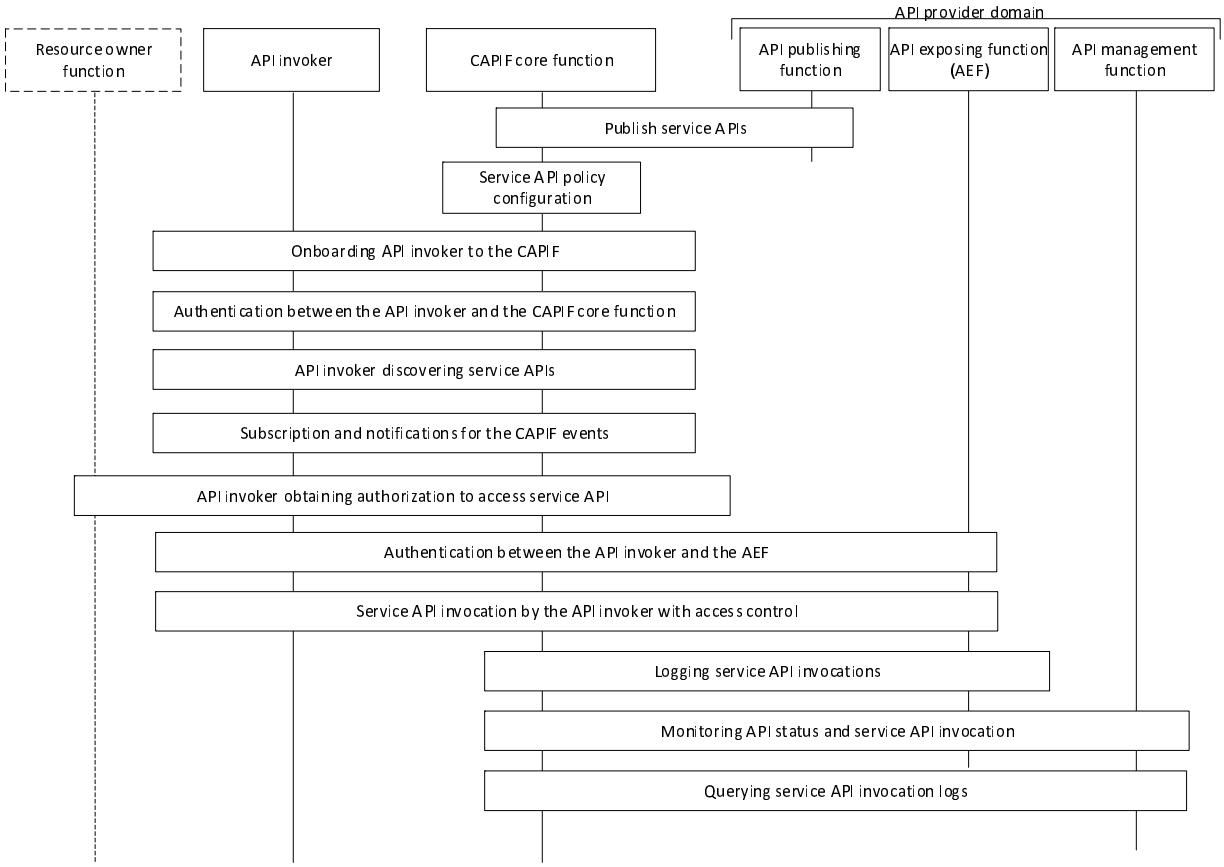


Figure A-1: Overview of CAPIF operations

The CAPIF defines the functional entities in subclause 6.3.

The CAPIF defines the reference points between the functional entities in subclause 6.4.

The following operations require the communication between the CAPIF entities:

1. Publishing service APIs: the API provider utilizes the API publishing function over CAPIF-4 reference point to publish the service APIs on the CAPIF core function, as specified in subclause 8.3 of this specification;

2. Discovering service APIs: the API invoker discovers the service APIs over CAPIF-1/CAPIF-1e reference points, as specified in subclause 8.7 of this specification;

3. API event subscription and notification: the API invoker subscribes to and receive service API event notifications over CAPIF-1/CAPIF-1e reference points, as specified in subclause 8.8 of this specification;

4. Authenticating with CAPIF: the API invoker authenticates itself over CAPIF-1/CAPIF-1e reference points, as specified in subclause 8.10 of this specification;

5. Authorizing with CAPIF: the API invoker obtains service API authorization over CAPIF-1/CAPIF-1e reference points, as specified in subclause 8.11 of this specification. In RNAA scenarios, API authorization is based on the authorization information obtained from the resource owner, as specified in clause 8.31;

6. Topology hiding: the API provider, to hide the topology, utilizes the API exposing function over CAPIF-3 reference point, as specified in subclause 8.13 of this specification;

7. Authenticating the API invoker prior to service API invocation: the API provider, to authenticate the API invoker prior to the service API invocation, utilizes the API exposing function over CAPIF-2/CAPIF-2e and CAPIF-3, as specified in subclause 8.14 of this specification;

8. Authenticating the API invoker upon the service API invocation: the API provider, to authenticate the API invoker upon invocation of the service APIs, utilizes the API exposing function over CAPIF-2/CAPIF-2e and CAPIF-3, as specified in subclause 8.15 of this specification;

9. Authorizing API invoker: the API provider, to authorize the API invoker to access the service APIs, utilizes the API exposing function over CAPIF-2/CAPIF-2e and CAPIF-3, as specified in subclause 8.16 of this specification;

10. Access control: the API provider, to control the access of the service API by the API invoker based on policy or usage limits,

    - utilizes the API exposing function over CAPIF-2/CAPIF-2e and CAPIF-3, as specified in subclause 8.17 of this specification; or

    - in a cascaded deployment, utilizes API exposing functions over CAPIF-2/CAPIF-2e, as specified in subclause 8.18 of this specification;

11. Logging service: the API provider, to maintain the log of the API invocations at the CAPIF core function for services such as invocation history, utilizes the API exposing function over CAPIF-3, as specified in subclause 8.19 of this specification;

12. Service monitoring: the API provider, to facilitate monitoring such as API invoker's ID and IP address, utilizes the API management function over CAPIF-5, as specified in subclause 8.21 of this specification; and

13. Auditing: the API provider, for auditing, utilizes the API management function over CAPIF-5, as specified in subclause 8.22 of this specification.

NOTE: Charging aspects of CAPIF framework, i.e., charging architecture and information flows, are specified in 3GPP TS 32.254 [20].

Annex B (informative): CAPIF relationship with network exposure aspects of 3GPP systems

This annex provides the relationship of CAPIF with network exposure aspects of 3GPP systems. Any system exposing capabilities as service APIs can implement CAPIF. Generic model for CAPIF utilization by service API provider is included. Network exposure aspects of EPS and 5GS are considered for illustration.

NOTE: As there are no impacts on CAPIF's relationship with network exposure aspects of 3GPP systems due to deployment of 3rd party trust domain, it is not illustrated in the figures.

B.0 CAPIF utilization by service API provider

Figure B.0-1 illustrates the service API interaction with the CAPIF for utilizing framework aspects provided by the CAPIF.

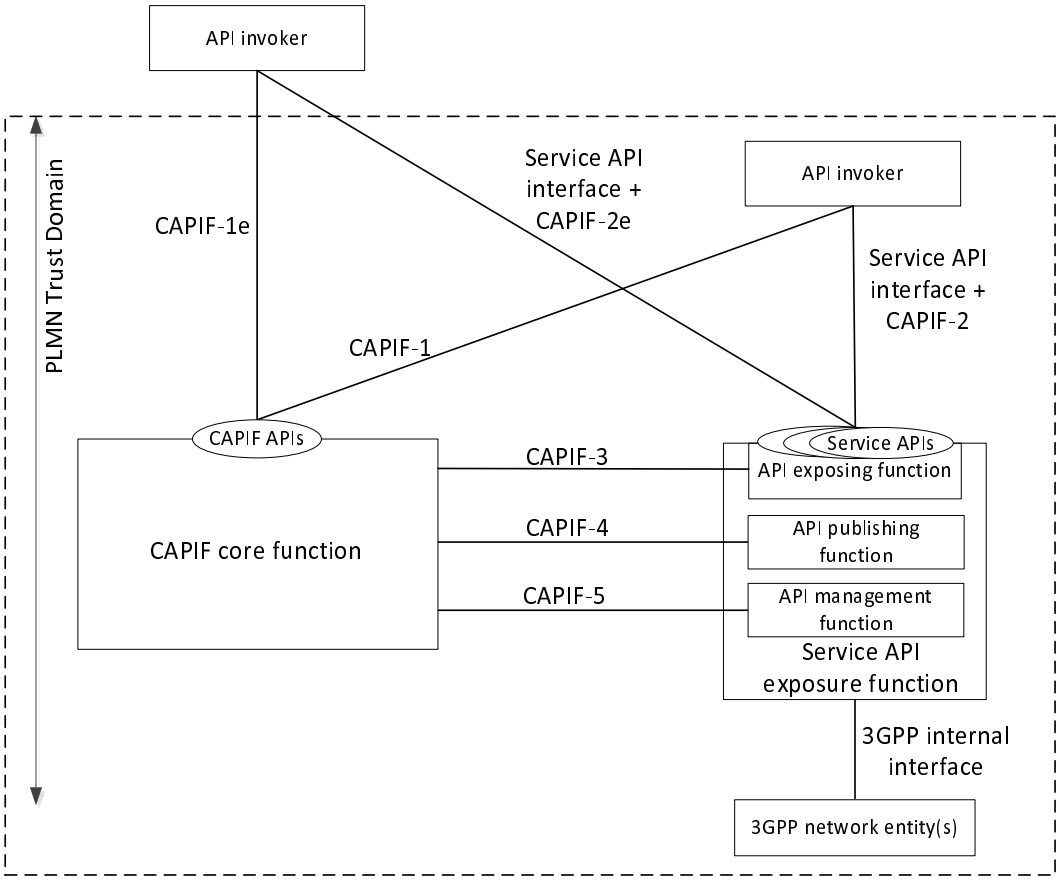


Figure B.0-1: CAPIF utilization by service API provider

The service API aspects of the 3GPP network services and capabilities such as subscriber management, mobility management, transport and other communication services can be exposed for consumption by external 3rd party applications (e.g. API invoker).

Framework aspects typically horizontal in nature caters to common functionality such as onboarding, offboarding, publishing, unpublishing, update service API, discovery, authentication, registration, authorization, logging, monitoring, configuration, topology hiding, that are required to provide service APIs to API invokers. Service APIs can utilize the

functions of the API provider domain (i.e. API exposing function, API publishing function, API management function) and interfaces CAPIF-3, CAPIF-4 and CAPIF-5 as specified in this specification.

The service API exposure function is connected to 3GPP network entity(s) via 3GPP internal interface(s). The API publishing function provides the service API information for publishing to the CAPIF core function.

For consuming service API, the API invoker interacts with the service API exposure function via service API interface and CAPIF-2/2e. While the service API interface is responsible for providing service aspects, CAPIF-2/2e supports service API by providing framework aspects such as authentication of the API invoker, authorization verification for the API invoker upon accessing the service API.

B.1 CAPIF relationship with 3GPP EPS network exposure

B.1.1 General

The table B.1.1-1 shows the relationship between CAPIF and EPS network exposure aspects. The details of SCEF and its role in exposing network capabilities of EPS to 3rd party applications are specified in 3GPP TS 23.682 [2]

Table B.1.1-1: CAPIF relationship with 3GPP EPS network exposure

Aspects	CAPIF	EPS network exposure
Entity providing the APIs to external or 3rd party applications	AEF	SCEF
Entity providing framework related services to the applications (discovery, authentication, authorization, etc)	CAPIF core function	SCEF
Entity representing the external or 3rd party applications	API invoker	SCS/AS
Entity providing framework related services to support the APIs operation and management (publish, policy enforcements)	CAPIF core function	SCEF
Interface/Reference point for exposing network capabilities as APIs	CAPIF-2 and CAPIF-2e (Do not include the service specific aspects)	T8
Interface/Reference point for exposing framework services as APIs to the applications	CAPIF-1 and CAPIF-1e	Not specified. (May be via T8)
Interface/Reference point for framework services to support the APIs operation and management	CAPIF-3, CAPIF-4 and CAPIF-5	Internal to SCEF

B.1.2 Deployment models

B.1.2.1 General

Based on the relationship captured in table B.1.1-1, the following deployment models for CAPIF are possible to enable EPS network exposure.

NOTE: The deployment models captured in subclause 7 are possible for the SCEF deployment compliant with CAPIF. Not all deployment models are illustrated in this subclause.

B.1.2.2 SCEF implements the CAPIF architecture

Figure B.1.2.2-1 illustrates the deployment model where SCEF implements the CAPIF architecture.

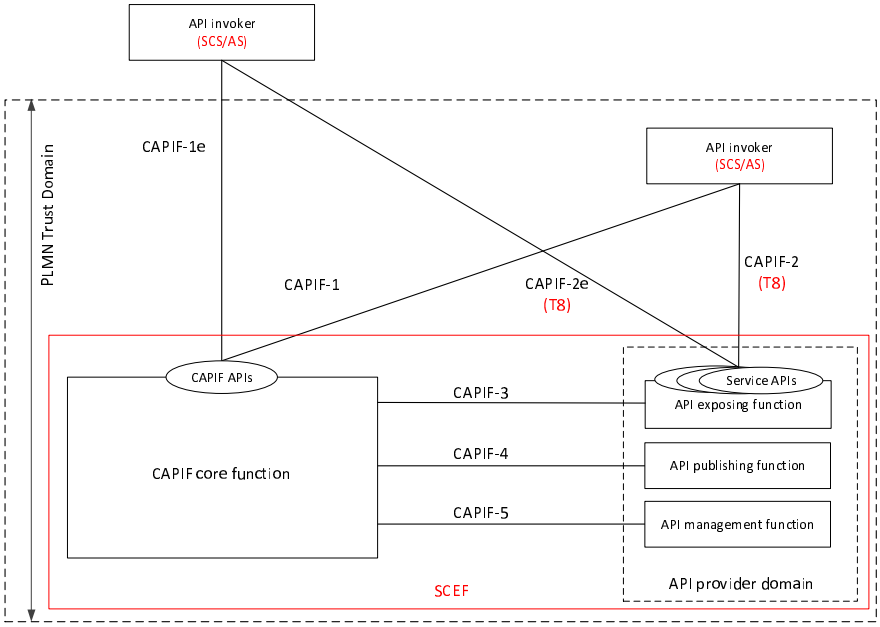


Figure B.1.2.2-1: SCEF implements the CAPIF architecture

The SCEF can implement the functionalities of the CAPIF core function, the API exposing function, the API publishing function and the API management function.

According to the CAPIF architecture, CAPIF-2 and CAPIF-2e consist of framework aspects and service specific aspects. The service specific aspects are out of scope of CAPIF. T8 can implement the service specific aspects of CAPIF-2 and CAPIF-2e, and can provide the service APIs exposed by SCEF (AEF) to the SCS/AS (API invoker).

The SCEF can additionally provide CAPIF-1 and CAPIF-1e (CAPIF APIs) to the SCS/AS (API invokers).

B.1.2.3 SCEF implements the service specific aspect compliant with the CAPIF architecture

Figure B.1.2.3-1 illustrates the deployment model where SCEF implements the service specific aspect compliant with the CAPIF architecture.

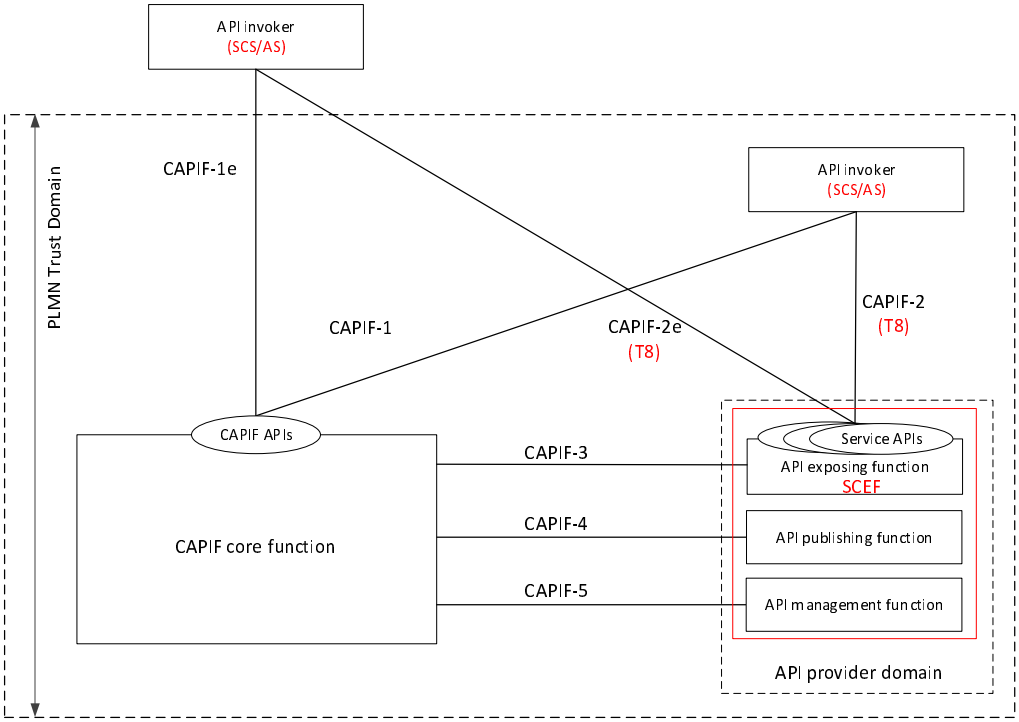


Figure B.1.2.3-1: SCEF implements the service specific aspect compliant with the CAPIF architecture

3GPP EPS can deploy the CAPIF core function along with the SCEF.

The SCEF can implement the functionalities of the API provider domain functions.

According to the CAPIF architecture, CAPIF-2 and CAPIF-2e consist of framework aspects and service specific aspects. The service specific aspects are out of scope of CAPIF. T8 can implement the service specific aspects of CAPIF-2 and CAPIF-2e, and can provide the service APIs exposed by SCEF (AEF) to the SCS/AS (API invoker).

The SCEF can implement the CAPIF-3 reference point/interface to the CAPIF core function.

B.1.2.4 Distributed deployment of the SCEF compliant with the CAPIF architecture

Figure B.1.2.4-1 illustrates the distributed deployment model where the SCEF implements the service specific aspect compliant with the CAPIF architecture.

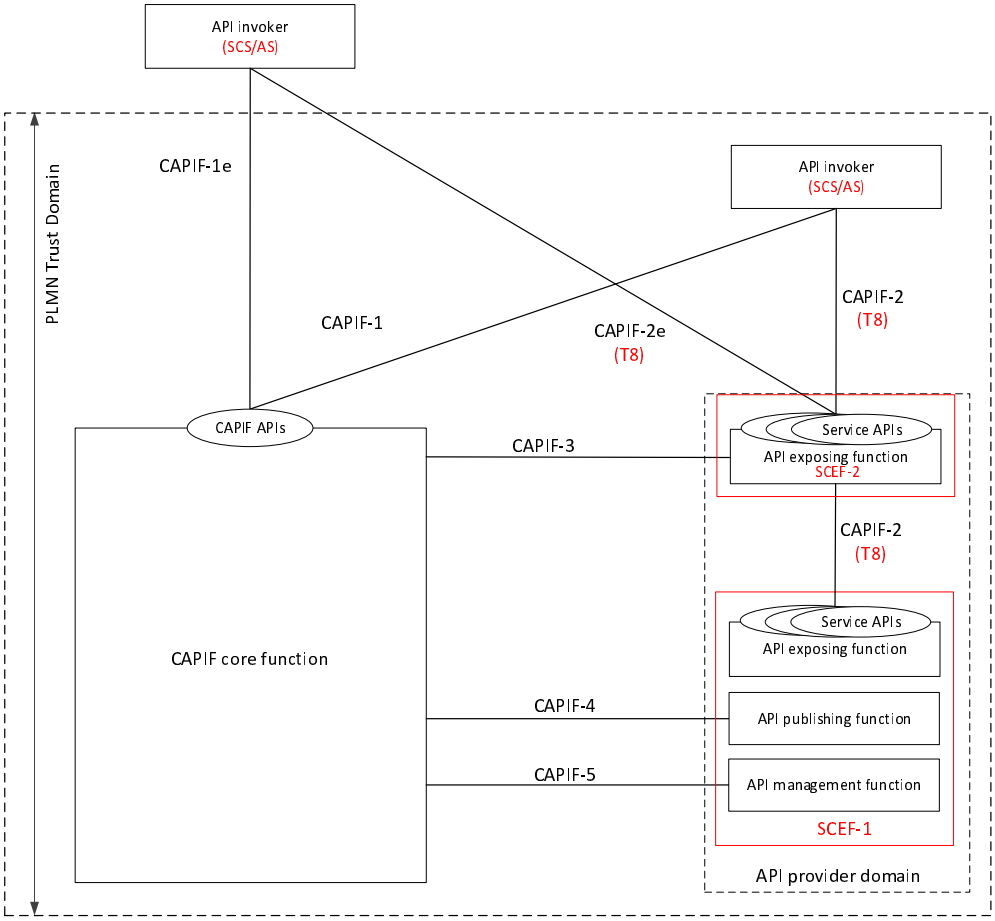


Figure B.1.2.4-1: Distributed deployment of SCEF compliant with the CAPIF architecture

The 3GPP EPS can deploy the CAPIF core function, the SCEF-2 (API exposing function as a gateway) along with the SCEF-1 as illustrated in subclause 7.3.

The SCEF can implement the functionalities of API provider domain functions.

According to the CAPIF architecture, CAPIF-2 or CAPIF-2e consists of framework aspects and service specific aspects. The service specific aspects are out of scope of the CAPIF. T8 can implement the service specific aspects of CAPIF-2 or CAPIF-2e and can provide the service APIs exposed by the SCEF-2 (AEF as a gateway) to the SCS/AS (API invoker).

The SCEF-2 can implement the CAPIF-3 reference point to the CAPIF core function and the SCEF-1 can implement the CAPIF-4 and CAPIF-5 reference points to the CAPIF core function.

B.2 CAPIF relationship with 3GPP 5GS network exposure

B.2.1 General

The table B.2.1-1 shows the relationship between CAPIF and 5GS network exposure aspects. The details of NEF and its role in exposing network capabilities of 5GS to 3rd party applications are specified in 3GPP TS 23.501 [3] and the details of NEF service operations are specified in 3GPP TS 23.502 [4].

Table B.2.1-1: CAPIF relationship with 3GPP 5GS network exposure

Aspects	CAPIF	5GS network exposure
Entity providing the APIs to external or 3rd partyapplications	AEF	NEF
Entity providing framework related services to the applications (discovery, authentication, authorization, etc)	CAPIF core function	NEF (Not specified yet)
Entity representing the external or 3rd party applications	API invoker	AF
Entity providing framework related services to support the APIs operation and management (publish, policy enforcements)	CAPIF core function	NEF (Not specified yet)
Interface/Reference point for exposing network capabilities as APIs	CAPIF-2 and CAPIF-2e (Do not include the service specific aspects)	Nnef
Interface/Reference point for exposing framework services as APIs to the applications	CAPIF-1 and CAPIF-1e	Nnef (Not specified yet)
Interface/Reference point for framework services to support the APIs operation and management	CAPIF-3, CAPIF-4 and CAPIF-5	Internal to NEF

B.2.2 Deployment models

B.2.2.1 General

Based on the relationship captured in table B.2.1-1, the following deployment models for CAPIF are possible to enable 5GS network exposure.

NOTE: The deployment models captured in subclause 7 are possible for the NEF deployment compliant with CAPIF. Not all deployment models are illustrated in this subclause.

B.2.2.2 NEF implements the CAPIF architecture

Figure B.2.2.2-1 illustrates the deployment model where the NEF implements the CAPIF architecture.

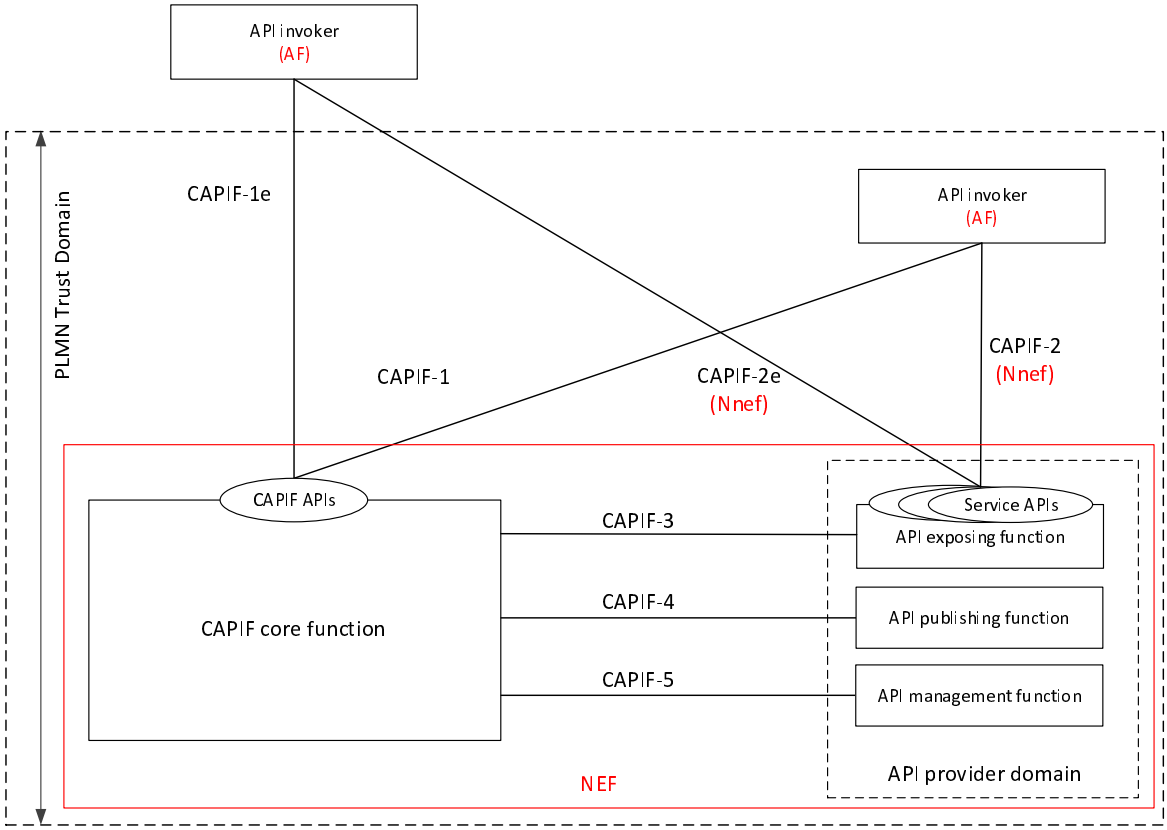


Figure B.2.2.2-1: NEF implements the CAPIF architecture

The NEF can implement the functionalities of the CAPIF core function, the API exposing function, the API publishing function and the API management function.

According to the CAPIF architecture, CAPIF-2 and CAPIF-2e consist of framework aspects and service specific aspects. The service specific aspects are out of scope of CAPIF. Nnef can implement the service specific aspects of CAPIF-2 and CAPIF-2e, and can provide the service APIs exposed by the NEF (AEF) to the AF (API invoker).

The NEF can additionally provide CAPIF-1 and CAPIF-1e (CAPIF APIs) to the AF (API invokers).

B.2.2.3 NEF implements the service specific aspect compliant with the CAPIF architecture

Figure B.2.2.3-1 illustrates the deployment model where the NEF implements the service specific aspect compliant with the CAPIF architecture.

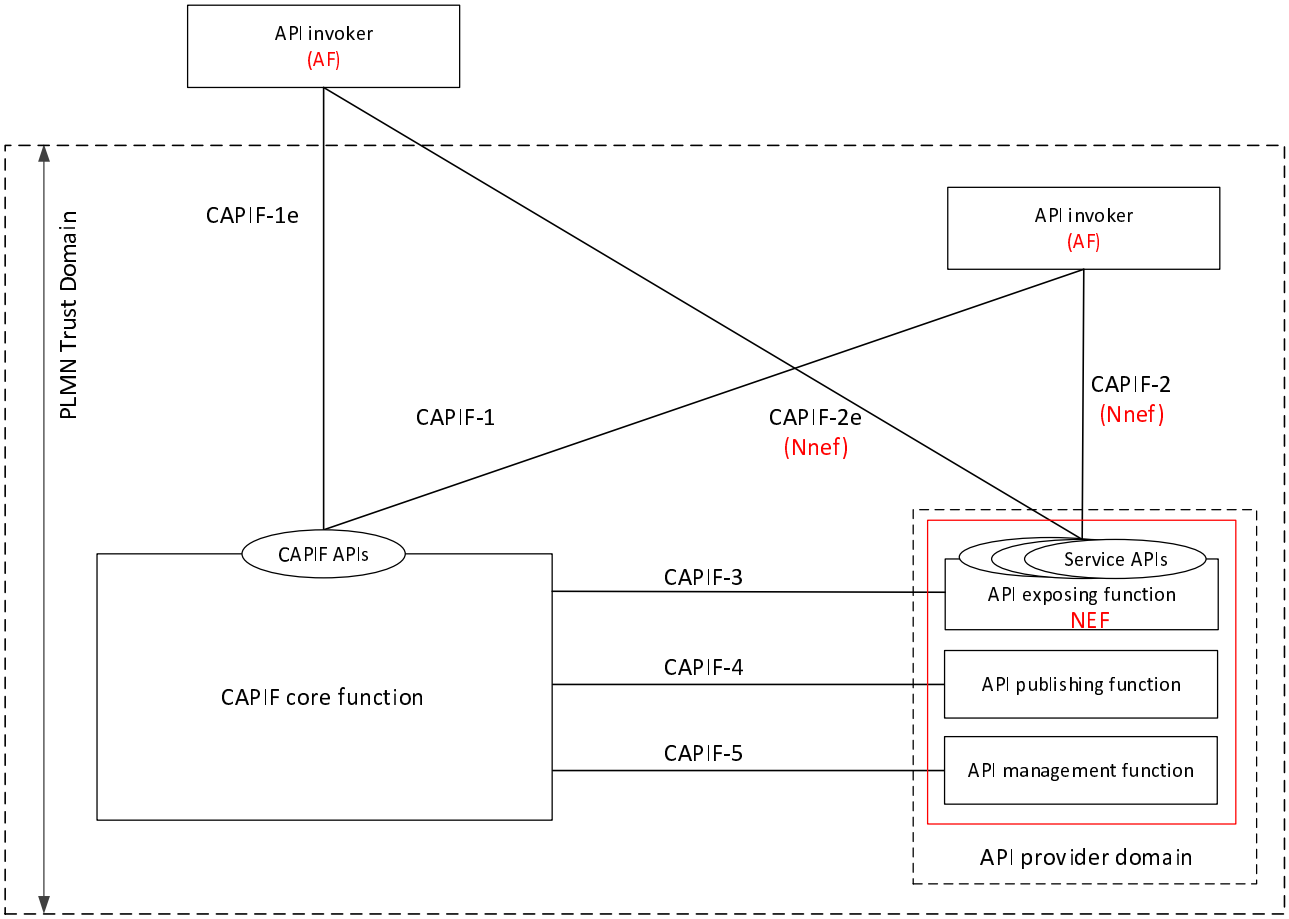


Figure B.2.2.3-1: NEF implements the service specific aspect compliant with the CAPIF architecture

3GPP 5GS can deploy the CAPIF core function along with the NEF.

The NEF can implement the functionalities of the API provider domain functions.

According to the CAPIF architecture, CAPIF-2 and CAPIF-2e consist of framework aspects and service specific aspects. The service specific aspects are out of scope of CAPIF. Nnef can implement the service specific aspects of CAPIF-2 and CAPIF-2e, and can provide the service APIs exposed by NEF (AEF) to the AF (API invoker).

The NEF can implement the CAPIF-3 reference point/interface to the CAPIF core function.

B.2.2.4 Distributed deployment of the NEF compliant with the CAPIF architecture

Figure B.2.2.4-1 illustrates the distributed deployment model where the NEF implements the service specific aspect compliant with the CAPIF architecture.

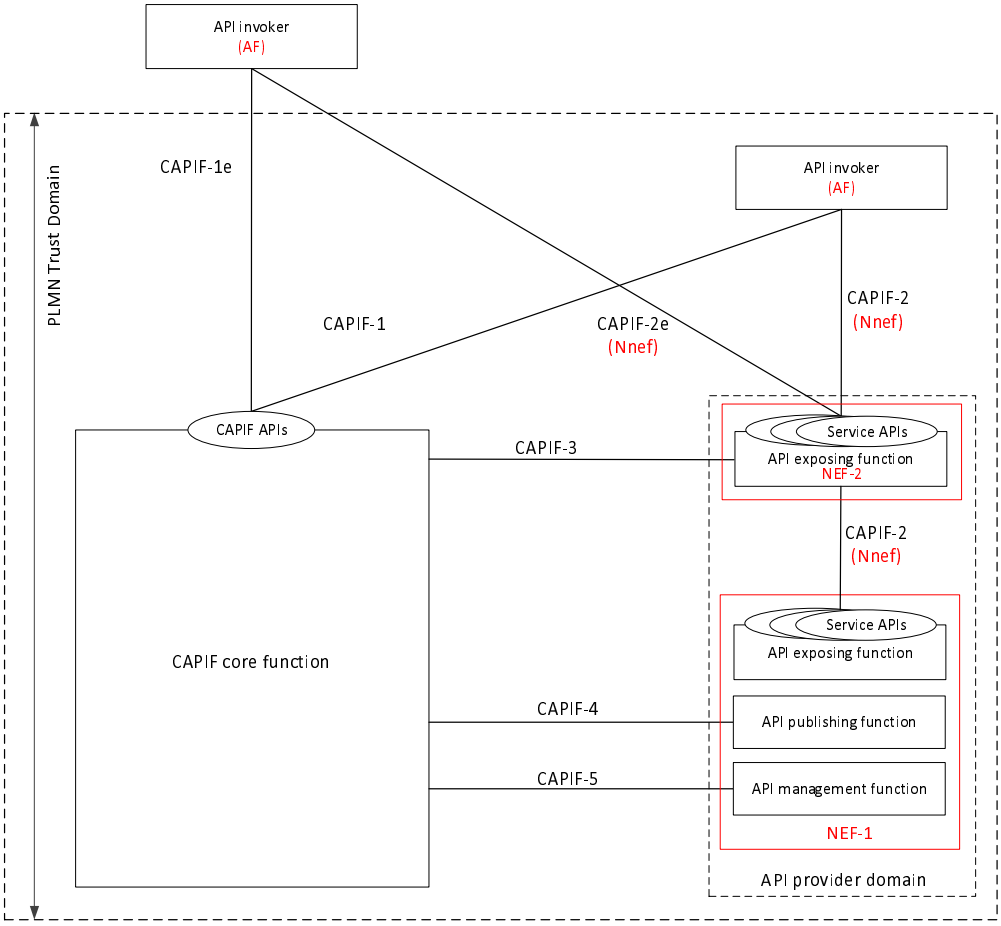


Figure B.2.2.4-1: Distributed deployment of NEF compliant with the CAPIF architecture

The 3GPP 5GS can deploy the CAPIF core function, the NEF-2 (API exposing function as a gateway) along with the NEF-1 as illustrated in subclause 7.3.

The NEF can implement the functionalities of API provider domain functions.

According to the CAPIF architecture, CAPIF-2 or CAPIF-2e consists of framework aspects and service specific aspects. The service specific aspects are out of scope of the CAPIF. Nnef can implement the service specific aspects of CAPIF-2 and CAPIF-2 or CAPIF-2e can provide the service APIs exposed by the NEF-2 (AEF as a gateway) to the AF (API invoker).

The NEF-2 (AEF) can implement the CAPIF-3 reference point to the CAPIF core function and the NEF-1 can implement the CAPIF-4 and CAPIF-5 reference points to the CAPIF core function.

B.3 Integrated deployment of 3GPP network exposure systems with the CAPIF

B.3.1 General

According to 3GPP TS 23.682 [2], when the CAPIF is supported, the SCEF supports the API provider domain functions. According to 3GPP TS 23.501 [3], when the CAPIF is supported,the NEF supports the API provider domain functions.

B.3.2 Deployment model

B.3.2.1 General

The SCEF and the NEF may be integrated with a single CAPIF core function to offer their respective service APIs to the API invokers. The following deployment model is possible for integrated deployment of the SCEF and the NEF with the CAPIF core function.

B.3.2.2 Integrated deployment of the SCEF and the NEF with the CAPIF

Figure B.3.2.2-1 illustrates integrated deployment of the SCEF and the NEF with the CAPIF.

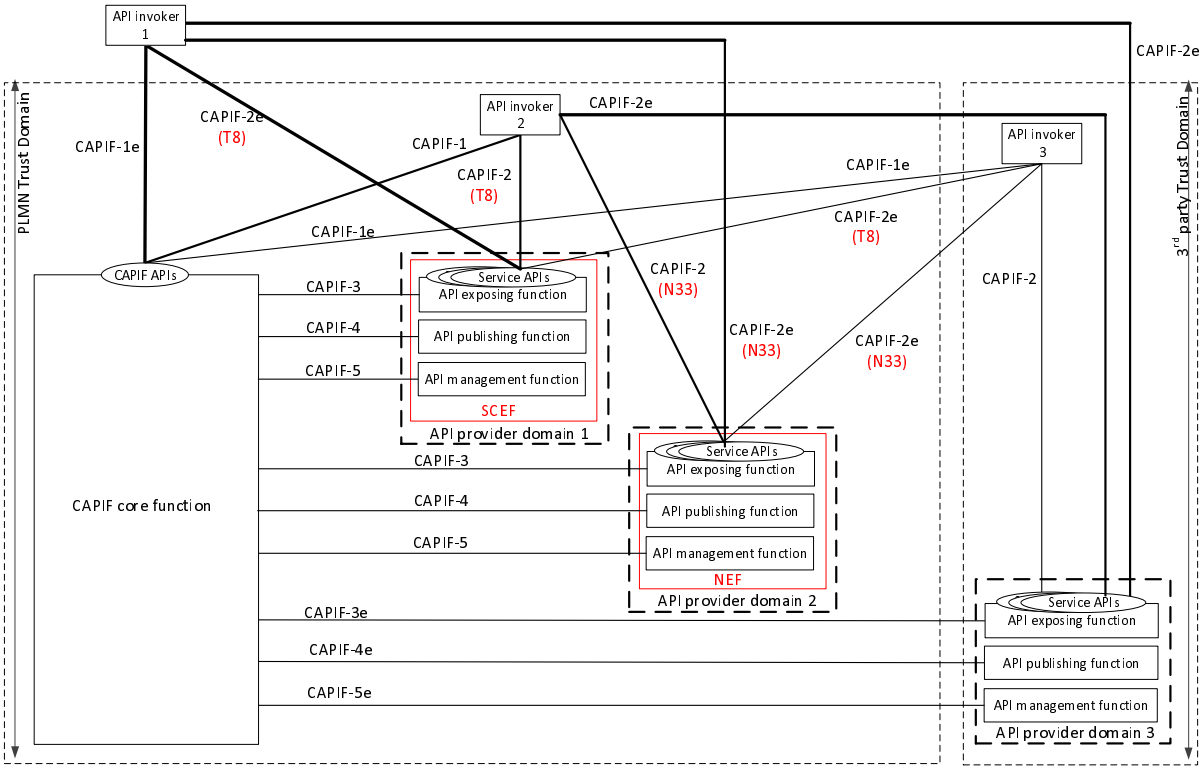


Figure B.3.2.2-1: Integrated deployment of the SCEF and the NEF with the CAPIF

The CAPIF core function, the SCEF and the NEF are deployed in the PLMN trust domain, where the CAPIF core function takes the role of a unified gateway and provides services to different API invokers. The API invokers obtains the T8 and N33 service API information and the corresponding entry point details from the CAPIF core function via CAPIF-1 or CAPIF-1e reference points.

The API invokers can interact independently with the SCEF, the NEF and the 3rd party API exposing functions via CAPIF-2 or CAPIF-2e reference points. In this case, T8 and N33 can be reused to implement the service specific aspects of CAPIF-2 or CAPIF-2e reference points for the corresponding service API interactions of the SCEF and the NEF respectively.

The SCEF and the NEF applies any service API access policy control to the interactions between the API invokers and the T8 and N33 service APIs respectively by communicating with the same CAPIF core function via the CAPIF-3 reference point.

Annex C (informative): CAPIF role in charging

C.1 General

This annex provides the information about the role of CAPIF in charging service API invocations. The common architecture for charging is illustrated in clause 4 of 3GPP TS 32.240 [6]. There are two charging mechanisms - offline charging and online charging. The role of CAPIF in both these charging mechanims is illustrated for informational purpose in this subclause. The charging management for the exposure of northbound APIs when the AEF is within the PLMN is specified in 3GPP TS 32.254 [20].

The API invocations are subjected to charging (online, offline) as described in clause 4.4 of 3GPP TS 32.254 [20].

C.2 CAPIF role in online charging

The API invocations are subjected to online charging as described in clause 5.3 of 3GPP TS 32.254 [20].

C.3 CAPIF role in offline charging

The API invocations are subjected to offline charging as described in clause 5.2 of 3GPP TS 32.254 [20].

Annex D (informative): CAPIF relationship with external API frameworks

This annex provides the relationship of CAPIF with the OMA Network APIs and the ETSI MEC API framework. The relationship of CAPIF with these external API frameworks is illustrated in the table D-1. "Yes" means that the external API framework supports the CAPIF functionality, "No" means that the API framework does not support the CAPIF functionality, and "Partial" means that it provides a mechanism that partially supports the CAPIF functionality.

Table D-1: CAPIF relationship with external API frameworks

CAPIF functionalities	OMA Network APIs		ETSI MEC API framework	
	Supported	Reference	Supported	Reference
Publish and discover service API information	Partial (see NOTE)	OMA-TS-NGSI_Registration_and_Discovery [11]	Yes	ETSI GS MEC 011 [7]
Topology hiding of the service	Yes	Individual API exposing function	Yes	Individual API exposing function
API invoker authentication to access service APIs	Partial	OMA-ER_Autho4API [9]	Partial	ETSI GS MEC 009 [8]
API invoker authorization to access service APIs	Partial	OMA-ER_Autho4API [9]	Partial	ETSI GS MEC 009 [8]
Lifecycle management of service APIs	No		No	
Monitoring service API invocations	No		No	
Logging API invoker onboarding and service API invocations	No		No	
Auditing service API invocations	No		No	
Onboarding API invoker to CAPIF	No		No	
CAPIF authentication of API invokers	No		No	
Service API access control	Partial	OMA-ER_Autho4API [9]	Partial	ETSI GS MEC 009 [8]
Secure API communication	Yes	OMA-ER_Autho4API [9]	Yes	ETSI GS MEC 009 [8]
Policy configuration	No		No	
API protocol stack model	Partial	for REST: OMA-TS_REST_NetAPI_Common [10]	Partial	for REST: ETSI GS MEC 009 [8]
API security protocol	Partial	OMA-ER_Autho4API [9]	Partial	ETSI GS MEC 009 [8]
CAPIF support for service APIs from multiple providers	No		No	
NOTE: OMA-TS-NGSI_Registration_and_Discovery [11] is only applicable to a specific type of web services (OWSER using UDDI and WSDL).				

Annex E (normative): Configuration data for CAPIF

The configuration data is stored in the CAPIF core function and provided by the CAPIF administrator.

The configuration data for CAPIF is specified in table E-1.

Table E-1: Configuration data for CAPIF

Reference	Parameter description
Subclause 4.2.2	List of published service API discovery restrictions
	> Service API identification
	> API invoker identity information
	> Service API category
Subclause 4.3.2	List of service API authorization restrictions per API invoker
	> API Invoker identity information
	> Service API identification
	> Per resource, operation authorization information. (see NOTE 2)
	> Network slice identification
Subclause 4.7.2	List of service API log storage durations
	> Service API identification
	> Service API log storage duration (in hours) (see NOTE 1)
Subclause 4.7.4	List of API invoker interactions log storage durations
	> Service API identification
	API invoker interactions log storage duration (in hours) (see NOTE 1)
Subclause 4.10	List of access control policy (for an API provider domain) per API invoker and optionally per network slice
	> Volume limit on service API invocations (total number of invocations allowed)
	> Time limit on service API invocations (The time range of the day during which the service API invocations are allowed)
	> Rate limit on service API invocations (allowed service API invocations per second)
	> Service API identification
	> API invoker identity information
	> Network Slice Info
Subclause 8.34.3	Group context information
	> Group identifier
	> List of UE IDs (group members)
	> UE identifier of the Group Resource Owner
	> Authorization information related to the list of supported applications
	>> Application identifier
	>> Purpose
	>> Scope
NOTE 1: If no value is set for the duration, the duration is assumed to be unlimited. NOTE 2: The API invoker request for access authorization to specific resources or operations of a Service API is specified in 3GPP TS 33.122 [12].	

Annex F (informative): Examples of API invoker roles in CAPIF

The figure F-1 provides examples of API invoker roles in CAPIF and illustrates the usage of CAPIF capabilities.

Application Management Client: As specified in clause 3.1, the application developers utilize the CAPIF APIs using an application management client as an API invoker to obtain the service APIs information to implement the application program. Such application programs are hosted on cloud, edge or on a UE.

BSS/OSS System: The BSS/OSS system of MNO enables the business relationship with the Application Service Providers (the consumers of the service APIs exposed by the MNO's exposure platform). The BSS/OSS system as an API invoker invoke the CAPIF APIs as per its business logic.

Channel Aggregator Platform: As specified in clause 3.1, the Channel Aggregator aggregates the APIs from one or more southbound CAPIF providers with the intention to re-expose such APIs or expose value-added APIs developed using the APIs from the southbound CAPIF providers to the northbound side Application management client or Hosted applications. The Channel Aggregator Platform as API invoker invokes the CAPIF APIs and service APIs as per its business logic.

Hosted Applications: As specified in clause 3.1, the hosted applications (which are programmed to utilize the service APIs) as an API invoker invoke the CAPIF APIs and service APIs as per the application business logic.

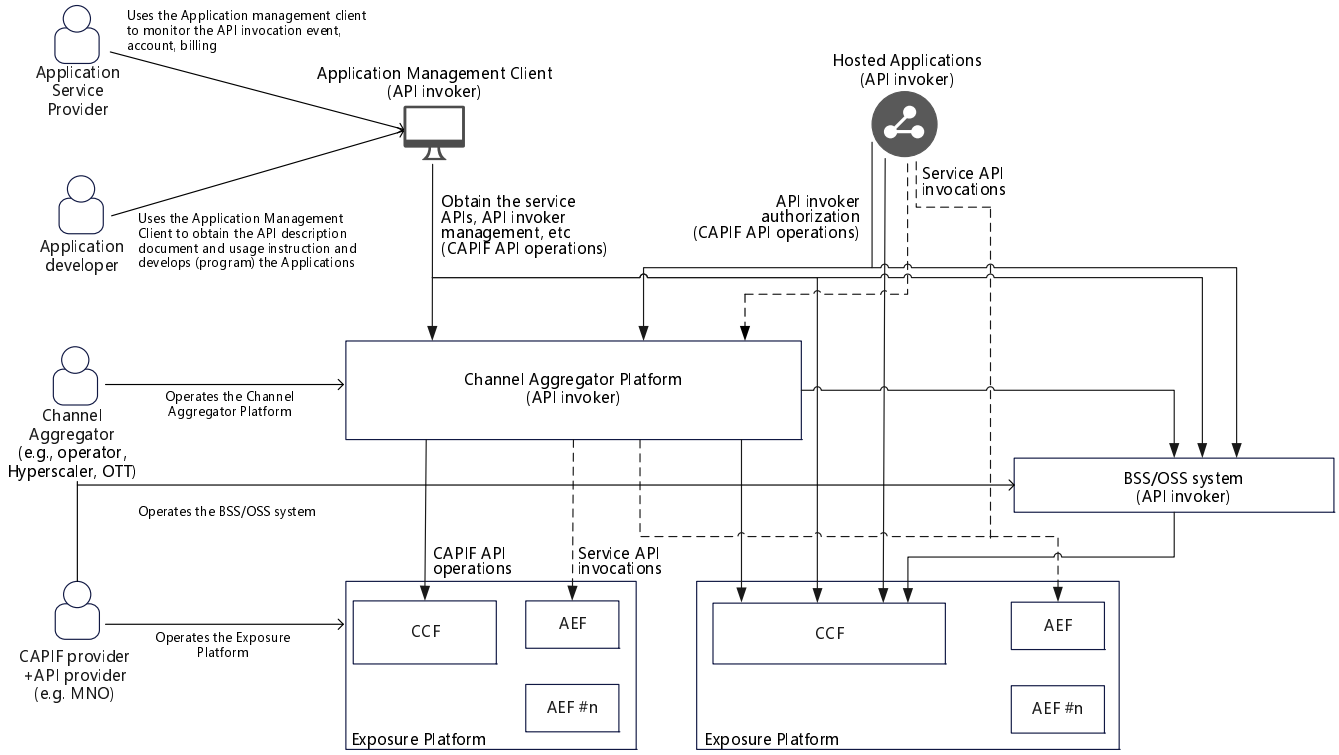


Figure F-1: Examples of API invoker roles in CAPIF

Annex G (informative): API invoker with a frontend and backend component

In clause 6.3.2 it is highlighted that the API invoker can be either an application on a server or an application on a UE. When considering the UE, the application can comprise of a frontend component on a UE plus a server-side backend component. The motivation for such an arrangement is to address security vulnerabilities associated with issuing access tokens to client-side entities (i.e., an API invoker as an application on a UE), e.g., persistent token theft.

When the API invoker is deployed with a backend component in the network, the backend component is responsible for authentication and authorization procedures towards the CCF Authorization Function. The backend component can proxy service API discovery and invocation requests initiated by the API invoker frontend. This arrangement is depicted in Figure G-1.

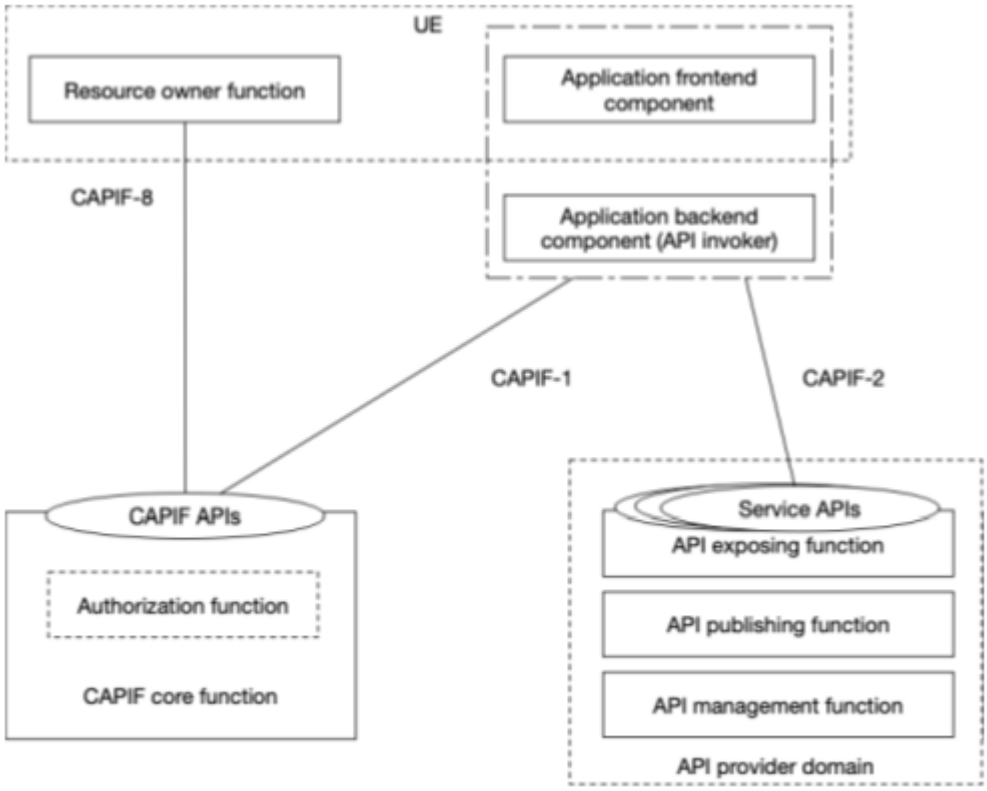


Figure G-1: High level functional RNAA architecture for CAPIF supporting an application with a frontend / backend component

NOTE: The information exchanged and the security aspects between the application frontend component and application backend component is not specified in this release of the specification.

Annex H (informative): AEF instantiation

As illustrated in figure H-1, CAPIF architecture supports AEFs from multiple API providers (e.g. MNO, 3rd party). The instances of such AEFs are managed by the respective management systems of the API provider (e.g. PLMN management system, 3rd party management system).

The AMF monitors the events reported by the CAPIF core function. These events can be utilized to trigger AEF instantiation by the PLMN management system or 3rd party management system. The AMF can utilize the services of PLMN or 3rd party management systems (e.g. trigger instantiation of corresponding AEFs) via the APIs exposed by the PLMN or 3rd party management systems. It is upto implementation of CAPIF to utilize the services of the PLMN and/or 3rd party management systems for AEF instantiation.

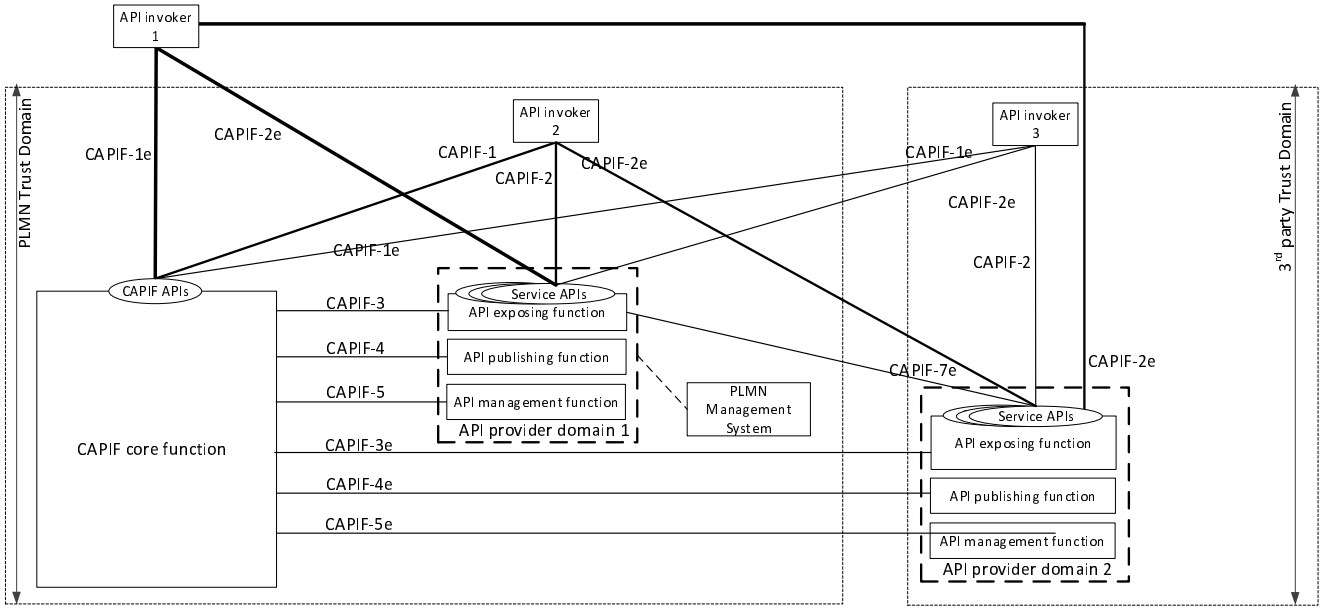


Figure H-1: AEF instantiation

Annex I (informative): Change history

Change history							
Date	Meeting	TDoc	CR	Rev	Cat	Subject/Comment	New version
2017-10	SA6#19	S6-171274				TS skeleton	0.0.0
2017-10	SA6#19					Implementation of the following p-CRs approved by SA6: S6-171444; S6-171343; S6-171445; S6-171446; S6-171466; S6-171448; S6-171348; S6-171449; S6-171359; S6-171467; S6-171451; S6-171452; S6-171362; S6-171463; S6-171356; S6-171355; S6-171453; S6-171454; S6-171455; S6-171464; S6-171468; S6-171350; S6-171349; S6-171407.	0.1.0
2017-12	SA6#20					Implementation of the following p-CRs approved by SA6: S6-171630; S6-171631; S6-171633; S6-171648; S6-171650; S6-171658; S6-171659; S6-171692; S6-171693; S6-171694; S6-171695; S6-171698; S6-171699; S6-171700; S6-171702; S6-171704; S6-171705; S6-171706; S6-171711; S6-171712; S6-171713; S6-171819; S6-171820; S6-171821; S6-171822; S6-171823; S6-171848; S6-171855; S6-171865; S6-171876.	0.2.0
2017-12	SA#78	SP-170901				Submitted to SA#78 for approval	1.0.0
2018-01	SA#78	SP-170901				MCC Editorial update for publication after TSG SA approval (SA#78)	15.0.0
2018-04	SA#79	SP-180156	0001	1	F	Use of specific ETSI and OMA references	15.1.0
2018-04	SA#79	SP-180156	0002		F	Corrections for CAPIF-1e and CAPIF-2e	15.1.0
2018-04	SA#79	SP-180156	0003		F	Miscellaneous corrections to procedures and information flows	15.1.0
2018-04	SA#79	SP-180156	0004	1	F	Addition of offboarding to functional entities and reference points description	15.1.0
2018-04	SA#79	SP-180156	0005	1	D	Editorial corrections	15.1.0
2018-04	SA#79	SP-180156	0006	2	B	Solution to EN on revoking authorization based on access control	15.1.0
2018-04	SA#79	SP-180156	0007	3	F	Configuration items for CAPIF	15.1.0
2018-04	SA#79	SP-180156	0008	3	F	Update to CAPIF relationship with 3GPP EPS and 5GS	15.1.0
2018-04	SA#79	SP-180156	0009	1	F	Solution to EN on policy synchronization	15.1.0
2018-04	SA#79	SP-180156	0010	2	F	CAPIF utilization by service APIs	15.1.0
2018-04	SA#79	SP-180156	0011	1	F	Proposal for definition for PLMN trust domain	15.1.0
2018-06	SA#80	SP-180374	0013	1	F	Correction for the details of service API information	15.2.0
2018-06	SA#80	SP-180374	0014	1	F	Correction for usage of service API identification information	15.2.0
2018-06	SA#80	SP-180374	0019	2	D	Editorial correction of TS 23.222 (CAPIF stage2)	15.2.0
2018-06	SA#80	SP-180375	0012	2	B	Architecture functional model to support multiple API providers	16.0.0
2018-06	SA#80	SP-180375	0015	1	B	Service API publish and discovery requirements for 3rd party API providers	16.0.0
2018-06	SA#80	SP-180375	0016	1	B	Charging requirements for 3rd party API providers	16.0.0
2018-06	SA#80	SP-180375	0017	1	B	OAM requirements for 3rd party API providers	16.0.0
2018-06	SA#80	SP-180375	0018	2	B	CAPIF interconnection requirements	16.0.0
2018-06	SA#80	SP-180375	0020	2	F	Updating representation of deployment models	16.0.0
2018-09	SA#81	SP-180675	0021	2	B	Integrated CAPIF with 3GPP EPS and 5GS network exposure	16.1.0
2018-09	SA#81	SP-180675	0022	1	C	Enhancement to the functional model deployments	16.1.0
2018-09	SA#81	SP-180675	0023	2	B	Enhancement to reference points for eCAPIF	16.1.0
2018-09	SA#81	SP-180674	0029	1	A	Update API naming convention	16.1.0
2018-09	SA#81	SP-180674	0030	2	A	Alignment of APIs	16.1.0
2018-09	SA#81	SP-180674	0031	1	A	Alignment to SA3 CAPIF TS	16.1.0
2018-09	SA#81	SP-180674	0032	1		Alignment to SA3 authentication procedure	16.1.0
2018-09	SA#81	SP-180675	0033	3	B	Functional architecture for CAPIF interconnection	16.1.0
2018-12	SA#82	SP-181176	0034	3	B	Topology hiding enhancement	16.2.0
2018-12	SA#82	SP-181176	0035	2	B	API publish and API discover for CAPIF interconnection	16.2.0
2018-12	SA#82	SP-181176	0036	1	C	Architectural requirements for identities	16.2.0
2018-12	SA#82	SP-181176	0038	2	B	Architectural requirements for provider domain entities interaction	16.2.0
2018-12	SA#82	SP-181176	0039	2	B	Update API invoker API list	16.2.0
2018-12	SA#82	SP-181175	0043	2	A	API invoker's onboarding response rel16	16.2.0
2019-03	SA#83	SP-190072	0044	2	F	Update procedures with topology hidding	16.3.0
2019-03	SA#83	SP-190072	0045	2	B	API sharing for CCF interconnection	16.3.0
2019-03	SA#83	SP-190072	0046	2	B	API invocation request routing with topology hiding	16.3.0
2019-03	SA#83	SP-190072	0048	1	C	Interactions between API exposing functions	16.3.0
2019-03	SA#83	SP-190072	0049	1	B	Service API discovery involving multiple CCFs	16.3.0
2019-03	SA#83	SP-190072	0050	2	B	Multiple CCFs deployment in a PLMN trust domain	16.3.0
2019-03	SA#83	SP-190072	0051	2	B	Service API discover for CAPIF interconnection	16.3.0
2019-03	SA#83	SP-190072	0052	1	B	Architectural requirements for registration of API provider domain functions	16.3.0
2019-03	SA#83	SP-190072	0053	2	B	Procedures for registration of API provider domain functions	16.3.0
2019-03	SA#83	SP-190072	0054	1	B	Updates to AEF procedures for 3rd party trust domain	16.3.0
2019-03	SA#83	SP-190072	0055	1	B	Updates to APF procedures for 3rd party trust domain	16.3.0
2019-03	SA#83	SP-190072	0056	1	B	Updates to AMF procedures for 3rd party trust domain	16.3.0
2019-03	SA#83	SP-190072	0057	-	B	Updates to CAPIF events procedures for 3rd party trust domain	16.3.0
2019-06	SA#84	SP-190483	0058	1	F	Clarification to routing rule of service API invocation	16.4.0
2019-06	SA#84	SP-190483	0059	3	F	Functional model update with reference points	16.4.0

2019-06	SA#84	SP-190483	0060	2	B	Update to service API publish for CAPIF interconnection	16.4.0
2019-06	SA#84	SP-190483	0061	2	B	Serving area and domain of service API for CAPIF interconnection	16.4.0
2019-06	SA#84	SP-190483	0062	1	B	3rd party trust domain with network exposure and charging aspects of 3GPP systems	16.4.0
2019-06	SA#84	SP-190483	0063	1	B	Interface based representation of CAPIF architecture	16.4.0
2019-09	SA#85	SP-190828	0064	1	F	Clarification and alignment with publish request information flows	16.5.0
2019-12	SA#86	SP-191107	0065		F	Correction on usage of service API information in access control message	16.6.0
2020-03	SA#87-E	SP-200112	0066	1	F	Shared CAPIF provider domain info in interconnection	16.7.0
2020-03	SA#87-E	SP-200116	0067	2	B	Serving area information for service APIs to support edge applications	17.0.0
2020-07	SA#88-E	SP-200337	0069		A	Add consumer for discover and publish service APIs	17.1.0
2020-07	SA#88-E	SP-200337	0071		A	Add obtaining routing info service API	17.1.0
2020-07	SA#88-E	SP-200337	0073	1	A	Correct API topology hiding	17.1.0
2020-07	SA#88-E	SP-200337	0075	1	A	Correction for CAPIF interconnection IEs	17.1.0
2020-09	SA#89-E	SP-200840	0077	3	F	Correction for API routing information	17.2.0
2020-12	SA#90-E	SP-200997	0078	2	B	Support AEF location and API invoker interface for edge application	17.3.0
2021-04	SA#91-E	SP-210183	0079	4	F	Clarification of Service-based interfaces interaction within CAPIF	17.4.0
2021-06	SA#92-E	SP-210482	0082		A	API provider management API	17.5.0
2022-06	SA#96	SP-220471	0084	1	A	Corrections to API invoker onboarding/offboarding in TS 23.222	17.6.0
2022-09	SA#97	SP-220918	0089		A	Corrections to API invoker onboarding/offboarding in TS 23.222	17.7.0
2022-12	SA#98-e	SP-221250	0090	1	B	Additional CAPIF architectural requirements for SNA	18.0.0
2022-12	SA#98-e	SP-221250	0091	2	B	CAPIF business relationship updates for SNA	18.0.0
2022-12	SA#98-e	SP-221250	0092	2	B	CAPIF functional model updates for SNA	18.0.0
2022-12	SA#98-e	SP-221250	0093	2	B	API invoker obtaining authorization from resource owner	18.0.0
2022-12	SA#98-e	SP-221250	0094	1	B	Discover a proper AEF with owner information	18.0.0
2022-12	SA#98-e	SP-221250	0095	2	B	Reducing resource owner consent inquiry in a nested API invocation	18.0.0
2022-12	SA#98-e	SP-221239	0096	2	B	CAPIF extensibility as requested by ETSI ISG MEC	18.0.0
2023-03	SA#99	SP-230295	0098		B	Discover proper AEF in interconnection	18.1.0
2023-03	SA#99	SP-230296	0099	1	B	Solve CAPIF extensibility EN	18.1.0
2023-03	SA#99	SP-230295	0100	1	B	API invoker clarification	18.1.0
2023-03	SA#99	SP-230295	0101	1	D	Modify a terminology for SNA	18.1.0
2023-03	SA#99	SP-230286	0102	2	B	New IE(Service KPI) in Service API publish request	18.1.0
2023-03	SA#99	SP-230295	0103	2	B	Discover proper AEF with IP information	18.1.0
2023-03	SA#99	SP-230292	0104		B	Support onboarding expiration	18.1.0
2023-03	SA#99	SP-230296	0105	2	F	Resolving editor's notes about TS reference	18.1.0
2023-03	SA#99	SP-230295	0106	2	B	Adding descriptions of new functional entities and reference points	18.1.0
2023-06	SA#100	SP-230714	0109	2	B	Support CAPIF in SNPN	18.2.0
2023-06	SA#100	SP-230712	0111	4	B	Service API status monitoring	18.2.0
2023-06	SA#100	SP-230713	0112		B	Clarification that RNAA is for both 4G and 5G	18.2.0
2023-06	SA#100	SP-230713	0113	2	B	SNAAPP alignment with SA3	18.2.0
2023-06	SA#100	SP-230713	0114	2	B	Overview of CAPIF operations for RNAA scenarios	18.2.0
2023-06	SA#100	SP-230714	0115	3	B	CAPIF add service procedure for update of subscriptions	18.2.0
2023-06	SA#100	SP-230714	0116	5	B	Alignment among CAPIF provider (trust) domains	18.2.0
2023-12	SA#102	SP-231547	0122	2	A	Editorial corrections	18.3.0
2023-12	SA#102	SP-231569	0125	1	F	Solve EN related to SA3	18.3.0
2023-12	SA#102	SP-231569	0126	2	F	Add response to RNAA procedural flows and correct cross-references	18.3.0
2023-12	SA#102	SP-231569	0128	2	F	Clarify how to monitor service API status when the APF is unable to update service API status	18.3.0
2023-12	SA#102	SP-231569	0129	2	F	Add CAPIF words to Abbreviations	18.3.0
2023-12	SA#102	SP-231569	0133	6	F	CAPIF Architecture alignment with SA3 RNAA aspects	18.3.0
2023-12	SA#102	SP-231546	0136		A	API Description Correction	18.3.0
2023-12	SA#102	SP-231569	0137	1	F	API invoker authorization corrections	18.3.0
2023-12	SA#102	SP-231545	0141		A	Security API corrections	18.3.0
2023-12	SA#102	SP-231569	0147	2	F	Editorial corrections regarding RNAA	18.3.0
2023-12	SA#102	SP-231569	0149	2	F	Corrections for CAPIF revocation of API Invoker's authorization based on RNAA	18.3.0
2023-12	SA#102	SP-231569	0150	1	F	Corrections for CAPIF deployment models supporting RNAA	18.3.0
2023-12	SA#102	SP-231570	0131	2	B	Slice-based service API exposure	19.0.0
2023-12	SA#102	SP-231570	0144	1	F	Service API unpublish for CAPIF interconnection	19.0.0
2023-12	SA#102	SP-231570	0145	1	F	Service API retrieval for CAPIF interconnection	19.0.0
2023-12	SA#102	SP-231570	0146	3	F	Service API update for CAPIF interconnection	19.0.0
2024-03	SA#103	SP-240317	0151	1	D	Align cross-references to the appropriate subclauses of TS 33.122	19.1.0
2024-03	SA#103	SP-240317	0152	1	F	Responsibilities of CAPIF API provider domain functions	19.1.0
2024-03	SA#103	SP-240310	0154	2	A	Consistent use of term "resource owner"	19.1.0
2024-03	SA#103	SP-240317	0157	2	D	Editorial correction for topology hiding	19.1.0
2024-03	SA#103	SP-240317	0158	1	F	Correction cardinality IE service API unpublish for CAPIF interconnection	19.1.0
2024-06	SA#104	SP-240755	0162	1	A	Corrections to Deregister_API_Provider operation	19.2.0
2024-06	SA#104	SP-240755	0167	2	A	Correction for Discover service APIs	19.2.0
2024-06	SA#104	SP-240764	0169	1	A	Add missing function to resource owner function	19.2.0

2024-06	SA#104	SP-240756	0174	3	A	Alignment of "API type" with "API category" terminology	19.2.0
2024-06	SA#104	SP-240755	0180	2	A	Correction for Service API discovery involving multiple CCFs	19.2.0
2024-06	SA#104	SP-240764	0181	1	A	Reducing authorization information inquiry in a nested API invocation	19.2.0
2024-06	SA#104	SP-240764	0185	3	A	Correction to clause 6.2.3	19.2.0
2024-09	SA#105	SP-241213	0188	2	A	Correction to clause 8.3.2.1	19.3.0
2024-09	SA#105	SP-241213	0190	2	A	Correction to Interconnection API publish	19.3.0
2024-09	SA#105	SP-241213	0192	2	A	Correction to Interconnection service API discover	19.3.0
2024-09	SA#105	SP-241217	0194	2	A	Rel-19 Add to Definitions and Abbreviations	19.3.0
2024-09	SA#105	SP-241217	0196	1	A	Rel-19 Correction on Resource owner	19.3.0
2024-09	SA#105	SP-241228	0197	3	F	Update the CAPIF business relationships	19.3.0
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2024-12	SA#106	SP-241714	0205	4	B	Requesting of collective resource owner authorization	19.4.0
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2024-12	SA#106	SP-241712	0218	1	A	API invoker obtaining authorization from resource owner	19.4.0
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2025-03	SA#107	SP-250208	0265	2	B	Group Information Provisioning in CAPIF to support RNAA	19.5.0
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2025-06	SA#108	SP-250596	0267	1	F	Enhanced AEF Capabilities Description	19.6.0
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2025-06	SA#108	SP-250596	0271	1	F	Remove the Note in Functional model description to support RNAA	19.6.0
2025-06	SA#108	SP-250596	0272	1	F	Remove the limitation in the introduction section of the RNAA architectural requirements	19.6.0
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2025-06	SA#108	SP-250596	0278	1	F	Corrections to RNAA deployments	19.6.0
2025-06	SA#108	SP-250617	0280	1	A	Correction to Preconditions of nested API authorization	19.6.0
2025-06	SA#108	SP-250594	0281	1	F	Correction to access control policy	19.6.0

2025-06	SA#108	SP-250596	0282	2	F	CCF obtaining RO authorization information	19.6.0
2025-06	SA#108	SP-250596	0283	3	F	Correction on Revocation on CAPIF interconnection	19.6.0
2025-06	SA#108	SP-250596	0284	3	F	AEF1 interaction with CCF for token exchange	19.6.0
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2025-06	SA#108	SP-250596	0287	1	F	Correction on Obtain service API authorization for RNAA	19.6.0
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2025-06	SA#108	SP-250596	0303	1	F	Enhanced API invoker obtaining authorization for service API access in CAPIF interconnection	19.6.0
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2025-06	SA#108	SP-250596	0305	1	F	Enhanced Open Service API discover Procedure	19.6.0
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2025-06	SA#108	SP-250596	0307	2	F	Revocation of authorization	19.6.0
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2025-06	SA#108	SP-250596	0309	1	F	Completion of RNAA functional model	19.6.0
2025-06	SA#108	SP-250596	0310	1	F	Obtaining collective authorization	19.6.0
2025-09	SA#109	SP-251052	0314	1	F	Proposal for AEF instantiation support in CAPIF	19.7.0
2025-09	SA#109	SP-251066	0315	2	F	Corrections to Revocation of authorization	19.7.0
2025-09	SA#109	SP-251052	0316	1	F	Solving ENs on granularity of service authorization	19.7.0
2025-09	SA#109	SP-251052	0317	1	F	Charging the invocation of service APIs	19.7.0
2025-09	SA#109	SP-251052	0318	1	F	Solving Editor's Note on Open Discovery	19.7.0
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2025-09	SA#109	SP-251052	0320	1	F	Solving Editor's Notes on Resource Owner authorization	19.7.0
2025-09	SA#109	SP-251052	0321	2	F	Solving ENs in authorization related to network slice	19.7.0
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2026-01	SA#110	SP-251473	0326	1	F	Correction to CAPIF-7/7e interfaces	19.8.0
2026-01	SA#110	SP-251473	0327	2	F	Completion of Open Discovery	19.8.0
2026-01	SA#110	SP-251473	0328	1	F	Completion of Update API invoker's API list	19.8.0
2026-01	SA#110	SP-251471	0336	2	F	Correction to Obtain API Invoker Information	19.8.0
2026-03	SA#111	SP-260193	0337	1	F	Correction of the Event Reporting Information	19.9.0
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2026-07	SA#112	SP-260576	0352	1	A	Corrections to RNAA	19.10.0

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